

# Traceability Matrix

REMEMBER GORDON! – Rulebook  $\rightleftharpoons$  Implementation Mapping

Generated from `docs/traceability.toml`

## Overview

| Implemented | Descriptive | Implicit | Out-of-scope |
|-------------|-------------|----------|--------------|
| 85          | 32          | 4        | 3            |

Total mappings: 124 · Total impl sites: 252

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 1/3 implemented

# §1 – Introduction

## §1 – Introduction

**descriptive**  
manual page unknown

### Introduction

## §1.1 – General Comments

**implemented**  
manual page 1

General Comments

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is a simulation of the final battle in Great Britain’s two-year campaign to reassert her presence in the Sudan (1896–1898). Fought September 2nd, 1898, Omdurman finally broke the back of the fanatical Dervish rebellion and gained Britain a million square miles of desolate territory and two million impoverished subjects. With two players, one assumes the role of Herbert Kitchener, Sirdar (CIC) of the Anglo-Egyptian army; the other player becomes the Khalifa, Abdullah the Taiasha, absolute ruler of the Dervishes. The game is also suited for multi-player participation, with each player assuming command of one or more Dervish tribes or Anglo-Egyptian brigades.

While “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is not, strictly speaking, a beginner’s game, the mechanics of play should be familiar to players of modest experience. It is suggested that the bonus game, FALL OF KHARTOUM, and the historical scenario be played first to familiarize players with the game system prior (see manual for full text)

| File                                                                                                  | Symbol                   | Code Snippet                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/unit_profiles.rs:194</a><br>GH:omdurman-rules/src/unit_profiles.rs:194 | <b>command_owns_unit</b> | <pre>192   /// treat *nobody-claims-it* units as the faction's communal pool, so `false` 193   /// here means "any member of the faction may act on this unit". 194   pub fn command_owns_unit(scope: &amp;omdurman_types::CommandScope, identity: &amp;UnitIdentity) -&gt; bool { 195       match scope { 196           omdurman_types::CommandScope::Army =&gt; false,</pre> |

Covered by tests:

- [omdurman-rules::src::unit\\_profiles::command\\_owns\\_unit\\_follows\\_tribes\\_and\\_brigades](#)
- [omdurman-net::src::lib::start\\_game\\_commands\\_round\\_trip](#)
- [omdurman-net::src::lib::legacy\\_start\\_game\\_without\\_commands\\_still\\_loads](#)
- [omdurman-net::src::lib::brigade\\_scope\\_display\\_and\\_membership](#)
- [omdurman-app::src::tests::replayed\\_start\\_game\\_stages\\_commands\\_and\\_resets\\_ready](#)

## §1.2 – Game Scale

**descriptive**  
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Game Scale

Each hexagon of the mapsheet represents approximately 400–440 yards of real terrain and each day turn is the equivalent of two hours of real time. Each counter of infantry and cavalry represents between 400 and 700 men, and each of the gunboats present at the battle has its own counter. The upper echelon of command is represented by individual leader counters for the Anglo-Egyptian force; and leaders plus their retinues for the Dervish army.

## §2 – Game Components

### §2 – Game Components

descriptive

manual page unknown

#### Game Components

### §2.1 – The Game Maps

descriptive

manual page unknown

#### The Game Maps

The Omdurman battle map represents approximately 100 square miles of real territory and includes the area north of Omdurman in which the historical battle took place as well as the dominant terrain features that influenced the course of the battle. Note that the mapsheet also contains the Turn Record Track, Turn Sequence, and Terrain Effects Chart at the top; and the Combat Tables and Howitzer Fire Scattergram in the lower right corner. The large letters “A”, “D”, “Y”, etc. are set-up hexes for the historical scenario only (9.2) and should be ignored in the campaign game. Similarly, the hexsides of the Zariba exist only in the historical scenario and should be considered clear terrain in the campaign game. Note, however, that the Anglo-Egyptian player may “construct” the Zariba in the campaign game if desired (see 5.3). All full hexes of the Omdurman game map are playable, including the seven hexes of the Howitzer Fire Scattergram.

The mini-map for the bonus game, FALL OF KHARTOUM, represents that city as it appeared in 1885. The portion (see manual for full text)

See also: §5.3, §9.2

### §2.2 – Play Aids

descriptive

manual page unknown

#### Play Aids

Certain charts and tables are needed to play the game. The Terrain Effects Chart lists all terrain found on the mapsheet and the effect of each type on movement and combat. The Combat Tables describe the range effects on various weapon types and includes the Combat Results Table. Also note the Line of Sight Table on the back of this rulebook. It tells players when certain terrain types block line of sight, thus preventing direct fire attacks on enemy units. Players should become familiar with these various charts and tables prior to the beginning of play.

(see manual for full text)

### §2.3 – The Units

implemented

manual page unknown

#### The Units

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|

|                                                                                   |                         |                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:872</a><br>GH:omdurman-types/src/lib.rs:872 | <b>UnitKind</b>         | <pre> 870   /// `Some(UnitKind::Marker)` or `None`. 871   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 872   pub enum UnitKind { 873       /// Foot infantry (\$2.3): fire / melee / movement. 874       Infantry { </pre>     |
| <a href="#">omdurman-rules/src/lib.rs:798</a><br>GH:omdurman-rules/src/lib.rs:798 | <b>UnitProfile</b>      | <pre> 796   /// print no melee value). 797   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 798   pub struct UnitProfile { 799       pub kind: UnitKind, 800       pub identity: UnitIdentity, </pre>                                                  |
| <a href="#">omdurman-rules/src/lib.rs:16</a><br>GH:omdurman-rules/src/lib.rs:16   | <b>BrigadeId</b>        | <pre> 14   15   use omdurman_types::{ 16       BrigadeId, BrigadeNationality, DayNight, DervishTribe, Faction, HexCoord, HexsideRef, Player, 17       SetupLetter, UnitKind, 18   }; </pre>                                                                                  |
| <a href="#">omdurman-types/src/lib.rs:679</a><br>GH:omdurman-types/src/lib.rs:679 | <b>SpriteAnnotation</b> | <pre> 677   /// as an optional overlay over the compiled `sprite_data` fallback. 678   #[derive(Serialize, Deserialize, Clone, Debug, PartialEq)] 679   pub struct SpriteAnnotation { 680       pub color: SpriteColor, 681       pub faction: Option&lt;Faction&gt;, </pre> |

Covered by tests:

- [omdurman-rules::src::unit\\_profiles::british\\_army\\_row\\_zero\\_specials\\_classify\\_by\\_counter](#)
- [omdurman-rules::src::unit\\_profiles::egyptian\\_army\\_row\\_zero\\_specials\\_classify\\_by\\_counter](#)
- [omdurman-rules::src::unit\\_profiles::tribe\\_stats\\_come\\_from\\_annotation](#)
- [omdurman-rules::src::unit\\_profiles::section\\_owner\\_dervish\\_sections](#)
- [omdurman-rules::src::unit\\_profiles::section\\_owner\\_green\\_sections\\_are\\_dervish](#)

## §2.4 – Game Parts Inventory

**out-of-scope**

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### Game Parts Inventory

Your complete copy of “REMEMBER GORDON!” — THE BATTLE OF OMDURMAN includes:

- One 22 × 28 Battle of Omdurman mapsheet
- One 8½ × 11 bonus game: FALL OF KHARTOUM mapsheet
- One Rules Booklet
- One die-cut Unit Counter Sheet
- One Campaign Game Order of Appearance Card
- One ten-sided die
- One game box

## §2.31 – Dervish weapon types

**implemented**

manual page unknown

Dervish artillery, gunboats, and forts fire on the “artillery” line of the Dervish Range Effects Table; Jehadia and Danagla units fire on the “rifles” line as does the Isa Zachneih unit. All other Dervish units (including leaders) are armed with spears and swords.

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|

|                                                                                                       |                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:476</a><br>GH:omdurman-rules/src/lib.rs:476                     | <b>WeaponClass</b>      | <pre> 474       Serialize, Deserialize, Clone, Copy, PartialEq, Eq, 475       PartialOrd, Ord, Hash, Debug, strum::Display, 476   pub enum WeaponClass { 477       /// Dervish spears and swords -- no ranged fire at 478       all. 479       Melee, </pre>                                                                                                                                                                                            |
| <a href="#">omdurman-rules/src/unit_profiles.rs:579</a><br>GH:omdurman-rules/src/unit_profiles.rs:579 | <b>dervish_tribe</b>    | <pre> 577   /// Resolve a Dervish tribal foot counter (§2.31): 578   /// Isa Zachneih fire on the rifles line; every other 579   tribe is spear-armed. 580   pub fn dervish_tribe(tribe: DervishTribe) -&gt; 581   Option&lt;Classification&gt; { 582       // §2.31: "Jehadia and Danagla units fire on the 583       'rifles' line as does the 584       // Isa Zachneih unit. All other Dervish units 585       (including leaders) are armed </pre> |
| <a href="#">omdurman-rules/src/unit_profiles.rs:341</a><br>GH:omdurman-rules/src/unit_profiles.rs:341 | <b>khalifa_abdullah</b> | <pre> 339   /// battle (§9.322). All three are interchangeable, 340   so they share the 341   `DervishArtillery` identity. 342   pub fn khalifa_abdullah(col: u32, row: u32) -&gt; 343   Option&lt;Classification&gt; { 344       let artillery =    { 345           Some(Classification { </pre>                                                                                                                                                       |

Covered by tests: [omdurman-rules::src::unit\\_profiles::dervish\\_weapon\\_class\\_follows\\_the\\_rifles\\_line](#)

## §2.32 – Anglo-Egyptian weapon types

**implemented**

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All Anglo-Egyptian units (except gunboats, Maxims, artillery, and leaders) are armed with rifles. Maxims fire on the “Maxims” line of the Anglo-Egyptian Range Effects Table, and artillery and old gunboats fire on the “Artillery” line. New type (named) gunboats may fire on the “Howitzer”, “Artillery”, and “Maxims” lines of the Range Effects Table. (See 6.52 for the fire capabilities of the “Friendlies”).

**\*\*Sample Dervish Units\*\*** (printed on counters): combat unit (Combat / Melee / Movement values, plus Tribe identifier); Leader (e.g. OSMAN DIGNA); Camel unit (e.g. Danagla, 4-6-12); Fort.

**\*\*Sample Anglo-Egyptian Units\*\*** (printed on counters): Cavalry (e.g. 21 Lancers); Artillery (e.g. 32 Battery); Old Gunboat (e.g. LORD KITCHENER, 0-0-15); New Gunboat — named (e.g. Sultan, with artillery and howitzer factor, plus movement downstream / movement upstream values); Maxim Guns (fire twice per turn); Infantry (Fire Combat Factor / Melee / Movement, plus Battalion ID and Brigade ID — e.g. “2B” = 2nd British Brigade, “3E” = 3rd Egyptian Brigade).

(see manual for full text)

See also: §6.52

| File                                                                              | Symbol              | Code Snippet                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:428</a><br>GH:omdurman-rules/src/lib.rs:428 | <b>GunboatId</b>    | <pre> 426   /// fire; "old" gunboats do not (rulebook §2.32). 427   #[derive(Serialize, Deserialize, Clone, Copy, 428   PartialEq, Eq, Hash, Debug, strum::Display)] 429   pub enum GunboatId { 430       /// One of the five new-type named gunboats with 431       howitzer capability. 432       Named(NamedGunboat), </pre> |
| <a href="#">omdurman-rules/src/lib.rs:447</a><br>GH:omdurman-rules/src/lib.rs:447 | <b>NamedGunboat</b> | <pre> 445   /// The five named gunboats with howitzer capability 446   (rulebook §6.64, §2.32). 447   #[derive(Serialize, Deserialize, Clone, Copy, 448   PartialEq, Eq, Hash, Debug, strum::Display)] 449   pub enum NamedGunboat { 450       Sultan, 451       Melik, </pre>                                                  |

|                                                                                   |                                   |                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:460</a><br>GH:omdurman-rules/src/lib.rs:460 | <b>OldGunboat</b>                 | <pre>458   /// in the Maxim Second Fire and Howitzer subphase (\$6.42). 459   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 460   pub enum OldGunboat { 461       LordKitchener, 462       Tamai,</pre> |
| <a href="#">omdurman-rules/src/lib.rs:460</a><br>GH:omdurman-rules/src/lib.rs:460 | <b>GunboatId::Old</b>             | <pre>458   /// in the Maxim Second Fire and Howitzer subphase (\$6.42). 459   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 460   pub enum OldGunboat { 461       LordKitchener, 462       Tamai,</pre> |
| <a href="#">omdurman-rules/src/lib.rs:434</a><br>GH:omdurman-rules/src/lib.rs:434 | <b>GunboatId::Dervish-Gunboat</b> | <pre>432       Old(OldGunboat), 433       /// A Dervish gunboat (\$9.111, \$10.14). 434       DervishGunboat(u8), 435   } 436  </pre>                                                                                                                |

Covered

by

tests:

omdurman-rules::src::effects::tests::old\_gunboat\_lacks\_howitzer

omdurman-rules::src::effects::tests::old\_gunboat\_rejected\_from\_howitzer\_subphase

0/1 implemented

## | §3 – Getting Started

### §3 – Getting Started

out-of-scope

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#### Getting Started

Spread out the mapsheet on a table. It should lie flat if you backfold it against the scored lines. The Dervish player should sit next to the west edge of the map and the Anglo-Egyptian player opposite him on the east edge. Read through the rules once, looking over the various charts as they are referred to in the various sections. Next, select a scenario and punch out only those unit counters needed to play. Later on, the rest of the unit counters should be punched out, sorted and stored by unit type.

(see manual for full text)

■ 1/1 implemented

## §4 – Turn Sequence

### §4 – Turn Sequence

implemented

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#### Turn Sequence

“REMEMBER GORDON!” — THE BATTLE OF OMDURMAN is played in “Game Turns”, each of which has two “Player Turns”. The player moving first will vary according to the scenario being played. In the Campaign Game, for example, the Anglo-Egyptian player moves first.

### A) Anglo-Egyptian Player Turn:

1. Anglo-Egyptian Movement Phase
2. Fire Combat Phase a. Dervish Defensive Fire b. Anglo-Egyptian Offensive Fire
  1. Direct Fire Subphase
  2. Maxim Second Fire and Howitzer Fire Subphase

(see manual for full text)

| File                                                                                                  | Symbol                | Code Snippet                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:252</a><br>GH:omdurman-rules/src/lib.rs:252                     | <b>GameTurnIndex</b>  | <pre>250   /// One-based Game Turn index (1, 2, ... up to the 251   scenario length) (rulebook §4). 252   #[derive(Serialize, Deserialize, Clone, Copy, 253   PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 254   pub struct GameTurnIndex(u8); 255   256   impl GameTurnIndex {</pre>                                                                             |
| <a href="#">omdurman-rules/src/lib.rs:274</a><br>GH:omdurman-rules/src/lib.rs:274                     | <b>Phase</b>          | <pre>272   /// etc. 273   #[derive(Serialize, Deserialize, Clone, Copy, 274   PartialEq, Eq, Debug, Default)] 275   pub enum Phase { 276       /// Pre-game deployment (§9.2/§9.3/§10): fixed 277       units are placed, each side 278       /// deploys its order of battle within its legal 279       zone, and river</pre>                                     |
| <a href="#">omdurman-rules/src/effects/state.rs:5</a><br>GH:omdurman-rules/src/effects/state.rs:5     | <b>GameState</b>      | <pre>3   /// All mutable state of a game in progress (rulebook 4   §4). 5   #[derive(Serialize, Deserialize, Clone, Debug)] 6   pub struct GameState { 7       pub scenario: Scenario, 8       pub current_turn: GameTurnIndex,</pre>                                                                                                                              |
| <a href="#">omdurman-rules/src/effects/state.rs:141</a><br>GH:omdurman-rules/src/effects/state.rs:141 | <b>GameState::new</b> | <pre>139   impl GameState { 140       /// Create a fresh game state for a given scenario 141       (rulebook §4). 142       pub fn new(scenario: Scenario) -&gt; Self { 143           let first = scenario_turn(scenario, 144   GameTurnIndex::new(1)); 145           // First player to *move* per scenario: 146           Campaign -- Anglo-Egyptian moves</pre> |
| <a href="#">omdurman-rules/src/effects/effect.rs:17</a><br>GH:omdurman-rules/src/effects/effect.rs:17 | <b>AdvancePhase</b>   | <pre>15       /// At end-of-turn, disrupted units recover and 16       per-turn tracking is 17       /// cleared. 18       AdvancePhase, 19       // -- Movement 20       -----</pre>                                                                                                                                                                              |
| <a href="#">omdurman-rules/src/effects/dispatch.rs:98</a><br>GH:omdurman-                             | <b>advance_phase</b>  | <pre>96   97   /// Advance the game state to the next phase (rulebook 98   §4).</pre>                                                                                                                                                                                                                                                                              |

|                                                                                                             |                      |                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/effects/<br>dispatch.rs:98                                                                        |                      | <pre> 98   pub fn advance_phase(state: &amp;mut GameState) -&gt; Result&lt;(), RuleError&gt; { 99       let old_phase = state.phase; 100       #[cfg(not(feature = "kani"))] </pre>                                                                                                                |
| omdurman-<br>rules/src/effects/<br>dispatch.rs:204<br>GH:omdurman-<br>rules/src/effects/<br>dispatch.rs:204 | end_player_turn      | <pre> 202   203   /// End the current player's turn: recover disrupted units, switch active player, advance turn index (rulebook §4). 204   pub fn end_player_turn(state: &amp;mut GameState) -&gt; Result&lt;(), RuleError&gt; { 205       #[cfg(not(feature = "kani"))] 206       debug!( </pre> |
| omdurman-<br>rules/src/<br>lib.rs:63<br>GH:omdurman-<br>rules/src/lib.rs:63                                 | GameTurnIndex::value | <pre> 61   pub const ALL: &amp;'static [Self] = &amp;[\$(Self:: \$variant,)+]; 62   63   pub fn value(self) -&gt; u16 { 64       match self { 65           \$(Self::\$variant =&gt; \$value,)+ </pre>                                                                                              |
| omdurman-<br>rules/src/effects/<br>state.rs:133<br>GH:omdurman-<br>rules/src/effects/<br>state.rs:133       | PendingMelee         | <pre> 131   /// resolution after the reaction window is deterministic and host-ordered (rulebook §7.5). 132   #[derive(Serialize, Deserialize, Clone, Debug)] 133   pub struct PendingMelee { 134       pub attack: MeleeAttack, 135       pub attacker_roll: DieRoll, </pre>                      |

Covered by tests:

- omdurman-rules::src::effects::tests::both\_ready\_auto\_advances\_out\_of\_setup
- omdurman-rules::src::effects::tests::fire\_combat\_wrong\_phase\_rejected
- omdurman-rules::src::effects::tests::new\_game\_starts\_in\_setup
- omdurman-rules::src::turn\_track::scenario\_turn\_dispatches\_correctly
- omdurman-rules::src::effects::tests::turn\_advances\_through\_phases
- omdurman-rules::src::unit\_profiles::game\_turn\_marker\_cell\_returns\_none

18/23 implemented

## §5 – Movement Phase

### §5 – Movement Phase (general)

descriptive

manual page unknown

#### Movement Phase

### §5.1 – General Rules

descriptive

manual page unknown

#### General Rules

### §5.2 – Movement Restrictions

descriptive

manual page unknown

#### Movement Restrictions

### §5.3 – Constructing the Zariba

implemented

manual page unknown

#### Constructing the Zariba

The Zariba trench and thorn hedge hexsides are built and in place in the historical scenario only. These hexsides are considered clear terrain in the campaign game. The Anglo-Egyptian player may, however, find it useful to construct this defensive position during the campaign game. The Zariba hexsides may only be built in their position as displayed on the mapsheet. Construction procedure is as follows: any Anglo-Egyptian infantry unit that begins and ends the Anglo-Egyptian player turn adjacent to (and on the Nile side of) Zariba hexsides has constructed all Zariba hexsides to which he is adjacent. The constructing unit may neither fire offensively nor melee attack during the turn of construction. Use a blank counter to denote units constructing Zariba hexsides. See 9.23 for defensive benefits and movement restrictions of Zariba hexsides.

(see manual for full text)

See also: §9.23

| File                                                                              | Symbol                     | Code Snippet                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:832</a><br>GH:omdurman-rules/src/lib.rs:832 | <b>constructing_zariba</b> | <pre>830       /// Set while the unit is building Zariba hexsides --        neither offensive 831       /// fire nor melee allowed that turn (§5.3). 832       pub constructing_zariba: bool, 833       /// Set when the Royal Engineers are committed to a demolition this turn 834       /// (§6.53) -- neither offensive fire nor melee allowed that turn.</pre> |
| <a href="#">omdurman-rules/src/effects/effect.rs:169</a><br>GH:omdurman-          | <b>ConstructZariba</b>     | <pre>167   168       /// Begin constructing a Zariba hexside (rulebook §5.3). 169       ConstructZariba {</pre>                                                                                                                                                                                                                                                     |

|                                                                                                           |                               |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/effects/<br>effect.rs:169                                                                       |                               | <pre> 170           unit_ids: Vec&lt;UnitId&gt;, 171           hexside: HexsideRef, </pre>                                                                                                                                                                                                                                                                                                                                              |
| <a href="#">omdurman-rules/src/effects/dispatch.rs:40</a><br>GH:omdurman-rules/src/effects/dispatch.rs:40 | <b>apply_construct_zariba</b> | <pre> 38           GameEffect::RecoverUnit { unit_id } =&gt;               apply_recover_unit(state, *unit_id), 39           GameEffect::ConstructZariba { unit_ids,               hexside } =&gt; { 40               apply_construct_zariba(state, unit_ids,                   *hexside) 41           } 42           GameEffect::Demolition { unit_id, target } =&gt;               apply_demolition(state, *unit_id, *target), </pre> |

Covered by tests: [omdurman-rules::src::effects::tests::construct\\_zariba\\_marks\\_builders\\_and\\_records\\_hexside](#)

## §5.4 – Zones of Control

**descriptive**

manual page unknown

Zones of Control

## §5.5 – Stacking

**descriptive**

manual page unknown

Stacking

## §5.11 – Movement allowances printed on units

**implemented**

manual page unknown

The movement allowances of the various unit types are printed directly on the units (see 2.3). A unit may move up to this printed movement allowance, paying varying costs for different terrain types (see the Terrain Effects Chart).

See also: §2.3

| File                                                                              | Symbol                   | Code Snippet                                                                                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:131</a><br>GH:omdurman-rules/src/lib.rs:131 | <b>MovementAllowance</b> | <pre> 129           /// is a named variant. 130           #[derive(Serialize, Deserialize, Clone, Copy,               PartialEq, Eq, Hash, Debug)] 131           pub enum MovementAllowance { 132               /// Immobile (forts, wrecked gunboats). 133               Immobile = 0, </pre>                                                                              |
| <a href="#">omdurman-rules/src/lib.rs:809</a><br>GH:omdurman-rules/src/lib.rs:809 | <b>UnitMovement</b>      | <pre> 807           /// Movement allowance -- uniform for land units, split               for gunboats (rulebook §5.11, §5.24, §5.25). 808           #[derive(Serialize, Deserialize, Clone, Copy,               PartialEq, Eq, Debug)] 809           pub enum UnitMovement { 810               Land(MovementAllowance), 811               Gunboat(GunboatMovement), </pre> |
| <a href="#">omdurman-types/src/lib.rs:302</a><br>GH:omdurman-types/src/lib.rs:302 | <b>HexDirection</b>      | <pre> 300           /// (`+q`, `+q+r`, `+r`, `-q`, `-q-r`, `-r` for pointy-               top hexes) (rulebook §5.11, §5.24). 301           #[derive(Serialize, Deserialize, Clone, Copy,               PartialEq, Eq, Hash, Debug, Default)] 302           pub enum HexDirection { 303               #[default] 304               East = 0, </pre>                         |

|                                                                                     |                                  |                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/lib.rs:173<br>GH:omdurman-rules/src/lib.rs:173                   | <b>MovementPoints</b>            | <pre> 171       Serialize, Deserialize, Clone, Copy, PartialEq, Eq, 172       PartialOrd, Ord, Hash, Debug, Default, 173   pub struct MovementPoints(i16); 174   175   impl MovementPoints { </pre>                                                                                                                                                                   |
| omdurman-rules/src/terrain_chart.rs:21<br>GH:omdurman-rules/src/terrain_chart.rs:21 | <b>terrain_effects_chart</b>     | <pre> 19   /// 20   /// Source: printed Terrain Effects Chart on the     mapsheet. 21   pub fn terrain_effects_chart(terrain: Terrain) -&gt;     TerrainEntry { 22       match terrain { 23           Terrain::Clear { .. } =&gt; TerrainEntry { </pre>                                                                                                               |
| omdurman-rules/src/terrain_chart.rs:77<br>GH:omdurman-rules/src/terrain_chart.rs:77 | <b>movement_cost</b>             | <pre> 75   /// Convenience: get the movement cost for a terrain     type (rulebook §5.11, Terrain Effects Chart). 76   /// Returns `None` for impassable terrain (Nile). 77   pub fn movement_cost(terrain: Terrain) -&gt;     Option&lt;MovementAllowance&gt; { 78       terrain_effects_chart(terrain).movement_cost 79   } </pre>                                  |
| omdurman-rules/src/terrain_chart.rs:85<br>GH:omdurman-rules/src/terrain_chart.rs:85 | <b>movement_cost_with_road</b>   | <pre> 83   /// underlying terrain; without a road it's the     terrain's own cost. The road is 84   /// a movement overlay only -- combat/LOS still use the     underlying terrain. 85   pub fn movement_cost_with_road(terrain: Terrain, road:     bool) -&gt; Option&lt;MovementAllowance&gt; { 86       if road { 87           Some(MovementAllowance::One) </pre> |
| omdurman-types/src/lib.rs:498<br>GH:omdurman-types/src/lib.rs:498                   | <b>Terrain::has_road</b>         | <pre> 496   497       /// Whether this hex has any road touching it. 498   pub fn has_road(self) -&gt; bool { 499       !matches!(self.road(), Road::None) 500   } </pre>                                                                                                                                                                                             |
| omdurman-types/src/lib.rs:435<br>GH:omdurman-types/src/lib.rs:435                   | <b>Terrain::passable_by_land</b> | <pre> 433   434       /// Whether this terrain may be entered by land     units (rulebook §5.11). 435   pub fn passable_by_land(self) -&gt; bool { 436       !self.is_nile() 437   } </pre>                                                                                                                                                                           |
| omdurman-types/src/lib.rs:503<br>GH:omdurman-types/src/lib.rs:503                   | <b>Terrain::is_crossroad</b>     | <pre> 501   502       /// Whether roads converge at this hex's centre. 503   pub fn is_crossroad(self) -&gt; bool { 504       matches!(self.road(), Road::Crossroad) 505   } </pre>                                                                                                                                                                                   |
| omdurman-hexmap/src/map.rs:17<br>GH:omdurman-hexmap/src/map.rs:17                   | <b>GameMap::roads</b>            | <pre> 15   pub hexes: HashMap&lt;HexCoord, HexData&gt;, 16   pub hexsides: HashMap&lt;HexsideRef, HexsideKind&gt;, 17   pub roads: HashSet&lt;HexsideRef&gt;, 18   pub excluded: HashSet&lt;HexCoord&gt;, 19   pub overlay: OverlayParams, </pre>                                                                                                                     |

Proven by: [omdurman-rules::src::terrain\\_chart::movement\\_column\\_matches\\_the\\_printed\\_chart](#)  
[omdurman-types::src::lib::terrain\\_road\\_state\\_round\\_trips](#)

Covered by tests: [omdurman-rules::src::terrain\\_chart::clear\\_terrain\\_no\\_bonus](#) [omdurman-rules::src::terrain\\_chart::nile\\_is\\_impassable](#)  
[omdurman-rules::src::terrain\\_chart::rough\\_movement\\_and\\_defense](#)  
[omdurman-rules::src::terrain\\_chart::swamp\\_movement\\_and\\_defense](#)  
[omdurman-rules::src::terrain\\_chart::hilltop\\_movement\\_and\\_defense](#)  
[omdurman-rules::src::terrain\\_chart::huts\\_movement\\_and\\_defense](#)  
[omdurman-rules::src::terrain\\_chart::movement\\_cost\\_convenience\\_matches\\_chart](#)  
[omdurman-rules::src::terrain\\_chart::movement\\_cost\\_with\\_road\\_always\\_one](#)  
[omdurman-rules::src::effects::tests::land\\_unit\\_may\\_not\\_enter\\_nile](#)  
[omdurman-rules::src::terrain\\_chart::movement\\_cost\\_without\\_road\\_matches\\_terrain](#)

omdurman-rules::src::effects::tests::movement\_cost\_for\_uses\_terrain  
omdurman-rules::src::effects::tests::movement\_cost\_for\_road\_costs\_one  
omdurman-rules::src::terrain\_chart::road\_gives\_crossroad      omdurman-rules::src::terrain\_chart::terrain\_movement\_costs\_in\_bounds  
omdurman-rules::src::terrain\_chart::terrain\_chart\_road\_always\_costs\_one

## §5.12 – Move up to allowance, hex by hex (cumulative MP cap)

implemented

manual page unknown

A player may move as many or as few of his units as desired during each movement phase, limited only by the units' movement allowance, the terrain costs paid in moving from hex to hex, and enemy zones of control (see 5.4).

See also: §5.4

| File                                                                                                    | Symbol               | Code Snippet                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state.rs:665</a><br>GH:omdurman-rules/src/effects/state.rs:665   | <b>can_move_unit</b> | <pre>663       /// the same `RuleError` the `MoveUnit` effect 664       would on rejection. Lets the 665       /// UI gate input without mutating or duplicating 666       the rules. 667       pub fn can_move_unit(&amp;self, unit_id: UnitId, cost: MovementPoints) -&gt; Result&lt;(), RuleError&gt; {         self.can_move_unit_to(unit_id, None, cost)     }</pre> |
| <a href="#">omdurman-rules/src/effects/state.rs:1369</a><br>GH:omdurman-rules/src/effects/state.rs:1369 | <b>mp_spent</b>      | <pre>1367   1368       /// Movement points `unit_id` has already spent 1369       this turn (§5.11/§5.12). 1370       pub fn mp_spent(&amp;self, unit_id: UnitId) -&gt; i16 { 1371           self.mp_spent_this_turn.get(&amp;unit_id).copied().unwrap_or(0)     }</pre>                                                                                                  |

Covered by tests: omdurman-rules::src::effects::tests::cumulative\_move\_cost\_may\_not\_exceed\_allowance  
omdurman-rules::src::effects::tests::wrong\_faction\_move\_is\_rejected  
omdurman-rules::src::effects::tests::wrong\_faction\_gunboat\_move\_is\_rejected

## §5.13 – No MP accumulation between turns

implemented

manual page unknown

A unit may never accumulate movement points from turn to turn, nor may a unit transfer unused movement points to other units. A unit's unused movement points in any given turn are considered lost.

| File                                                                                                        | Symbol                 | Code Snippet                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/dispatch.rs:204</a><br>GH:omdurman-rules/src/effects/dispatch.rs:204 | <b>end_player_turn</b> | <pre>202   203       /// End the current player's turn: recover disrupted 204       units, switch active player, advance turn index (rulebook 205       §4). 206       pub fn end_player_turn(state: &amp;mut GameState) -&gt; Result&lt;(), RuleError&gt; {         #[cfg(not(feature = "kani"))]         debug!(</pre> |

Covered by tests: omdurman-rules::src::effects::tests::unused\_movement\_points\_do\_not\_carry\_over

## §5.21 – Friendlies transport via gunboat

implemented

manual page unknown

In general, naval transport missions are not allowed, i.e. gunboats may not carry any land units. The sole exception is that the Anglo-Egyptian player may transport the surviving units of the “Friendlies” brigade from the east bank of the Nile to the west bank after, and only after, the Dervish east bank unit (Isa Zachneih) has been eliminated. The transport is accomplished in the following sequence: a) on any turn that a “Friendlies” unit and any Anglo-Egyptian gunboat start their turn adjacent, that unit may load onto (i.e. stack with) the gunboat; b) during the Anglo-Egyptian player’s next turn the gunboat may move to any Nile hex adjacent to a west bank hex (up to the gunboat’s movement allowance); c) on the Anglo-Egyptian player’s third turn the “Friendlies” unit may disembark and move normally, paying the normal terrain cost for the first hex entered. The gunboat may also move normally that turn.

(see manual for full text)

| File                                                                                    | Symbol                            | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/lib.rs:629<br>GH:omdurman-rules/src/lib.rs:629                       | <b>is_friendlylies</b>            | <pre> 627       /// "Friendlies" units obey several special rules           (\$5.21, \$5.23, \$6.52, 628       /// \$9.14 victory conditions).           pub fn is_friendlylies(&amp;self) -&gt; bool { 629           matches!( 630               self, 631               self, </pre>                                                                                                                                                   |
| omdurman-rules/src/effects/state.rs:2063<br>GH:omdurman-rules/src/effects/state.rs:2063 | <b>friendlylies_transport_off</b> | <pre> 2061       /// regardless of selection. Pairs with           [`GameEffect::FriendlyliesTransport`] 2062       /// so the UI can offer exactly the action the           engine would accept. 2063       pub fn friendlylies_transport_offer(&amp;self, selected:           Option&lt;UnitId&gt;) -&gt; Option&lt;FriendlyliesAction&gt; { 2064           match self.friendlylies_transport { 2065               None =&gt; { </pre> |
| omdurman-rules/src/lib.rs:829<br>GH:omdurman-rules/src/lib.rs:829                       | <b>loaded_on</b>                  | <pre> 827       pub disrupted: bool, 828       /// `Some(gunboat)` after a "Friendlies" unit loads           onto a gunboat (\$5.21). 829       pub loaded_on: Option&lt;UnitId&gt;, 830       /// Set while the unit is building Zariba hexsides           -- neither offensive 831       /// fire nor melee allowed that turn (\$5.3). </pre>                                                                                          |
| omdurman-rules/src/effects/effect.rs:196<br>GH:omdurman-rules/src/effects/effect.rs:196 | <b>FriendlyliesTransport</b>      | <pre> 194   195       /// Load/disembark the "Friendlies" brigade via           gunboat (rulebook \$5.21). 196       FriendlyliesTransport(crate::FriendlyliesAction), 197   198       // -- Optional rules           ----- </pre>                                                                                                                                                                                                       |
| omdurman-rules/src/effects/river.rs:12<br>GH:omdurman-rules/src/effects/river.rs:12     | <b>apply_friendlylies_transpo</b> | <pre> 10       /// unit is freed (a disembarking `MoveUnit` should           follow, costed by 11       /// terrain). 12       pub fn apply_friendlylies_transport( 13           state: &amp;mut GameState, 14           action: FriendlyliesAction, </pre>                                                                                                                                                                              |
| omdurman-rules/src/lib.rs:1077<br>GH:omdurman-rules/src/lib.rs:1077                     | <b>FriendlyliesAction</b>         | <pre> 1075       /// tracks each unit-gunboat pair independently. 1076       #[derive(Serialize, Deserialize, Clone, Copy,           PartialEq, Eq, Debug)] 1077       pub enum FriendlyliesAction { 1078           /// Turn N (the load turn): unit and gunboat               started adjacent; unit 1079           /// loads onto (stacks with) the gunboat. </pre>                                                                    |
| omdurman-rules/src/lib.rs:1097<br>GH:omdurman-rules/src/lib.rs:1097                     | <b>TransportState</b>             | <pre> 1095       /// third turn. 1096       #[derive(Serialize, Deserialize, Clone, Copy,           PartialEq, Eq, Debug)] 1097       pub enum TransportState { 1098           /// Turn N (the load turn): unit and gunboat               started adjacent; unit 1099           /// loads onto (stacks with) the gunboat. </pre>                                                                                                         |

Covered by tests: `omdurman-rules::src::effects::tests::friendlies_transport_offer_load_requires_prerequisites`  
`omdurman-rules::src::effects::tests::friendlies_transport_offer_follows_state_machine`

## §5.22 – Land units may never enter a Nile River hex

implemented

manual page unknown

With the exception of 5.21, land units may never enter a Nile River hex. Only gunboats may enter and move along Nile River hexes.

See also: §5.21

| File                                                                                                  | Symbol                    | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state.rs:682</a><br>GH:omdurman-rules/src/effects/state.rs:682 | <b>can_move_unit_to</b>   | <pre>680       /// 681       /// [<code>hex_in_enemy_zoc</code>]: Self::hex_in_enemy_zoc 682       pub fn can_move_unit_to( 683           &amp;self, 684           unit_id: UnitId,</pre>                                                                                                                                                                                                          |
| <a href="#">omdurman-rules/src/effects/state.rs:353</a><br>GH:omdurman-rules/src/effects/state.rs:353 | <b>in_deployment_zone</b> | <pre>351       /// plan / UI rather than this hex predicate.           Documented, not silently 352       /// dropped. 353       pub fn in_deployment_zone(&amp;self, player: Player,           hex: HexCoord, is_boat: bool) -&gt; bool { 354           // No board attached -&gt; permissive (unit tests,               unbound session). 355           if self.board.terrain.is_empty() {</pre> |

Covered by tests: `omdurman-rules::src::effects::tests::campaign_deployment_is_boat_land_exclusive`  
`omdurman-rules::src::effects::tests::fok_ae_gunboat_deploys_only_on_nile`  
`omdurman-rules::src::effects::tests::fok_ae_land_unit_rejected_on_nile`  
`omdurman-rules::src::effects::tests::deploy_via_real_sprite_resolution_matches_engine`  
`omdurman-rules::src::effects::tests::fok_dervish_land_unit_rejected_on_nile`  
`omdurman-rules::src::effects::tests::retreat_before_melee_may_not_land_on_nile`

## §5.23 – Walled city entry restrictions

implemented

manual page unknown

Only certain units may enter the walled portion of Omdurman. For the Dervish player these are the Khalifa unit, the three Dervish artillery units, and the Taiasha units (the Khalifa's bodyguard). Any Anglo-Egyptian units that can get to the walled city may enter it (except gunboats and "Friendlies"). Units entering and/or exiting the walled city may only do so through a gate or breach hexside.

| File                                                                              | Symbol             | Code Snippet                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:152</a><br>GH:omdurman-types/src/lib.rs:152 | <b>HexsideRef</b>  | <pre>150       /// data by [<code>HexsideRef</code>]. 151       #[derive(Serialize, Deserialize, Clone, Copy,           PartialEq, Eq, Hash, Debug)] 152       pub struct HexsideRef { 153           pub a: HexCoord, 154           pub b: HexCoord,</pre>                                        |
| <a href="#">omdurman-types/src/lib.rs:187</a><br>GH:omdurman-types/src/lib.rs:187 | <b>HexsideKind</b> | <pre>185           strum::EnumIter, 186       )] 187       pub enum HexsideKind { 188           /// City wall (Khartoum, walled city of Omdurman).               Blocks LOS, blocks 189           /// movement except across gates/breaches (§5.23),               blocks ZOC into the city</pre> |

|                                                                                     |                              |                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-types/src/lib.rs:258<br>GH:omdurman-types/src/lib.rs:258                   | <b>blocks_movement</b>       | <pre> 256       /// `omdurman-rules`). The trench *end* variants are therefore intentionally 257       /// not blocking. 258       pub fn blocks_movement(self) -&gt; bool { 259           matches!( 260               self, </pre>                                                                                                                                                                                                  |
| omdurman-rules/src/lib.rs:655<br>GH:omdurman-rules/src/lib.rs:655                   | <b>may_enter_walled_city</b> | <pre> 653       /// Taiasha bodyguard may enter. Anglo-Egyptian: any unit that can reach the 654       /// walled city *except* gunboats and "Friendlies". 655       pub fn may_enter_walled_city(&amp;self) -&gt; bool { 656           match self { 657               // §5.23 Dervish: Khalifa, artillery, Taiasha. </pre>                                                                                                         |
| omdurman-rules/src/board.rs:311<br>GH:omdurman-rules/src/board.rs:311               | <b>is_walled_city</b>        | <pre> 309       /// are its seeds). The set is derived once from the board data, replacing 310       /// the older "at least two of six hexsides are Wall/Gate/Breach" heuristic. 311       pub fn is_walled_city(&amp;self, hex: HexCoord) -&gt; bool { 312           // Membership in the precomputed enclosed area (see `walled_city`). 313           // Palace/Tomb hexes are always part of it (the they are the seeds). </pre> |
| omdurman-rules/src/effects/error.rs:90<br>GH:omdurman-rules/src/effects/error.rs:90 | <b>WalledCityEntry</b>       | <pre> 88   89       #[error("unit {0:?} is not eligible to enter the walled city of Omdurman at {1:?} (§5.23)")] 90       WalledCityEntry(UnitId, HexCoord), 91   92       #[error("movement cost {cost:?} exceeds allowance {allowance:?}")] </pre>                                                                                                                                                                                 |

Proven

by:

[omdurman-types::src::lib::hexside\\_ref\\_is\\_order\\_independent](#) [omdurman-types::src::lib::hexside\\_blocking\\_classifiers\\_are\\_exact](#)

Covered

by

tests:

[omdurman-rules::src::effects::tests::can\\_move\\_rejects\\_wall\\_hexside](#) [omdurman-rules::src::effects::tests::can\\_move\\_allows\\_gate\\_hexside](#)  
[omdurman-rules::src::effects::tests::walled\\_city\\_entry\\_allows\\_khalifa](#)  
[omdurman-rules::src::effects::tests::walled\\_city\\_entry\\_rejects\\_unauthorized\\_dervish](#)  
[omdurman-rules::src::effects::tests::walled\\_city\\_entry\\_rejects\\_ae\\_gunboat](#)  
[omdurman-rules::src::effects::tests::walled\\_city\\_entry\\_not\\_enforced\\_for\\_fok](#)  
[omdurman-rules::src::board\\_data::campaign\\_walled\\_city\\_is\\_enclosed\\_by\\_walls](#)

## §5.24 – Gunboat upstream/downstream movement

**implemented**

manual page unknown

Note that gunboats have two movement allowances separated by a slash, e.g. 10/16. The smaller number is the movement allowance when moving upstream, i.e. against the current (the direction of the current is indicated by arrows in the Nile). The larger number is the movement allowance when moving downstream, i.e. with the current. Gunboats may combine movement in both directions, but if they move even one hex upstream, their upstream movement allowance is their maximum movement allowance for that turn, and may not be exceeded.

(see manual for full text)

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|

|                                                                                   |                        |                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:552</a><br>GH:omdurman-rules/src/lib.rs:552 | <b>GunboatMovement</b> | <pre> 550   /// the turn. 551   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 552   pub struct GunboatMovement { 553       pub upstream: MovementAllowance, 554       pub downstream: MovementAllowance, </pre> |
| <a href="#">omdurman-types/src/lib.rs:957</a><br>GH:omdurman-types/src/lib.rs:957 | <b>is_boat</b>         | <pre> 955   956       /// Gunboats use the split upstream/downstream movement allowance (§5.24). 957       pub fn is_boat(self) -&gt; bool { 958           matches!(self, UnitKind::Gunboat { .. }) 959       } </pre>                 |

Proven by: [omdurman-types::src::lib::unit\\_kind\\_fire\\_and\\_movement\\_capability\\_law\\_is\\_exact](#)  
[omdurman-types::src::lib::hex\\_direction\\_from\\_index\\_is\\_total\\_mod\\_six](#)  
[omdurman-types::src::lib::nile\\_flow\\_rotation\\_composes\\_and\\_never\\_leaves\\_the\\_nile](#)

Covered by tests: [omdurman-rules::src::unit\\_profiles::boat\\_annotation\\_yields\\_split\\_gunboat\\_movement](#)  
[omdurman-rules::src::effects::tests::gunboat\\_upstream\\_cap\\_is\\_sticky\\_across\\_moves](#)

## §5.25 – Dervish forts may not move

**implemented**

manual page unknown

Dervish forts may not move in any way once placed.

| File                                                                              | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:813</a><br>GH:omdurman-rules/src/lib.rs:813 | <b>Immobile</b>               | <pre> 811   Gunboat(GunboatMovement), 812       /// Forts may not move once placed (§5.25). 813       Immobile, 814   } 815   </pre>                                                                                                                                                                        |
| <a href="#">omdurman-types/src/lib.rs:914</a><br>GH:omdurman-types/src/lib.rs:914 | <b>UnitKind::Fort</b>         | <pre> 912       /// Permanent emplacement (§6.54): fire (artillery) / melee (defensive). 913       /// May not move once placed (§5.25). 914       Fort { fire: i32, melee: i32 }, 915       /// Dervish leader (§6.51): fire / melee / movement. May melee attack (§7.4). 916       DervishLeader { </pre> |
| <a href="#">omdurman-rules/src/lib.rs:813</a><br>GH:omdurman-rules/src/lib.rs:813 | <b>UnitMovement::Immobile</b> | <pre> 811   Gunboat(GunboatMovement), 812       /// Forts may not move once placed (§5.25). 813       Immobile, 814   } 815   </pre>                                                                                                                                                                        |

Covered by tests: [omdurman-rules::src::effects::tests::forts\\_are\\_never\\_advance\\_eligible](#)

## §5.26 – Units stop on entering enemy ZOC

**implemented**

manual page unknown

Units must stop their movement immediately upon entering an enemy zone of control (see 5.4).

See also: §5.4

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|

|                                                                                         |                         |                                              |                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------|-------------------------|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects/state.rs:682<br>GH:omdurman-rules/src/effects/state.rs:682   | <b>can_move_unit_to</b> | 680  <br>681  <br>682  <br>683  <br>684      | ///<br>/// [`hex_in_enemy_zoc`]: Self::hex_in_enemy_zoc<br>pub fn can_move_unit_to(<br>&self,<br>unit_id: UnitId,                                                               |
| omdurman-rules/src/effects/state.rs:1495<br>GH:omdurman-rules/src/effects/state.rs:1495 | <b>hex_in_enemy_zoc</b> | 1493  <br>1494  <br>1495  <br>1496  <br>1497 | /// does not extend into or out of a Nile hex.<br>With no board loaded these<br>/// reduce to the plain adjacency rule.<br>pub fn hex_in_enemy_zoc(<br>&self,<br>hex: HexCoord, |

Covered by tests: omdurman-rules::src::effects::tests::unit\_entering\_enemy\_zoc\_may\_move\_no\_further\_that\_turn  
omdurman-rules::src::effects::tests::zoc\_transit\_check\_uses\_the\_actual\_path

## §5.41 – All units except AE leaders exert ZOC

implemented

manual page unknown

All units except Anglo-Egyptian leaders exert a zone of control (hereafter called a ZOC) into their six adjacent hexes (exception: Gunboats exert a ZOC only against enemy gunboats). Disrupted units have no ZOC.

| File                                                                                                    | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state.rs:1742</a><br>GH:omdurman-rules/src/effects/state.rs:1742 | <b>unit_projects_zoc_rule</b> | <pre>1740   /// * A fort projects ZOC out of its hex even when         unoccupied (§5.44), 1741   /// modelled by the fort unit projecting normally. 1742   pub fn unit_projects_zoc_rule( 1743       unit: &amp;UnitPlacement, 1744       mover_player: Player,</pre>                                                                                                                    |
| <a href="#">omdurman-rules/src/lib.rs:868</a><br>GH:omdurman-rules/src/lib.rs:868                       | <b>ZocReason</b>              | <pre>866   /// Used by the engine when answering "is this hex in         an enemy ZOC?". 867   #[derive(Serialize, Deserialize, Clone, Copy,         PartialEq, Eq, Debug)] 868   pub enum ZocReason { 869       /// Normal ZOC: any non-disrupted unit other than         an Anglo-Egyptian 870       /// leader (§5.41) projects ZOC into each of its         six adjacent hexes.</pre> |
| <a href="#">omdurman-rules/src/effects/state.rs:1477</a><br>GH:omdurman-rules/src/effects/state.rs:1477 | <b>unit_projects_zoc</b>      | <pre>1475       /// §5.44) need the game map, which the engine         does not hold; the app layers 1476       /// those on top. This is the position/kind/         disruption core of the rule. 1477       pub fn unit_projects_zoc( 1478           &amp;self, 1479           unit: &amp;UnitPlacement,</pre>                                                                           |
| <a href="#">omdurman-rules/src/effects/state.rs:1536</a><br>GH:omdurman-rules/src/effects/state.rs:1536 | <b>zoc_hexes</b>              | <pre>1534       /// ZOC covers a given hex; this function returns         *which* hexes a 1535       /// specific unit covers. 1536       pub fn zoc_hexes( 1537           &amp;self, 1538           unit: &amp;UnitPlacement,</pre>                                                                                                                                                      |

Proven by: omdurman-rules::src::effects::unit\_projects\_zoc\_matches\_manual\_clauses  
omdurman-types::src::lib::player\_opponent\_is\_a\_fixed\_point\_free\_involution

Covered by tests: omdurman-rules::src::effects::tests::zoc\_hexes\_empty\_for\_anglo\_egyptian\_leader  
omdurman-rules::src::effects::tests::zoc\_hexes\_normal\_unit\_projects\_six\_adjacent\_minus\_exclusions  
omdurman-rules::src::effects::tests::zoc\_hexes\_empty\_for\_disrupted\_unit  
omdurman-rules::src::effects::tests::zoc\_hexes\_matches\_hex\_in\_enemy\_zoc omdurman-rules::src::effects::tests::zoc\_hexes\_excludes\_nile  
omdurman-rules::src::effects::tests::zoc\_hexes\_excludes\_khor

## §5.42 – No MP cost to enter/leave enemy ZOC

implemented

manual page unknown

There is no movement point cost to enter or leave an enemy ZOC.

| File                                                                                                  | Symbol                   | Code Snippet                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state.rs:856</a><br>GH:omdurman-rules/src/effects/state.rs:856 | <b>movement_cost_for</b> | <pre>854      /// 855      /// §5.42: entering or leaving an enemy ZOC adds no MP cost. 856      pub fn movement_cost_for( 857          &amp;self, 858          unit: &amp;UnitPlacement,</pre> |

Covered by tests: [omdurman-rules::src::effects::tests::entering\\_enemy\\_zoc\\_costs\\_no\\_extra\\_mp](#)

## §5.43 – Units stop when entering enemy ZOC

implemented

manual page unknown

All units must stop when they enter an enemy ZOC and may move no further that turn. In their next movement phase they may withdraw or, if desired, move directly into another enemy ZOC.

| File                                                                                                    | Symbol                  | Code Snippet                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state.rs:682</a><br>GH:omdurman-rules/src/effects/state.rs:682   | <b>can_move_unit_to</b> | <pre>680      /// 681      /// [`hex_in_enemy_zoc`]: Self::hex_in_enemy_zoc 682      pub fn can_move_unit_to( 683          &amp;self, 684          unit_id: UnitId,</pre>                                                                 |
| <a href="#">omdurman-rules/src/effects/state.rs:1495</a><br>GH:omdurman-rules/src/effects/state.rs:1495 | <b>hex_in_enemy_zoc</b> | <pre>1493      /// does not extend into or out of a Nile hex. With no board loaded these 1494      /// reduce to the plain adjacency rule. 1495      pub fn hex_in_enemy_zoc( 1496          &amp;self, 1497          hex: HexCoord,</pre> |

Covered by tests: [omdurman-rules::src::effects::tests::unit\\_entering\\_enemy\\_zoc\\_may\\_move\\_no\\_further\\_that\\_turn](#)

## §5.44 – ZOC limitations (walls, khor, fort, Nile, Zariba)

implemented

manual page unknown

ZOCs do not extend into or out of a Nile River hex (exception: Gunboats, see 5.41). ZOCs do not extend across a khor, into a fort, or into a hex inside the walled city across a wall hexside. ZOCs do extend out of a fort (even if unoccupied), and from a walled city hex into an adjacent non-walled-city hex across a wall hexside. ZOCs also extend out of (but not into) a walled city hex across a gate hexside. ZOCs extend both ways across a breach hexside. ZOCs also extend out of, but not into, a hut or building hex. In the historical scenario ZOCs extend out of, but not into, the Zariba across a Zariba hexside (also in the campaign game if the Zariba is constructed).

(see manual for full text)

See also: §5.41

| File                                                                                                    | Symbol                         | Code Snippet                                                                                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:868</a><br>GH:omdurman-rules/src/lib.rs:868                       | <b>ZocReason</b>               | <pre>866   /// Used by the engine when answering "is this hex in 867   an enemy ZOC?". 867   #[derive(Serialize, Deserialize, Clone, Copy, 868   PartialEq, Eq, Debug)] 868   pub enum ZocReason { 869       /// Normal ZOC: any non-disrupted unit other than 870       an Anglo-Egyptian 870       /// leader (§5.41) projects ZOC into each of its 870       six adjacent hexes.</pre> |
| <a href="#">omdurman-types/src/lib.rs:192</a><br>GH:omdurman-types/src/lib.rs:192                       | <b>Wall</b>                    | <pre>190       /// (§5.44), blocks melee (§7.2), blocks advance- 191       after-combat (§6.82). 191       #[default] 192       Wall, 193       /// Gate hexside in a wall. ZOC extends *out of* 194       the walled city through 194       /// gates but not into it (§5.44). Melee may be 194       made through a gate (§7.2).</pre>                                                  |
| <a href="#">omdurman-types/src/lib.rs:201</a><br>GH:omdurman-types/src/lib.rs:201                       | <b>Khor</b>                    | <pre>199       /// Khor -- gully/wadi. ZOCs do not extend across 200       (§5.44); advance after 201       /// combat may not cross (§6.82). 201       Khor, 202       /// Crest line. Blocks LOS unless the firer is on 203       the higher side 203       /// (§6.3 note 7).</pre>                                                                                                    |
| <a href="#">omdurman-rules/src/lib.rs:881</a><br>GH:omdurman-rules/src/lib.rs:881                       | <b>ZocReason::Zariba</b>       | <pre>879       /// Zariba hexside ZOC behaviour in the historical 880       scenario / when the 881       /// Zariba is constructed (§5.44). 881       Zariba, 882   } 883  </pre>                                                                                                                                                                                                        |
| <a href="#">omdurman-rules/src/effects/state.rs:1477</a><br>GH:omdurman-rules/src/effects/state.rs:1477 | <b>unit_projects_zoc</b>       | <pre>1475       /// §5.44) need the game map, which the engine 1476       does not hold; the app layers 1476       /// those on top. This is the position/kind/ 1477       disruption core of the rule. 1477       pub fn unit_projects_zoc( 1478           &amp;self, 1479           unit: &amp;UnitPlacement,</pre>                                                                     |
| <a href="#">omdurman-types/src/lib.rs:275</a><br>GH:omdurman-types/src/lib.rs:275                       | <b>HexsideKind::blocks_zoc</b> | <pre>273       /// cannot express; those are left to the caller. 274       This predicate captures the 274       /// symmetric "does not extend across" cases. 275       pub fn blocks_zoc(self) -&gt; bool { 276           matches!( 277               self,</pre>                                                                                                                       |
| <a href="#">omdurman-rules/src/effects/state.rs:1495</a><br>GH:omdurman-rules/src/effects/state.rs:1495 | <b>hex_in_enemy_zoc</b>        | <pre>1493       /// does not extend into or out of a Nile hex. 1494       With no board loaded these 1494       /// reduce to the plain adjacency rule. 1495       pub fn hex_in_enemy_zoc( 1496           &amp;self, 1497           hex: HexCoord,</pre>                                                                                                                                 |

Proven by: [omdurman-rules::src::effects::unit\\_projects\\_zoc\\_matches\\_manual\\_clauses](#)  
[omdurman-types::src::lib::hexside\\_blocking\\_classifiers\\_are\\_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::zoc\\_hexes\\_excludes\\_nile](#) [omdurman-rules::src::effects::tests::zoc\\_hexes\\_excludes\\_khor](#)  
[omdurman-rules::src::effects::tests::zoc\\_hexes\\_normal\\_unit\\_projects\\_six\\_adjacent\\_minus\\_exclusions](#)

## §5.51 – Stacking limit (4 units + leaders, gunboats isolated)

**implemented**

manual page unknown

No more than four units may occupy a hex, with the exception of leaders and gunboats. All leader units are free stacking, i.e. they may stack in addition to the four-unit-per-hex stacking limitation. Gunboats may not stack with any other unit (Exception: 5.21). Players may move through friendly units at no additional cost in movement points. The stacking limitation applies only at the end of the movement phase and during combat.

See also: §5.21

| File                                                                                                    | Symbol                   | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:893</a><br>GH:omdurman-rules/src/lib.rs:893                       | <b>OverLimit</b>         | <pre> 891       /// (§6.51: a Dervish unit occupying his hex       eliminates him). 892       #[error("hex stack exceeds the four-unit limit       [\$5.51]")] 893       OverLimit, 894       /// "Gunboats may not stack with any other unit"       (§5.51, exception §5.21). 895       #[error("gunboats may not stack with non-gunboat       units [\$5.51]")] </pre>                             |
| <a href="#">omdurman-rules/src/lib.rs:896</a><br>GH:omdurman-rules/src/lib.rs:896                       | <b>GunboatStack</b>      | <pre> 894       /// "Gunboats may not stack with any other unit"       (§5.51, exception §5.21). 895       #[error("gunboats may not stack with non-gunboat       units [\$5.51]")] 896       GunboatStack, 897       /// "Units of different Dervish tribes may not       stack together" (§5.52). 898       /// The Dervish artillery (§9.322's three guns) is       not a tribe but is not </pre> |
| <a href="#">omdurman-rules/src/lib.rs:906</a><br>GH:omdurman-rules/src/lib.rs:906                       | <b>EnemyCohabitation</b> | <pre> 904       /// lone Anglo-Egyptian leader is exempt (§6.51). 905       #[error("enemy units may not share a hex; melee,       not movement, engages them [\$5.51, §7.1]")] 906       EnemyCohabitation, 907       /// "If Dervish leaders elect to stack, they may       only stack with units of 908       /// their command (i.e. colour)" (§5.53). </pre>                                    |
| <a href="#">omdurman-rules/src/effects/state.rs:1428</a><br>GH:omdurman-rules/src/effects/state.rs:1428 | <b>check_stacking</b>    | <pre> 1426       /// Dervish artillery is its own group) may not       stack together. 1427       /// * §5.53 -- a Dervish leader may stack only       with units of its command. 1428       pub fn check_stacking( 1429           &amp;self, 1430           mover: &amp;UnitPlacement, </pre>                                                                                                       |
| <a href="#">omdurman-rules/src/effects/state.rs:1653</a><br>GH:omdurman-rules/src/effects/state.rs:1653 | <b>stacking_rule</b>     | <pre> 1651       /// artillery as its own group) may not share a hex. 1652       /// * §5.53: a Dervish leader stacks only with units       of its command. 1653       pub fn stacking_rule(occupants: &amp;[&amp;UnitPlacement]) -&gt;       Result&lt;(), crate::StackingError&gt; { 1654           use crate::StackingError; 1655   </pre>                                                        |

Proven by:

- [omdurman-rules::src::effects::stacking\\_rule\\_cohabitation\\_is\\_exact](#)
- [omdurman-rules::src::effects::stacking\\_rule\\_is\\_symmetric](#)
- [omdurman-rules::src::effects::stacking\\_rule\\_leaders\\_are\\_free\\_stacking](#)
- [omdurman-rules::src::effects::stacking\\_rule\\_limit\\_is\\_four\\_counted\\_units](#)
- [omdurman-rules::src::effects::stacking\\_rule\\_gunboat\\_never\\_shares](#)

Covered by tests:

- [omdurman-rules::src::effects::tests::stacking\\_over\\_limit\\_rejected](#)
- [omdurman-rules::src::effects::tests::mid\\_move\\_stacking\\_allows\\_pass\\_through](#)
- [omdurman-rules::src::effects::tests::mid\\_move\\_stacking\\_rejects\\_over\\_limit\\_destination](#)
- [omdurman-rules::src::effects::tests::validate\\_stacking\\_invariants\\_clean\\_state](#)
- [omdurman-rules::src::effects::tests::validate\\_stacking\\_invariants\\_catches\\_stacking\\_violation](#)
- [omdurman-rules::src::effects::tests::validate\\_stacking\\_invariants\\_allows\\_leaders\\_stacking](#)
- [omdurman-rules::src::effects::tests::deploy\\_rejects\\_enemy\\_cohabitation\\_during\\_setup](#)
- [omdurman-rules::src::effects::tests::validate\\_stacking\\_invariants\\_catches\\_enemy\\_cohabitation](#)
- [omdurman-rules::src::effects::tests::ae\\_garrison\\_stacks\\_freely\\_under\\_gordon](#)
- [omdurman-rules::src::effects::tests::melee\\_mandatory\\_advance\\_leaders\\_are\\_free\\_stacking](#)

## §5.52 – Different Dervish tribes may not stack together

implemented

manual page unknown

The units of different Dervish tribes may not stack together, even if they are the same color (e.g. although both are green, Mulazmin and Jehadia units may not stack with each other).

| File                                                                              | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:902</a><br>GH:omdurman-rules/src/lib.rs:902 | <b>DervishTribeMix</b>        | <pre>900       /// [`UnitIdentity::dervish_stacking_group`]). 901       #[error("Dervish units of different tribes may not stack [\$5.52]")] 902       DervishTribeMix, 903       /// A unit may never share a hex with enemy units (\$5.51, \$7.1); only the 904       /// lone Anglo-Egyptian leader is exempt (\$6.51).</pre>                                                                     |
| <a href="#">omdurman-rules/src/lib.rs:707</a><br>GH:omdurman-rules/src/lib.rs:707 | <b>dervish_stacking_group</b> | <pre>705       /// `None` for every unit the §5.52 law does not constrain: Dervish 706       /// leaders, forts and gunboats, and all Anglo- Egyptian units. 707       pub fn dervish_stacking_group(&amp;self) -&gt; Option&lt;DervishStackingGroup&gt; { 708           match self { 709               UnitIdentity::DervishTribal { tribe } =&gt; Some(DervishStackingGroup::Tribe(*tribe)),</pre> |

Proven by: [omdurman-rules::src::effects::stacking\\_rule\\_group\\_purity\\_is\\_exact](#)

Covered by tests: [omdurman-rules::src::unit\\_profiles::green\\_sections\\_are\\_mulazmin\\_tribal\\_units](#)  
[omdurman-rules::src::effects::tests::deploy\\_rejects\\_dervish\\_tribe\\_mix](#)  
[omdurman-rules::src::effects::tests::deploy\\_rejects\\_hadendowa\\_on\\_dervish\\_gun](#)  
[omdurman-rules::src::effects::tests::dervish\\_guns\\_stack\\_with\\_guns\\_and\\_their\\_leader](#)  
[omdurman-rules::src::effects::tests::validate\\_stacking\\_invariants\\_clean\\_state](#)

## §5.53 – Leader stacking with command colour only

implemented

manual page unknown

Leader units are not required to stack. If Dervish leaders elect to stack, however, they may only stack with units of their command (i.e. color). For example, Sheik El Din may only stack with Mulazmins or Jehadias.

**\*\*5.54) Anglo-Egyptian Brigade Integrity:\*\*** All British, Sudanese, and Egyptian infantry units have their brigade designation printed in the upper right corner (e.g. “2B” = 2nd British Brigade; “3E” = 3rd Egyptian Brigade, etc.). In any combat phase in which all four infantry battalions belonging to any Anglo-Egyptian infantry brigade are stacked in the same hex they are said to have brigade integrity. Stacks having brigade integrity receive a +1 modifier to their fire combat die roll provided they all fire at the same enemy occupied hex. This modifier is in addition to the normal +1 bonus given to all Anglo-Egyptian direct fire attacks (see 6.24).

(see manual for full text)

See also: §5.54, §6.24

| File                                                                              | Symbol                              | Code Snippet                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:910</a><br>GH:omdurman-rules/src/lib.rs:910 | <b>DervishLeaderCommandMismatch</b> | <pre>908       /// their command (i.e. colour)" (§5.53). 909       #[error("Dervish leader may only stack with units of their own command [\$5.53]")] 910       DervishLeaderCommandMismatch, 911   } 912  </pre> |

Proven by: [omdurman-rules::src::effects::stacking\\_rule\\_leaders\\_are\\_free\\_stacking](#)

Covered by tests: [omdurman-rules::src::effects::tests::dervish\\_leader\\_stacks\\_only\\_with\\_command\\_colour](#)  
[omdurman-rules::src::unit\\_profiles::command\\_owns\\_unit\\_follows\\_tribes\\_and\\_brigades](#)

## §5.54 – Anglo-Egyptian Brigade Integrity

[implemented](#)

manual page unknown

| File                                                                                | Symbol                                | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:916</a><br>GH:omdurman-rules/src/lib.rs:916   | <b>BrigadeIntegrity</b>               | <pre>914   /// stack contains all four battalions of a single       Anglo-Egyptian brigade. 915   #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Debug)] 916   pub enum BrigadeIntegrity { 917       None, 918       Integrated(BrigadeId),</pre>                                                                                                                                                  |
| <a href="#">omdurman-rules/src/lib.rs:771</a><br>GH:omdurman-rules/src/lib.rs:771   | <b>brigade_integrity</b>              | <pre>769   /// Only a full stack of four battalions qualifies.       Three or fewer may still 770   /// stack and fire, but they receive no brigade-       integrity bonus. 771   pub fn brigade_integrity(identities: &amp;[UnitIdentity]) -       &gt; BrigadeIntegrity { 772       let Some(brigade) = identities.first().and_then( i        i.brigade()) else { 773           return BrigadeIntegrity::None;</pre> |
| <a href="#">omdurman-rules/src/lib.rs:916</a><br>GH:omdurman-rules/src/lib.rs:916   | <b>FireModifier::BrigadeIntegrity</b> | <pre>914   /// stack contains all four battalions of a single       Anglo-Egyptian brigade. 915   #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Debug)] 916   pub enum BrigadeIntegrity { 917       None, 918       Integrated(BrigadeId),</pre>                                                                                                                                                  |
| <a href="#">omdurman-rules/src/lib.rs:329</a><br>GH:omdurman-rules/src/lib.rs:329   | <b>BattalionOrdinal</b>               | <pre>327       /// brigade integrity requires all four stacked in       one hex (§5.54). 328       #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Hash, Debug, strum::Display)] 329       pub enum BattalionOrdinal { 330           First = 1, 331           Second = 2,</pre>                                                                                                                     |
| <a href="#">omdurman-types/src/lib.rs:1023</a><br>GH:omdurman-types/src/lib.rs:1023 | <b>BrigadeId</b>                      | <pre>1021   /// same field for uniform handling. 1022   #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 1023   pub struct BrigadeId { 1024       pub number: u8, 1025       pub nationality: BrigadeNationality,</pre>                                                                                                                                               |

Proven by: [omdurman-rules::src::lib::brigade\\_integrity\\_requires\\_all\\_four\\_distinct\\_battalions\\_of\\_one\\_brigade](#)

Covered by tests: [omdurman-rules::src::unit\\_profiles::brigade\\_designation\\_ignored\\_for\\_non\\_infantry](#)  
[omdurman-rules::src::unit\\_profiles::printed\\_brigade\\_designation\\_overrides\\_column](#)  
[omdurman-rules::src::unit\\_profiles::section\\_owner\\_anglo\\_egyptian\\_sections](#)  
[omdurman-rules::src::lib::brigade\\_integrity\\_four\\_battalions\\_returns\\_integrated](#)  
[omdurman-rules::src::unit\\_profiles::ae\\_infantry\\_fourth\\_battalion\\_from\\_col\\_3](#)  
[omdurman-rules::src::lib::brigade\\_integrity\\_empty\\_slice](#) [omdurman-rules::src::lib::brigade\\_integrity\\_friendlys\\_returns\\_none](#)  
[omdurman-rules::src::lib::brigade\\_integrity\\_three\\_battalions\\_returns\\_none](#)  
[omdurman-rules::src::lib::unit\\_identity\\_brigade\\_and\\_battalion\\_accessors](#)  
[omdurman-rules::src::unit\\_profiles::ae\\_infantry\\_brigade\\_number\\_three\\_from\\_col\\_7](#)  
[omdurman-rules::src::lib::brigade\\_integrity\\_non\\_infantry\\_returns\\_none](#)  
[omdurman-rules::src::lib::brigade\\_integrity\\_mixed\\_brigades\\_returns\\_none](#)  
[omdurman-rules::src::unit\\_profiles::ae\\_infantry\\_third\\_battalion\\_from\\_col\\_2](#)  
[omdurman-rules::src::effects::tests::brigade\\_integrity\\_modifier\\_is\\_engine\\_derived](#)



## §6 – Fire Combat Phase

### §6 – Fire Combat Phase

descriptive

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Fire Combat Phase

### §6.1 – General Rules

descriptive

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General Rules

### §6.2 – How To Have Fire Combat

descriptive

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How To Have Fire Combat

### §6.3 – Line of Sight Table

implemented

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Line of Sight Table

This table is located on the back of this rulebook and should be self-explanatory. Locate the terrain type the firing unit is in and cross-index it with the terrain type the target unit is in. Terrain types in the intersecting box block line of sight, with exceptions as footnoted. Also study the “Special LOS Notes” given and remember that (with the exception of howitzer fire — see 6.64) you can’t fire at anything you can’t see!

See also: §6.64

| File                                                                                          | Symbol              | Code Snippet                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/los_table.rs:57</a><br>GH:omdurman-rules/src/los_table.rs:57   | <b>LosLevel</b>     | <pre>55       serde::Serialize, serde::Deserialize, Clone, Copy, 56       PartialEq, Eq, PartialOrd, Ord, Hash, Debug, 57   )] 58   pub enum LosLevel { 59       Ground, 59       Rough,</pre>                                                                                         |
| <a href="#">omdurman-rules/src/los_table.rs:81</a><br>GH:omdurman-rules/src/los_table.rs:81   | <b>LosFeature</b>   | <pre>79   /// authored RON spellings. 80   #[derive(serde::Serialize, serde::Deserialize, Clone, 81   Copy, PartialEq, Eq, Hash, Debug)] 82   pub enum LosFeature { 83       /// A hex containing units (gunboats/forts excluded 83       Units,</pre>                                 |
| <a href="#">omdurman-rules/src/los_table.rs:103</a><br>GH:omdurman-rules/src/los_table.rs:103 | <b>LosCondition</b> | <pre>101   /// A positional condition from the LOS table Detail 102   footnotes. 103   #[derive(serde::Serialize, serde::Deserialize, Clone, 104   Copy, PartialEq, Eq, Hash, Debug)] 105   pub enum LosCondition { 106       /// (1) Blocks only if the ray passes through more</pre> |

|                                                                                               |                                |                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------------|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                               |                                | than two such features.<br><b>105</b>       MoreThanTwo,                                                                                                                                                                                                                                                                                                         |
| <a href="#">omdurman-rules/src/los_table.rs:183</a><br>GH:omdurman-rules/src/los_table.rs:183 | <b>los_level</b>               | <b>181</b>       ///<br><b>182</b>       /// For all other units, the level is derived from the terrain at `hex`.<br><b>183</b>       pub fn los_level_for_unit(<br><b>184</b>           kind: UnitKind,<br><b>185</b>           hex: HexCoord,                                                                                                                  |
| <a href="#">omdurman-rules/src/los_table.rs:183</a><br>GH:omdurman-rules/src/los_table.rs:183 | <b>los_level_for_unit</b>      | <b>181</b>       ///<br><b>182</b>       /// For all other units, the level is derived from the terrain at `hex`.<br><b>183</b>       pub fn los_level_for_unit(<br><b>184</b>           kind: UnitKind,<br><b>185</b>           hex: HexCoord,                                                                                                                  |
| <a href="#">omdurman-rules/src/los_table.rs:223</a><br>GH:omdurman-rules/src/los_table.rs:223 | <b>blocking_rules</b>          | <b>221</b>       /// always blocks. Indexing is in-bounds by construction (both enums have<br><b>222</b>       /// exactly three variants).<br><b>223</b>       pub fn blocking_rules(firer: LosLevel, target: LosLevel) -> &'static [BlockingRule] {<br><b>224</b>           crate::tables_data::LOS_CELLS[firer.index()][target.index()]<br><b>225</b>       } |
| <a href="#">omdurman-rules/src/los_table.rs:303</a><br>GH:omdurman-rules/src/los_table.rs:303 | <b>has_los</b>                 | <b>301</b>       /// `unit_level_at` closure returns the LOS level of blocking units<br><b>302</b>       /// (non-gunboat, non-fort per note a) in an intervening hex, or `None`.<br><b>303</b>       pub fn has_los(<br><b>304</b>           board: &crate::board::BoardInfo,<br><b>305</b>           from: HexCoord,                                           |
| <a href="#">omdurman-types/src/lib.rs:230</a><br>GH:omdurman-types/src/lib.rs:230             | <b>HexsideKind::blocks_los</b> | <b>228</b>       /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine<br><b>229</b>       /// in `omdurman_rules::los_table`, not by this predicate.<br><b>230</b>       pub fn blocks_los(self) -> bool {<br><b>231</b>           matches!(self, HexsideKind::Wall   HexsideKind::Crest)<br><b>232</b>       }                              |
| <a href="#">omdurman-types/src/lib.rs:230</a><br>GH:omdurman-types/src/lib.rs:230             | <b>Terrain::blocks_los</b>     | <b>228</b>       /// (LOS table conditions 2-4, 7) and note (e) are handled by the engine<br><b>229</b>       /// in `omdurman_rules::los_table`, not by this predicate.<br><b>230</b>       pub fn blocks_los(self) -> bool {<br><b>231</b>           matches!(self, HexsideKind::Wall   HexsideKind::Crest)<br><b>232</b>       }                              |
| <a href="#">omdurman-types/src/lib.rs:452</a><br>GH:omdurman-types/src/lib.rs:452             | <b>Terrain::is_los_trees</b>   | <b>450</b>       /// (\$6.3 note 1). Retained for compatibility; the full LOS engine<br><b>451</b>       /// checks `Terrain::Trees` directly.<br><b>452</b>       pub fn is_los_trees(self) -> bool {<br><b>453</b>           matches!(self, Terrain::Trees { .. })<br><b>454</b>       }                                                                       |

Proven

by: [omdurman-types::src::lib::line\\_between\\_forms\\_a\\_connected\\_ray](#) [omdurman-types::src::lib::hexside\\_blocking\\_classifiers\\_are\\_exact](#)  
[omdurman-rules::src::los\\_table::los\\_level\\_depends\\_only\\_on\\_the\\_ground](#)  
[omdurman-rules::src::los\\_table::los\\_level\\_overrides\\_hold\\_for\\_gunboats\\_and\\_forts](#)  
[omdurman-rules::src::los\\_table::blocking\\_grid\\_is\\_total\\_and\\_walls\\_block\\_ground\\_firers](#)

Covered by tests: [omdurman-rules::src::los\\_table::los\\_level\\_mapping](#) [omdurman-rules::src::los\\_table::blocking\\_rules\\_all\\_cells\\_covered](#)  
[omdurman-rules::src::los\\_table::has\\_los\\_empty\\_board\\_is\\_clear](#) [omdurman-rules::src::los\\_table::has\\_los\\_adjacent\\_clear](#)  
[omdurman-rules::src::los\\_table::has\\_los\\_howitzzer\\_bypasses](#)  
[omdurman-rules::src::los\\_table::has\\_los\\_wall\\_hexside\\_blocks\\_ground\\_to\\_ground](#)  
[omdurman-rules::src::los\\_table::has\\_los\\_gate\\_hexside\\_passes](#) [omdurman-rules::src::los\\_table::has\\_los\\_breach\\_hexside\\_passes](#)  
[omdurman-rules::src::los\\_table::has\\_los\\_rough\\_intervening\\_blocks\\_ground\\_to\\_ground](#)

```

omdurman-rules::src::los_table::has_los_two_tree_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::has_los_three_tree_hexes_block_ground_to_ground
omdurman-rules::src::los_table::has_los_two_hut_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::has_los_three_hut_hexes_block_ground_to_ground
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_clear_no_units
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_blocked_by_hilltop_unit
omdurman-rules::src::los_table::has_los_hilltop_to_hilltop_not_blocked_by_ground_unit
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_lower_level_passes
omdurman-rules::src::los_table::has_los_rough_to_rough_unit_at_same_level_blocks
omdurman-rules::src::los_table::has_los_rough_to_rough_hilltop_blocks
omdurman-rules::src::los_table::has_los_ground_to_hilltop_intervening_hilltop_blocks
omdurman-rules::src::los_table::has_los_building_blocks_like_huts_ground_to_ground
omdurman-rules::src::los_table::has_los_two_building_hexes_pass_ground_to_ground
omdurman-rules::src::los_table::los_level_for_unit_gunboat_is_rough
omdurman-rules::src::los_table::los_level_for_unit_fort_is_ground
omdurman-rules::src::los_table::los_level_for_unit_walled_city_adj_wall_is_rough
omdurman-rules::src::los_table::gunboat_firer_uses_rough_row_not_ground
omdurman-rules::src::los_table::los_reflexive_all_terrains
omdurman-rules::src::los_table::los_reflexive_hilltop
omdurman-rules::src::los_table::los_symmetric_ground_to_ground_no_units
omdurman-rules::src::los_table::los_howitzer_always_has_los
omdurman-rules::src::los_table::los_howitzer_same_hex
omdurman-rules::src::los_table::los_blocking_rules_match_reference_table
omdurman-rules::src::los_table::los_ground_to_ground_features_block_as_expected

```

## §6.4 – Fire Combat Sequence

implemented

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### Fire Combat Sequence

The sequence of fire combat resolution is the same for both defensive and offensive fire combat. During the Dervish player turn the Anglo-Egyptian player executes 6.41 AND 6.42 as defensive fire, after which the Dervish player executes 6.41 as offensive fire. During the Anglo-Egyptian player turn the Dervish player executes 6.41 as defensive fire, after which the Anglo-Egyptian player executes 6.41 AND 6.42 as offensive fire.

**\*\*6.41) Direct Fire Subphase (Dervish and Anglo-Egyptian players):\*\*** The firing player must first allocate all of his fire attacks, combining his units' direct fire combat factors in any manner he wishes. After all fire has been allocated, the firing player then resolves his attacks in any order he wishes.

**\*\*6.42) Maxim Second Fire and Howitzer Fire Subphase (Anglo-Egyptian player only):\*\*** Anglo-Egyptian named gunboats may now fire their artillery factor as howitzer fire (see 6.64) and all Maxim guns may fire a second time. Once

(see manual for full text)

See also: §6.41, §6.42, §6.64

| File                                                                                                  | Symbol              | Code Snippet                                                                                                                                                                                                                                                                                          |
|-------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:302</a><br>GH:omdurman-rules/src/lib.rs:302                     | <b>FireSubPhase</b> | <pre> 300   301   #[derive(Serialize, Deserialize, Clone, Copy, 302           PartialEq, Eq, Debug)] 303   pub enum FireSubPhase { 304       /// Direct fire (§6.41). Both sides participate in 305       this sub-phase. 306       DirectFire, </pre>                                                |
| <a href="#">omdurman-rules/src/effects/effect.rs:72</a><br>GH:omdurman-rules/src/effects/effect.rs:72 | <b>FireCombat</b>   | <pre> 70       /// - Firers marked as fired; target hex marked as 71       fired-at. 72       /// - Victory points awarded for eliminations. 73       FireCombat { attack: FireAttack, roll: DieRoll }, 74       /// Resolve a howitzer bombardment (two rolls: CRT 75       + impact scatter) </pre> |

Covered by tests: [omdurman-rules::src::effects::tests::turn\\_advances\\_through\\_phases](#)

§6.5 – Special Unit Capabilities

descriptive  
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Special Unit Capabilities

**\*\*6.51) Leader Units:\*\*** Dervish leader units have a fire factor, a melee factor, and a movement factor. They may thus attack, melee, and be eliminated like any other combat unit. Their special benefit is that they stack free.

Anglo-Egyptian leaders have a movement factor only. They are eliminated if a) they are alone in a hex when a Dervish unit occupies or passes through that hex, or b) if all of the combat units a leader is stacked with are eliminated in fire combat or melee. The special function of Anglo-Egyptian leaders is that at least one must survive to occupy the Mahdi’s tomb hex if it is to be taken from the Dervish player (see 9.14).

**\*\*6.52) Anglo-Egyptian “Friendlies” Brigade:\*\*** These units represent native volunteers in the Anglo-Egyptian army. They fire rifles on the Dervish Range Effects Table and melee with the Dervish melee modifier. They may not enter the walled city of Omdurman (see 5.23). They may be transferred to the west bank (see 5.21).

(see manual for full text)

See also: §5.21, §5.23, §5.44, §6.51, §6.52, §6.53, §6.54, §6.62, §6.63, §7.6, §9.14

§6.6 – Special Artillery Capabilities

descriptive  
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Special Artillery Capabilities

§6.7 – Defensive Fire

implemented  
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Defensive Fire

In Defensive Fire phase, all of the non-moving player’s units may fire at any of the moving player’s units in range, within the limitations imposed by the rules of combat (see 6.1 to 6.6). There is no advance after combat as a result of defensive fires.

See also: §6.1, §6.6

| File                                                                                    | Symbol                 | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects/state.rs:1932<br>GH:omdurman-rules/src/effects/state.rs:1932 | can_advance_after_comb | 1930   // player's unit, not artillery, adjacent to<br>`to`, and `to` now empty.<br>1931   // Wall/khor hexside restrictions are not<br>enforced (no hexside map data).<br>1932   pub fn can_advance_after_combat(&self, unit_id:<br>UnitId, to: HexCoord) -> Result<(), RuleError> {<br>1933   let unit = self.unit_or_err(unit_id)?;<br>1934   // §6.7: there is no advance after combat as a<br>result of defensive fire. |

Covered by tests: omdurman-rules::src::effects::tests::no\_advance\_after\_defensive\_fire  
omdurman-rules::src::effects::tests::defensive\_fire\_opens\_no\_advance\_window

§6.8 – Offensive Fire

descriptive  
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Offensive Fire

## §6.11 – Fire combat factor printed on units

implemented

manual page unknown

The fire combat factor of the various unit types is printed directly on the units and is a numerical expression of the unit's fire strength.

| File                                                                            | Symbol            | Code Snippet                                                                                                                                                                                                                                                                                               |
|---------------------------------------------------------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:86</a><br>GH:omdurman-rules/src/lib.rs:86 | <b>FireFactor</b> | <pre>84       /// Every possible value from the annotated counter 85       set is a named variant. 86       #[derive(Serialize, Deserialize, Clone, Copy, 87               PartialEq, Eq, Hash, Debug, strum::Display)] 88       pub enum FireFactor { 89           One = 1, 90           Three = 3,</pre> |

Covered by tests: [omdurman-rules::src::lib::fire\\_factor\\_sum\\_to\\_row](#)

## §6.12 – Fire combat is always voluntary

implicit

manual page unknown

Fire combat is always voluntary. A unit is never required to fire at enemy units merely because they are in range or adjacent.

## §6.13 – Fire factor is unitary (may not be divided)

implicit

manual page unknown

If a unit elects to fire, its fire combat factor at an enemy unit, that fire combat factor is unitary. A unit's fire combat factor may not be divided up to fire at enemy units on different hexes.

## §6.14 – Players may combine fire factors into one attack

implemented

manual page unknown

Players may combine fire during fire combat phase, i.e. they may fire at an enemy-occupied hex with as many friendly units as may legally do so, combining all of their fire combat factors into one attack. Note that in any given fire combat phase, however, a combat unit may only fire once and may only be fired at once (exceptions: Maxim guns and gunboats — see 6.4).

See also: §6.4

| File                                                                              | Symbol            | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:100</a><br>GH:omdurman-rules/src/lib.rs:100 | <b>sum_to_row</b> | <pre>98       impl FireFactor { 99           /// Sum multiple fire factors and return the 100           corresponding Combat Results Table row (rulebook §6.11). 101           pub fn sum_to_row&lt;'a&gt;(factors: impl 102               IntoIterator&lt;Item = &amp;'a FireFactor&gt;) -&gt; FireFactorRow { 103               let total: u16 = factors.into_iter().map( f  104                   f.value()).sum(); 105               crate::combat_results_table::FireFactorRow::from_total(total)</pre> |

Covered by tests: `omdurman-rules::src::effects::tests::unit_may_only_be_fired_at_once_per_phase`  
`omdurman-rules::src::effects::tests::gunboat_and_maxim_may_be_fired_at_repeatedly`  
`omdurman-rules::src::effects::tests::rejected_fire_attack_does_not_mark_firers_as_fired`

## §6.15 – May divide a stack to fire at different hexes

implicit

manual page unknown

Players may also divide a stack of units in order to fire at different enemy-occupied hexes. Anglo-Egyptian infantry units having brigade integrity, however, do not receive their +1 direct fire modifier unless they all fire at the same enemy-occupied hex (see 5.54).

See also: §5.54

## §6.16 – Halving fire strength rounds down, minimum 1

implemented

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When halving fire combat strength, always round down each individual unit. For example, an Egyptian brigade of four battalions, each having a printed strength of 9 fire factors, will fire a total of 16 factors when halved. However, a unit's firing strength is never reduced below one by halving.

| File                                                                              | Symbol           | Code Snippet                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:519</a><br>GH:omdurman-rules/src/lib.rs:519 | <b>RangeBand</b> | <pre>517   /// multiplied at a given distance (§6.22). 518   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 519   pub enum RangeBand { 520       Tripled, 521       Doubled,</pre> |

Proven by: `omdurman-rules::src::lib::range_band_halved_is_max_of_one_and_floor_half`  
`omdurman-rules::src::lib::range_band_apply_is_monotone_in_raw` `omdurman-rules::src::lib::disrupt_half_is_rounded_up`

Covered by tests: `omdurman-rules::src::lib::halving_rounds_down_and_never_below_one`

## §6.21 – First check LOS before firing

implemented

manual page unknown

When combat units wish to fire at enemy units, first check the Line of Sight Table to be sure the firing unit can see the target hex (exception: howitzer fire, see 6.64).

See also: §6.64

| File                                                                                          | Symbol                | Code Snippet                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/los_table.rs:223</a><br>GH:omdurman-rules/src/los_table.rs:223 | <b>blocking_rules</b> | <pre>221   /// always blocks. Indexing is in-bounds by construction (both enums have 222   /// exactly three variants). 223   pub fn blocking_rules(firer: LosLevel, target: LosLevel) -&gt; &amp;'static [BlockingRule] { 224       crate::tables_data::LOS_CELLS[firer.index()] [target.index()] 225   }</pre> |
| <a href="#">omdurman-rules/src/los_table.rs:303</a><br>GH:omdurman-                           | <b>has_los</b>        | <pre>301   /// `unit_level_at` closure returns the LOS level of blocking units 302   /// (non-gunboat, non-fort per note a) in an</pre>                                                                                                                                                                          |

|                                |  |                                                                                                                                 |
|--------------------------------|--|---------------------------------------------------------------------------------------------------------------------------------|
| rules/src/<br>los_table.rs:303 |  | intervening hex, or `None`.<br>303   pub fn has_los(<br>304       board: &crate::board::BoardInfo,<br>305       from: HexCoord, |
|--------------------------------|--|---------------------------------------------------------------------------------------------------------------------------------|

Covered by tests:  
omdurman-rules::src::effects::tests::can\_fire\_at\_rejects\_blocked\_los omdurman-rules::src::effects::tests::can\_fire\_at\_allows\_clear\_los

## §6.22 – Consult Range Effects Table

implemented

manual page unknown

Next consult the Range Effects Table to see if the firing unit's fire combat factor is tripled, doubled, normal, halved, or if the target hex is out of range. Add up the total number of fire combat factors firing at the enemy-occupied hex.

| File                                                                                | Symbol                | Code Snippet                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/range_effects.rs:34<br>GH:omdurman-rules/src/range_effects.rs:34 | ae_range_effects      | 32   /// Look up the range band for an Anglo-Egyptian weapon (\$6.22, §6.24).<br>33   /// Distances > 10 are out of range for all weapons.<br>34   pub fn ae_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand {<br>35       band_at(faction_rows(true), weapon, distance)<br>36   } |
| omdurman-rules/src/range_effects.rs:40<br>GH:omdurman-rules/src/range_effects.rs:40 | dervish_range_effects | 38   /// Look up the range band for a Dervish weapon (\$6.22).<br>39   /// Distances > 10 are out of range for all weapons.<br>40   pub fn dervish_range_effects(weapon: WeaponClass, distance: HexDistance) -> RangeBand {<br>41       band_at(faction_rows(false), weapon, distance)<br>42   }          |
| omdurman-rules/src/lib.rs:519<br>GH:omdurman-rules/src/lib.rs:519                   | RangeBand             | 517   /// multiplied at a given distance (§6.22).<br>518   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]<br>519   pub enum RangeBand {<br>520       Tripled,<br>521       Doubled,                                                                                                 |
| omdurman-rules/src/lib.rs:187<br>GH:omdurman-rules/src/lib.rs:187                   | HexDistance           | 185   /// (rulebook §6.22, §7.5).<br>186   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)]<br>187   pub struct HexDistance(u16);<br>188  <br>189   impl HexDistance {                                                                                         |

Proven by:  
omdurman-types::src::lib::adjacency\_iff\_distance\_one omdurman-rules::src::lib::fire\_factor\_row\_from\_total\_matches\_printed\_bands  
omdurman-rules::src::lib::range\_band\_multiplier\_arithmetic\_is\_exact  
omdurman-rules::src::range\_effects::range\_effects\_are\_out\_of\_range\_outside\_the\_printed\_ten\_hex\_distance\_window  
omdurman-rules::src::range\_effects::max\_day\_range\_is\_the\_last\_in\_range\_hex  
omdurman-rules::src::effects::fire::range\_band\_for\_routes\_to\_the\_right\_faction\_table

Covered by tests:  
omdurman-rules::src::range\_effects::ae\_rifles\_doubled\_at\_range\_1 omdurman-rules::src::range\_effects::ae\_rifles\_halved\_at\_range\_4  
omdurman-rules::src::range\_effects::ae\_howitzer\_range omdurman-rules::src::range\_effects::dervish\_rifles\_shorter\_range  
omdurman-rules::src::range\_effects::melee\_only\_range\_1 omdurman-rules::src::range\_effects::ae\_range\_effects\_artillery\_full  
omdurman-rules::src::range\_effects::ae\_range\_effects\_maxims\_match\_rifles  
omdurman-rules::src::range\_effects::ae\_range\_effects\_distance\_over\_10  
omdurman-rules::src::range\_effects::ae\_range\_effects\_howitzer\_halved\_4\_to\_10  
omdurman-rules::src::range\_effects::dervish\_range\_effects\_rifles omdurman-rules::src::range\_effects::dervish\_range\_effects\_artillery  
omdurman-rules::src::range\_effects::dervish\_range\_effects\_maxims\_and\_howitzer  
omdurman-rules::src::range\_effects::dervish\_range\_effects\_melee  
omdurman-rules::src::range\_effects::dervish\_range\_effects\_distance\_over\_10

omdurman-rules::src::effects::tests::fire\_combat\_eliminate\_target  
 omdurman-rules::src::range\_effects::range\_effects\_every\_cell\_ae\_artillery  
 omdurman-rules::src::range\_effects::range\_effects\_every\_cell\_ae\_howitzer  
 omdurman-rules::src::range\_effects::range\_effects\_every\_cell\_ae\_maxims  
 omdurman-rules::src::range\_effects::range\_effects\_every\_cell\_ae\_rifles  
 omdurman-rules::src::range\_effects::range\_effects\_every\_cell\_dervish\_artillery  
 omdurman-rules::src::range\_effects::range\_effects\_every\_cell\_dervish\_maxims\_howitzer  
 omdurman-rules::src::range\_effects::range\_effects\_every\_cell\_dervish\_rifles  
 omdurman-rules::src::range\_effects::range\_effects\_every\_cell\_dervish\_spears  
 omdurman-rules::src::effects::tests::can\_fire\_at\_gates\_phase\_range\_and\_player  
 omdurman-rules::src::effects::tests::mixed\_attack\_bands\_per\_firer  
 omdurman-rules::src::range\_effects::ae\_range\_effects\_monotone\_non\_increasing  
 omdurman-rules::src::range\_effects::ae\_howitzer\_has\_minimum\_range  
 omdurman-rules::src::range\_effects::dervish\_range\_effects\_monotone\_non\_increasing  
 omdurman-rules::src::range\_effects::range\_effects\_first\_range\_max\_effect\_last\_range\_oor

## §6.23 – Terrain defensive modifier

implemented

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Next check the Terrain Effects Chart to see if the enemy-occupied hex fired upon contains any terrain which gives the enemy units in that hex a defensive benefit. If so, apply this negative modifier to the roll of the ten-sided die and cross-index your net die roll on the Combat Results Table with the number of combat factors firing.

| File                                                                                | Symbol                       | Code Snippet                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/terrain_chart.rs:59<br>GH:omdurman-rules/src/terrain_chart.rs:59 | <b>defense_modifier</b>      | <pre> 57   58   /// Convenience: get the defense modifier for a terrain       type (rulebook §6.23, Terrain Effects Chart). 59   pub fn defense_modifier(terrain: Terrain) -&gt; i16 { 60       terrain_effects_chart(terrain).defense_modifier 61   }</pre> |
| omdurman-rules/src/lib.rs:937<br>GH:omdurman-rules/src/lib.rs:937                   | <b>FireModifier::Terrain</b> | <pre> 935       /// Negative modifier from the Terrain Effects       Chart applied to the 936       /// defender's hex (§6.23). 937       Terrain(i16), 938       /// -2 thorn-hedge defensive modifier (§9.231). 939       ZaribaThornHedge,</pre>          |

Proven by: omdurman-rules::src::lib::fire\_modifier\_keeps\_roll\_legal  
 omdurman-rules::src::terrain\_chart::defence\_column\_never\_helps\_the\_attacker

Covered by tests: omdurman-rules::src::terrain\_chart::clear\_terrain\_no\_bonus omdurman-rules::src::terrain\_chart::building\_gives\_minus\_3  
 omdurman-rules::src::terrain\_chart::palm\_grove\_gives\_minus\_2 omdurman-rules::src::terrain\_chart::rough\_movement\_and\_defense  
 omdurman-rules::src::terrain\_chart::swamp\_movement\_and\_defense  
 omdurman-rules::src::terrain\_chart::hilltop\_movement\_and\_defense  
 omdurman-rules::src::terrain\_chart::huts\_movement\_and\_defense  
 omdurman-rules::src::terrain\_chart::defense\_modifier\_convenience\_matches\_chart  
 omdurman-rules::src::terrain\_chart::terrain\_movement\_costs\_in\_bounds  
 omdurman-rules::src::terrain\_chart::terrain\_defense\_modifier\_non\_positive

## §6.24 – Anglo-Egyptian direct fire accuracy bonus and brigade integrity

implemented

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All Anglo-Egyptian direct fire attacks receive a +1 modifier to their die roll as an accuracy bonus. In addition, any stack of Anglo-Egyptian infantry having brigade integrity (see 5.54) receives a +1 modifier to their die roll if all four fire at the same enemy-occupied hex. These modifiers are cumulative.

See also: §5.54

| File                                                                                | Symbol                            | Code Snippet                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:931</a><br>GH:omdurman-rules/src/lib.rs:931   | <b>AngloEgyptianDirectFire</b>    | <pre>929   pub enum FireModifier { 930       /// +1 to all Anglo-Egyptian *direct* fire (§6.24). 931       AngloEgyptianDirectFire, 932       /// +1 brigade integrity, applied only if all four           battalions fire at 933       /// the same enemy-occupied hex (§5.54, §6.24).</pre>  |
| <a href="#">omdurman-rules/src/lib.rs:916</a><br>GH:omdurman-rules/src/lib.rs:916   | <b>BrigadeIntegrity</b>           | <pre>914   /// stack contains all four battalions of a single       Anglo-Egyptian brigade. 915   #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Debug)] 916   pub enum BrigadeIntegrity { 917       None, 918       Integrated(BrigadeId),</pre>                          |
| <a href="#">omdurman-rules/src/lib.rs:1031</a><br>GH:omdurman-rules/src/lib.rs:1031 | <b>FireModifier::die_modifier</b> | <pre>1029   impl MeleeModifier { 1030       /// Return the numeric die-roll modifier for this       melee bonus/penalty (rulebook §7.7, §9.232). 1031       pub fn die_modifier(self) -&gt; i16 { 1032           match self { 1033               MeleeModifier::DervishStandard =&gt; 2,</pre> |

Proven by: [omdurman-rules::src::lib::die\\_roll\\_apply\\_modifier\\_is\\_total](#)

Covered by tests: [omdurman-rules::src::effects::tests::brigade\\_integrity\\_modifier\\_is\\_engine\\_derived](#)  
[omdurman-rules::src::effects::tests::fire\\_modifiers\\_are\\_engine\\_derived\\_and\\_mismatches\\_rejected](#)

## §6.41 – Direct Fire Subphase

**implemented**

manual page unknown

| File                                                                              | Symbol            | Code Snippet                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:304</a><br>GH:omdurman-rules/src/lib.rs:304 | <b>DirectFire</b> | <pre>302   pub enum FireSubPhase { 303       /// Direct fire (§6.41). Both sides participate in       this sub-phase. 304       DirectFire, 305       /// Anglo-Egyptian only: Maxim second fire + named-       gunboat howitzer fire (§6.42). 306       MaximSecondAndHowitzer,</pre> |

Covered by tests: [omdurman-rules::src::effects::tests::disrupted\\_unit\\_cannot\\_fire](#) [omdurman-rules::src::effects::tests::fire\\_combat\\_elimimates\\_target](#)

## §6.42 – Maxim Second Fire and Howitzer Fire Subphase

**implemented**

manual page unknown

| File                                                                              | Symbol                        | Code Snippet                                                                                                                                                                        |
|-----------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:306</a><br>GH:omdurman-rules/src/lib.rs:306 | <b>MaximSecondAndHowitzer</b> | <pre>304       DirectFire, 305       /// Anglo-Egyptian only: Maxim second fire + named-       gunboat howitzer fire (§6.42). 306       MaximSecondAndHowitzer, 307   } 308  </pre> |
| <a href="#">omdurman-types/src/lib.rs:975</a><br>GH:omdurman-types/src/lib.rs:975 | <b>fires_twice</b>            | <pre>973       /// Maxim guns fire twice per turn -- once in the       Direct Fire Subphase and 974       /// again in the Maxim Second Fire Subphase       (rulebook §6.42).</pre> |

|  |  |                                                                                                                                           |
|--|--|-------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | <pre> 975           pub fn fires_twice(self) -&gt; bool { 976               matches!(self, UnitKind::Maxim { .. }) 977           } </pre> |
|--|--|-------------------------------------------------------------------------------------------------------------------------------------------|

Proven by: `omdurman-types::src::lib::unit_kind_fire_and_movement_capability Law is exact`

Covered by tests: `omdurman-rules::src::howitzer_scatter::howitzer_on_target_7_to_10`  
`omdurman-rules::src::howitzer_scatter::howitzer_scatters_below_7`  
`omdurman-rules::src::effects::tests::advance_window_bridges_fire_subphase_and_closes_at_melee`  
`omdurman-rules::src::effects::tests::fired_at_tracker_resets_at_maxim_subphase`

## §6.51 – Leader Units

**implemented**

manual page unknown

| File                                                                              | Symbol                    | Code Snippet                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:417</a><br>GH:omdurman-rules/src/lib.rs:417 | <b>BritishLeader</b>      | <pre> 415   // to claim the Mahdi's Tomb (§9.14). 416   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Hash, Debug, strum::Display)] 417   pub enum BritishLeader { 418       Kitchener, 419       Gatacre, </pre>                       |
| <a href="#">omdurman-types/src/lib.rs:922</a><br>GH:omdurman-types/src/lib.rs:922 | <b>BritishLeader</b>      | <pre> 920       }, 921       // Anglo-Egyptian leader (§6.51): movement only. 922       BritishLeader { movement: i32 }, 923       // Wall-breach marker placed by artillery fire (\$6.63). Not a combat unit. 924       Breach, </pre>                |
| <a href="#">omdurman-types/src/lib.rs:963</a><br>GH:omdurman-types/src/lib.rs:963 | <b>has_combat_factors</b> | <pre> 961   // British leaders print a movement factor only (\$6.51); other kinds carry 962   // fire and/or melee factors. Markers carry no stats. 963   pub fn has_combat_factors(self) -&gt; bool { 964       !matches!( 965           self, </pre> |

Proven by: `omdurman-types::src::lib::unit_kind_fire_and_movement_capability Law is exact`

Covered by tests: `omdurman-rules::src::unit_profiles::zero_factor_is_none_not_zero`  
`omdurman-rules::src::unit_profiles::kitchener_block_resolves_leaders_friendlys_camel_and_sudanese`  
`omdurman-rules::src::unit_profiles::dervish_leader_sections_resolve_leader_and_retinue_per_cell`  
`omdurman-rules::src::effects::tests::deploy_rejects_enemy_cohabitation_during_setup`  
`omdurman-rules::src::effects::tests::dervish_move_through_lone_ae_leader_hex_elimimates_the_leader`  
`omdurman-rules::src::effects::tests::dervish_move_onto_lone_ae_leader_hex_elimimates_the_leader`  
`omdurman-rules::src::effects::tests::ae_leader_with_combat_unit_is_not_overrun_by_dervish_move`  
`omdurman-rules::src::effects::tests::ae_leader_eliminated_with_last_combat_unit_in_fire_combat`

## §6.52 – Anglo-Egyptian Friendlys Brigade

**implemented**

manual page unknown

| File                                                                              | Symbol              | Code Snippet                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:629</a><br>GH:omdurman-rules/src/lib.rs:629 | <b>is_friendlys</b> | <pre> 627   // "Friendlys" units obey several special rules (\$5.21, §5.23, §6.52, 628   // §9.14 victory conditions). 629   pub fn is_friendlys(&amp;self) -&gt; bool { 630       matches!( 631           self, </pre> |

|                                                                                   |                   |                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:995</a><br>GH:omdurman-types/src/lib.rs:995 | <b>Friendlies</b> | <pre> 993       /// Native volunteer brigade -- the Shaggyeh 994       (\$6.52). Do not receive 995       /// brigade integrity (\$5.54 enumerates only 996       British/Egyptian/Sudanese). 997       Friendlies, 998   }</pre> |
|-----------------------------------------------------------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Proven by: [omdurman-rules::src::effects::fire::range\\_table\\_player\\_for\\_routes\\_friendlys\\_and\\_fok\\_to\\_the\\_dervish\\_table](#)

Covered by tests: [omdurman-rules::src::unit\\_profiles::friendlys\\_counters\\_score\\_by\\_bank\\_not\\_as\\_leaders](#)  
[omdurman-rules::src::effects::tests::friendlys\\_validate\\_and\\_resolve\\_on\\_dervish\\_table](#)

## §6.53 – Royal Engineers demolition

**implemented**

manual page unknown

| File                                                                                                    | Symbol                    | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:600</a><br>GH:omdurman-rules/src/lib.rs:600                       | <b>RoyalEngineers</b>     | <pre> 598       /// The Royal Engineers (\$6.53) -- a *specific* 599       unit, not a class, so we 600       /// model it explicitly. 601       RoyalEngineers, 602   }</pre>                                                                                                                                                                                                                                                    |
| <a href="#">omdurman-rules/src/lib.rs:835</a><br>GH:omdurman-rules/src/lib.rs:835                       | <b>demolishing</b>        | <pre> 833       /// Set when the Royal Engineers are committed to a 834       demolition this turn 835       /// (\$6.53) -- neither offensive fire nor melee 836       allowed that turn. 837       pub demolishing: bool, 838       /// Set when a gunboat has lost its engines to a 839       river mine (\$10.12, roll 840       /// 5-7): it may no longer move under power and 841       instead drifts two hexes per</pre> |
| <a href="#">omdurman-rules/src/effects/effect.rs:175</a><br>GH:omdurman-rules/src/effects/effect.rs:175 | <b>Demolition</b>         | <pre> 173       /// Royal Engineers demolition (rulebook \$6.53). 174       Demolition { 175           unit_id: UnitId, 176           target: DemolitionTarget, 177       }</pre>                                                                                                                                                                                                                                                 |
| <a href="#">omdurman-rules/src/effects/setup.rs:35</a><br>GH:omdurman-rules/src/effects/setup.rs:35     | <b>apply_demolition</b>   | <pre> 33       /// resolution happens at end of turn via 34       ['apply_resolve_demolition'], which 35       /// checks the engineer is still adjacent and 36       undisrupted. 37       pub fn apply_demolition( 38           state: &amp;mut GameState, 39           unit_id: UnitId,</pre>                                                                                                                                  |
| <a href="#">omdurman-rules/src/lib.rs:1061</a><br>GH:omdurman-rules/src/lib.rs:1061                     | <b>DemolitionTarget</b>   | <pre> 1059       /// disrupted or driven off. 1060       #[derive(Serialize, Deserialize, Clone, Copy, 1061               PartialEq, Eq, Debug)] 1062       pub enum DemolitionTarget { 1063           Fort(UnitId), 1064           WallHexside(HexsideRef),</pre>                                                                                                                                                                |
| <a href="#">omdurman-rules/src/effects/state.rs:2028</a><br>GH:omdurman-rules/src/effects/state.rs:2028 | <b>demolition_targets</b> | <pre> 2026       /// the rules would accept. Empty when the unit 2027       doesn't exist or has no 2028       /// adjacent target. 2029       pub fn demolition_targets(&amp;self, unit_id: UnitId) 2030       -&gt; Vec&lt;DemolitionTarget&gt; { 2031           let Ok(unit) = self.unit_or_err(unit_id) else 2032           { 2033               return Vec::new(); 2034           }</pre>                                    |

Covered by tests: `omdurman-rules::src::effects::tests::demolition_cancelled_when_engineer_disrupted`  
`omdurman-rules::src::effects::tests::demolition_cancelled_when_engineer_moved_away`  
`omdurman-rules::src::effects::tests::demolition_targets_finds_adjacent_fort_and_wall`

## §6.54 – Forts

implemented

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| File                                                                              | Symbol                          | Code Snippet                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:868</a><br>GH:omdurman-rules/src/lib.rs:868 | <b>ZocReason</b>                | <pre>866   /// Used by the engine when answering "is this hex in 867   an enemy ZOC?". 867   #[derive(Serialize, Deserialize, Clone, Copy, 868   PartialEq, Eq, Debug)] 868   pub enum ZocReason { 869       /// Normal ZOC: any non-disrupted unit other than 870       an Anglo-Egyptian 870       /// leader (§5.41) projects ZOC into each of its 870       six adjacent hexes.</pre> |
| <a href="#">omdurman-rules/src/lib.rs:875</a><br>GH:omdurman-rules/src/lib.rs:875 | <b>Fort</b>                     | <pre>873   GunboatVsGunboat, 874       /// Forts project ZOC out of, but not into, an 875       empty fort (§5.44, §6.54). 875       Fort, 876       /// Walled-city ZOC: extends out through walls and 877       gates but not in, 877       /// across a breach in both directions (§5.44).</pre>                                                                                       |
| <a href="#">omdurman-rules/src/lib.rs:823</a><br>GH:omdurman-rules/src/lib.rs:823 | <b>UnitState</b>                | <pre>821   /// rather than one big enum. 822   #[derive(Serialize, Deserialize, Clone, Copy, 823   PartialEq, Eq, Debug, Default)] 823   pub struct UnitState { 824       /// Reference table: "Disrupted units: no ZOC; may 825       not move; may not fire 825       /// offensively or defensively; may not melee; are 825       turned face up at the</pre>                          |
| <a href="#">omdurman-rules/src/lib.rs:976</a><br>GH:omdurman-rules/src/lib.rs:976 | <b>FireAttack</b>               | <pre>974   /// modifiers (rulebook §6). 975   #[derive(Serialize, Deserialize, Clone, Debug, 976   PartialEq, Eq)] 976   pub struct FireAttack { 977       pub firing_player: Player, 978       pub phase: Phase,</pre>                                                                                                                                                                   |
| <a href="#">omdurman-rules/src/lib.rs:991</a><br>GH:omdurman-rules/src/lib.rs:991 | <b>FireAttack::net_modifier</b> | <pre>989   impl FireAttack { 990       /// Sum of all fire modifiers applied to this 991       attack (rulebook §6.24). 991       pub fn net_modifier(&amp;self) -&gt; i16 { 992           // Saturating fold rather than `sum()`: the 993           modifier list is unbounded 993           // and arrives over the network, so a plain sum 993           can overflow `i16`.</pre>     |

Covered by tests: `omdurman-rules::src::effects::tests::retreat_before_melee_may_not_land_on_enemy_fort`

## §6.61 – Only artillery may fire at gunboats; 3+ to sink

implemented

manual page unknown

Only artillery may fire at gunboats. A result of 3 or more on the combat results table is required to sink a gunboat. Any other result is a miss.

| File | Symbol | Code Snippet |
|------|--------|--------------|
|------|--------|--------------|

|                                                                                   |                    |                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:476</a><br>GH:omdurman-rules/src/lib.rs:476 | <b>WeaponClass</b> | <pre>474   Serialize, Deserialize, Clone, Copy, PartialEq, Eq, 475   PartialOrd, Ord, Hash, Debug, strum::Display, 476   pub enum WeaponClass { 477       /// Dervish spears and swords -- no ranged fire at 478       all. 479       Melee,</pre> |
|-----------------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Covered by tests: [omdurman-rules::src::effects::tests::rifles\\_may\\_not\\_sink\\_a\\_gunboat](#)  
[omdurman-rules::src::effects::tests::artillery\\_sinks\\_gunboat\\_only\\_on\\_three\\_plus](#)

§6.62 – Only artillery may fire at forts; 2+ to destroy

**implemented**

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Only artillery may fire at forts. A result of 2 or more on the combat results table is required to eliminate a fort. Any other result is a miss. If the fort contains any enemy units at the instant it is destroyed, one unit is eliminated with the fort.

| File                                                                              | Symbol             | Code Snippet                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:476</a><br>GH:omdurman-rules/src/lib.rs:476 | <b>WeaponClass</b> | <pre>474   Serialize, Deserialize, Clone, Copy, PartialEq, Eq, 475   PartialOrd, Ord, Hash, Debug, strum::Display, 476   pub enum WeaponClass { 477       /// Dervish spears and swords -- no ranged fire at 478       all. 479       Melee,</pre> |

Covered by tests: [omdurman-rules::src::effects::tests::artillery\\_destroys\\_fort\\_on\\_two\\_or\\_better\\_only](#)

§6.63 – Only artillery may breach wall hexsides; 2+ required

**implemented**

manual page unknown

Only artillery may fire to breach a wall hexside of Khartoum or the walled city of Omdurman. A result of 2 or more on the combat results table is required to breach a wall. Any other result is a miss. The effect of the breach is to negate the wall hexside for line of sight purposes. Place a “BREACH” marker in an adjacent hex so that the arrow points to the breached hexside. If any enemy units are adjacent to the wall hexside at the instant it is breached, one enemy unit is eliminated.

**\*\*6.64) Howitzer fire:\*\*** Five units in the game have howitzer fire capability. These are the five named British gunboats. They may fire their artillery factor as direct fire during the Direct Fire Subphase (see 4 and 6.41) and may then fire the same artillery factor as howitzer fire during the Maxim Second Fire and Howitzer Subphase (see 4 and 6.42). Exception: no howitzer fire is allowed during night game turns. To fire howitzer fire, select any target hex between 4 and 10 hexes from the firing gunboat (ignoring the Line of Sight Table) and roll the ten-sided die twice. The first die roll is the Combat Results Table die roll and the second roll is the impact hex die roll. Refer to the Howitzer Fire Subphase on the marchest for the impact hex. The designated target hex is hit on a roll of 7-10. (see manual for full text)

See also: §6.41, §6.42, §6.64

| File                                                                              | Symbol        | Code Snippet                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:198</a><br>GH:omdurman-types/src/lib.rs:198 | <b>Breach</b> | <pre>196   /// Breach in a wall (artillery/§6.63 or Royal 197   Engineers/§6.53). ZOC both 198   /// ways; LOS no longer blocked across the hexside. 199   Breach, 200   /// Khor -- gully/wadi. ZOCs do not extend across 201   (\$5.44); advance after 202   /// combat may not cross (§6.82).</pre> |

|                                                                                                         |                               |                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/effect.rs:278</a><br>GH:omdurman-rules/src/effects/effect.rs:278 | <b>ArtilleryBreachWall</b>    | <pre> 276       /// pre-rolled d10 used for the CRT lookup; range/           LOS are re-derived by the 277       /// engine from the firers and `target`. 278       ArtilleryBreachWall { 279           firers: Vec&lt;UnitId&gt;, 280           target: HexsideRef,           </pre>             |
| <a href="#">omdurman-rules/src/effects/fire.rs:607</a><br>GH:omdurman-rules/src/effects/fire.rs:607     | <b>apply_artillery_breach</b> | <pre> 605       /// artillery's CRT roll -- the rulebook specifies the           same "2+ required" 606       /// threshold for both trigger styles. 607       pub fn apply_artillery_breach_wall(               state: &amp;mut GameState, 608           firers: &amp;[UnitId],           </pre> |
| <a href="#">omdurman-rules/src/effects/state.rs:1228</a><br>GH:omdurman-rules/src/effects/state.rs:1228 | <b>can_fire_at_wall</b>       | <pre> 1226       /// range band and resolving the CRT -- this method           only validates one 1227       /// firer at a time. 1228       pub fn can_fire_at_wall(               &amp;self, 1229           firer: UnitId,           </pre>                                                     |

Covered by tests: [omdurman-rules::src::unit\\_profiles::breach\\_marker\\_cell\\_returns\\_none](#)  
[omdurman-rules::src::effects::tests::artillery\\_breaches\\_wall\\_only\\_on\\_crt\\_two\\_or\\_better](#)  
[omdurman-rules::src::effects::tests::wall\\_breach\\_eliminate\\_one\\_adjacent\\_enemy\\_unit](#)

## §6.64 – Howitzer fire

**implemented**

manual page unknown

| File                                                                                                      | Symbol                     | Code Snippet                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/howitzer_scatter.rs:8</a><br>GH:omdurman-rules/src/howitzer_scatter.rs:8   | <b>ScatterHexDirection</b> | <pre> 6       /// the diagram away from the firing player. 7       #[derive(serde::Serialize, serde::Deserialize, Clone,           Copy, PartialEq, Eq, Hash, Debug)] 8       pub enum ScatterHexDirection { 9           UpperLeft, 10           UpperRight,           </pre>                                                                                    |
| <a href="#">omdurman-rules/src/howitzer_scatter.rs:38</a><br>GH:omdurman-rules/src/howitzer_scatter.rs:38 | <b>howitzer_scatter</b>    | <pre> 36       /// onto a hex-grid offset oriented away from the firer           (see 37       /// `GameState::howitzer_impact_hex`). 38       pub fn howitzer_scatter(impact_roll: DieRoll) -&gt;           ScatterHexDirection { 39             crate::tables_data::SCATTERGRAM[(impact_roll.value() - 1) as           usize] 40   }           </pre>          |
| <a href="#">omdurman-rules/src/lib.rs:428</a><br>GH:omdurman-rules/src/lib.rs:428                         | <b>GunboatId</b>           | <pre> 426       /// fire; "old" gunboats do not (rulebook §2.32). 427       #[derive(Serialize, Deserialize, Clone, Copy,           PartialEq, Eq, Hash, Debug, strum::Display)] 428       pub enum GunboatId { 429           /// One of the five new-type named gunboats with               howitzer capability. 430           Named(Gunboat),           </pre> |
| <a href="#">omdurman-rules/src/effects/effect.rs:88</a><br>GH:omdurman-rules/src/effects/effect.rs:88     | <b>HowitzerFire</b>        | <pre> 86       /// - CRT result applied to units at impact hex           (not the original target). 87       /// - Firers marked as fired. 88       HowitzerFire { 89           attack: FireAttack, 90           combat_results_table_roll: DieRoll,           </pre>                                                                                            |
| <a href="#">omdurman-rules/src/effects/fire.rs:13</a><br>GH:omdurman-rules/src/effects/fire.rs:13         | <b>apply_howitzer_fire</b> | <pre> 11   12       /// Validate and apply a howitzer fire attack (scatter           path) (rulebook §6.64). 13       pub fn apply_howitzer_fire(               state: &amp;mut GameState, 14           attack: &amp;FireAttack, 15             </pre>                                                                                                           |

|                                                                                         |                    |                                                                                                                                                                                                                                                                  |
|-----------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects/state.rs:1043<br>GH:omdurman-rules/src/effects/state.rs:1043 | <b>can_fire_at</b> | <pre> 1041       /// modifier in the [`FireAttack`] and is 1042       responsible for the LOS gate. 1043       /// (Howitzer fire ignores LOS entirely -- \$6.64.) 1044       pub fn can_fire_at( 1045           &amp;self, 1046           firer: UnitId, </pre> |
|-----------------------------------------------------------------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Proven

by: omdurman-rules::src::effects::opposite\_is\_an\_involution omdurman-rules::src::effects::step\_toward\_lands\_on\_an\_adjacent\_hex omdurman-rules::src::howitzer\_scatter::scatter\_is\_center\_exactly\_for\_rolls\_7\_to\_10

Covered

by

tests:

omdurman-rules::src::howitzer\_scatter::howitzer\_on\_target\_7\_to\_10  
omdurman-rules::src::howitzer\_scatter::howitzer\_scatters\_below\_7  
omdurman-rules::src::howitzer\_scatter::howitzer\_each\_miss\_gets\_its\_printed\_hex  
omdurman-rules::src::unit\_profiles::named\_and\_old\_gunboats\_resolve  
omdurman-rules::src::effects::tests::named\_gunboat\_has\_howitzer  
omdurman-rules::src::effects::tests::named\_gunboat\_may\_fire\_howitzer\_in\_second\_subphase  
omdurman-rules::src::effects::tests::named\_gunboat\_direct\_fire\_uses\_artillery\_weapon  
omdurman-rules::src::effects::tests::named\_gunboat\_no\_howitzer\_at\_night  
omdurman-rules::src::effects::tests::dervish\_gunboat\_lacks\_howitzer

## §6.81 – Moving player may fire with all capable units

implicit

manual page unknown

During Offensive Fire phase, the moving player may fire with all of his units capable of firing, up to their maximum range, within the limitations imposed by the rules of combat.

## §6.82 – Advance after combat (offensive fire)

implemented

manual page unknown

If an enemy-occupied hex is vacated as a result of offensive fire, friendly units may advance after combat into the vacated hex. To be eligible to advance, the friendly units must have participated in the attack and must have been adjacent to the vacated hex. Note that artillery may not advance, nor may units advance across a wall hexside (except at a gate or breach). Units may never advance after combat across a khor.

| File                                                                                      | Symbol                             | Code Snippet                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects/effect.rs:162<br>GH:omdurman-rules/src/effects/effect.rs:162   | <b>AdvanceAfterCombat</b>          | <pre> 160       /// **Postconditions:** Unit position moved to 161       `to`; `vacated_by_combat` 162       /// entry consumed. 163       AdvanceAfterCombat { unit_id: UnitId, to: 164       HexCoord }, 165       // -- Unit state changes 166       ----- </pre>           |
| omdurman-rules/src/effects/dispatch.rs:36<br>GH:omdurman-rules/src/effects/dispatch.rs:36 | <b>apply_advance_after_combat</b>  | <pre> 34       } 35       GameEffect::AdvanceAfterCombat { unit_id, to } =&gt; { 36           apply_advance_after_combat(state, *unit_id, 37           *to) 38       } 39       GameEffect::RecoverUnit { unit_id } =&gt; 40       apply_recover_unit(state, *unit_id), </pre> |
| omdurman-types/src/lib.rs:241<br>GH:omdurman-types/src/lib.rs:241                         | <b>blocks_advance_after_combat</b> | <pre> 239       /// Whether advance-after-combat may *not* cross 240       this side (\$6.82, \$7.6). 241       pub fn blocks_advance_after_combat(self) -&gt; bool { </pre>                                                                                                   |

|                                                                                                     |                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                     |                                   | <div>242   matches!(<br/>243   self,</div>                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <div>omdurman-rules/src/effects/state.rs:1932<br/>GH:omdurman-rules/src/effects/state.rs:1932</div> | <div>can_advance_after_comb</div> | <div>1930   /// player's unit, not artillery, adjacent to<br/>1931   and `to` now empty.<br/>1932   /// Wall/khor hexside restrictions are not<br/>1933   enforced (no hexside map data).<br/>1934   pub fn can_advance_after_combat(&amp;self, unit_id:<br/>UnitId, to: HexCoord) -&gt; Result&lt;(), RuleError&gt; {<br/>1935   let unit = self.unit_or_err(unit_id)?;<br/>1936   // §6.7: there is no advance after combat as a<br/>1937   result of defensive fire.</div> |

Proven by: omdurman-rules::src::effects::advance\_phase\_is\_atomic omdurman-types::src::lib::hexside\_blocking\_classifiers\_are\_exact

Covered by tests: omdurman-rules::src::effects::tests::can\_advance\_after\_combat\_rejects\_wall\_hexside  
omdurman-rules::src::effects::tests::can\_advance\_after\_combat\_rejects\_khor\_hexside  
omdurman-rules::src::effects::tests::advance\_requires\_combat\_vacated\_hex  
omdurman-rules::src::effects::tests::advance\_requires\_participation  
omdurman-rules::src::effects::tests::rejected\_advance\_phase\_keeps\_vacated\_windows

8/8 implemented

## §7 – Melee Phase

### §7 – Melee Phase (chapter)

implemented

manual page unknown

#### Melee Phase

| File                                                                                                  | Symbol              | Code Snippet                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state.rs:133</a><br>GH:omdurman-rules/src/effects/state.rs:133 | <b>PendingMelee</b> | <pre>131   /// resolution after the reaction window is       deterministic and host-ordered (rulebook §7.5). 132   #[derive(Serialize, Deserialize, Clone, Debug)] 133   pub struct PendingMelee {           pub attack: MeleeAttack, 134       pub attacker_roll: DieRoll,</pre> |

Covered by tests: [omdurman-rules::src::effects::tests::declared\\_melee\\_blocks\\_phase\\_advance](#)

### §7.1 – Melee strength printed on counter

implemented

manual page unknown

The melee strength of all units is printed on the counter. Note that gunboats have no melee strength. Gunboats may neither melee attack nor be melee attacked.

| File                                                                              | Symbol                       | Code Snippet                                                                                                                                                                                                                                                                                       |
|-----------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:110</a><br>GH:omdurman-rules/src/lib.rs:110 | <b>MeleeFactor</b>           | <pre>108       /// Every possible value from the annotated counter       set is a named variant. 109       #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Hash, Debug, strum::Display)] 110       pub enum MeleeFactor {               One = 1, 111           Three = 3,</pre> |
| <a href="#">omdurman-rules/src/lib.rs:121</a><br>GH:omdurman-rules/src/lib.rs:121 | <b>MeleeFactor::sum</b>      | <pre>119   impl MeleeFactor { 120       /// Sum multiple melee factors (rulebook §7.1). 121       pub fn sum&lt;'a&gt;(factors: impl IntoIterator&lt;Item =       &amp;'a MeleeFactor&gt;) -&gt; u16 { 122           factors.into_iter().map( f  f.value()).sum() 123       }</pre>                |
| <a href="#">omdurman-types/src/lib.rs:946</a><br>GH:omdurman-types/src/lib.rs:946 | <b>may_be_melee_attacked</b> | <pre>944   945       /// Gunboats neither attack nor are attacked in       melee (§7.1). 946       pub fn may_be_melee_attacked(self) -&gt; bool { 947           !matches!(self, UnitKind::Gunboat { .. }) 948       }</pre>                                                                       |

Proven by: [omdurman-types::src::lib::unit\\_kind\\_melee\\_capability\\_law\\_is\\_exact](#)

Covered by tests: [omdurman-rules::src::lib::melee\\_factor\\_values\\_and\\_sum](#) [omdurman-rules::src::lib::unit\\_kind\\_melee\\_capability](#)

### §7.2 – Melee adjacent only, not across wall hexsides

implemented

manual page unknown

Melee simulates the hand-to-hand fighting of the period. Units may melee attack adjacent enemy units only. Units may not melee attack across a wall hexside, but may melee attack through a gate or breach hexside.

| File                                                                                                    | Symbol              | Code Snippet                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:236</a><br>GH:omdurman-types/src/lib.rs:236                       | <b>blocks_melee</b> | <pre>234       /// Whether melee may *not* be made across this           side (§7.2). Gates and 235       /// breaches are passable to melee. 236       pub fn blocks_melee(self) -&gt; bool { 237           matches!(self, HexsideKind::Wall   HexsideKind::ZaribaThornHedge) 238       }</pre>                                                                |
| <a href="#">omdurman-rules/src/effects/state.rs:1318</a><br>GH:omdurman-rules/src/effects/state.rs:1318 | <b>can_melee</b>    | <pre>1316       /// that may be melee-attacked (gunboats may not           -- §7.1), and no wall or 1317       /// thorn-hedge hexside blocks the attack (§7.2). 1318       pub fn can_melee(&amp;self, attacker: UnitId, defender_hex: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 1319           let unit = self.unit_or_err(attacker)?; 1320       }</pre> |

Proven by: [omdurman-types::src::lib::hexside\\_blocking\\_classifiers\\_are\\_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::can\\_melee\\_gates\\_phase\\_adjacency\\_and\\_kind](#)  
[omdurman-rules::src::effects::tests::can\\_melee\\_rejects\\_wall\\_hexside](#)  
[omdurman-rules::src::effects::tests::can\\_melee\\_rejects\\_thorn\\_hedge\\_hexside](#)  
[omdurman-rules::src::effects::tests::can\\_melee\\_allows\\_gate\\_hexside](#)

## §7.3 – Simultaneous melee combat

**implemented**

manual page unknown

Melee combat is considered simultaneous, so that units eliminated by melee attacks still get a melee combat die roll.

| File                                                                                                    | Symbol                    | Code Snippet                                                                                                                                                                                                                     |
|---------------------------------------------------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/effect.rs:111</a><br>GH:omdurman-rules/src/effects/effect.rs:111 | <b>MeleeCombat</b>        | <pre>109       /// - Winner may advance into vacated hex (§7.6). 110       /// - Victory points awarded for eliminations. 111       MeleeCombat { 112           attack: MeleeAttack, 113           attacker_roll: DieRoll,</pre> |
| <a href="#">omdurman-rules/src/effects/melee.rs:4</a><br>GH:omdurman-rules/src/effects/melee.rs:4       | <b>apply_melee_combat</b> | <pre>2   3       /// Apply a simultaneous melee combat between two           adjacent hexes (rulebook §7). 4       pub fn apply_melee_combat( 5           state: &amp;mut GameState, 6           attack: &amp;MeleeAttack,</pre> |

Covered by tests: [omdurman-rules::src::effects::tests::melee\\_resolves\\_simultaneously](#)

## §7.4 – Who may melee attack / defend

**implemented**

manual page unknown

Only infantry, cavalry, camel units, and Dervish leaders may melee attack. All units (except gunboats — see 7.1) may melee defend.

See also: §7.1

| File                                                                                                    | Symbol                  | Code Snippet                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-types/src/lib.rs:935</a><br>GH:omdurman-types/src/lib.rs:935                       | <b>may_melee_attack</b> | <pre> 933       /// Rulebook §7.4 -- only infantry, cavalry, camel           and Dervish leaders may 934       /// melee *attack*. All others (except gunboats)           may melee *defend* (§7.1). 935       pub fn may_melee_attack(self) -&gt; bool { 936           matches!( 937               self, </pre>                                                      |
| <a href="#">omdurman-types/src/lib.rs:872</a><br>GH:omdurman-types/src/lib.rs:872                       | <b>UnitKind</b>         | <pre> 870       /// `Some(UnitKind::Marker)` or `None`. 871       #[derive(Serialize, Deserialize, Clone, Copy,           PartialEq, Eq, Hash, Debug, strum::Display)] 872       pub enum UnitKind { 873           /// Foot infantry (§2.3): fire / melee / movement. 874           Infantry { </pre>                                                                 |
| <a href="#">omdurman-types/src/lib.rs:714</a><br>GH:omdurman-types/src/lib.rs:714                       | <b>DervishTribe</b>     | <pre> 712           strum::EnumIter, 713       )] 714       pub enum DervishTribe { 715           Baggara, 716           Jaalin, </pre>                                                                                                                                                                                                                               |
| <a href="#">omdurman-rules/src/effects/state.rs:1318</a><br>GH:omdurman-rules/src/effects/state.rs:1318 | <b>can_melee</b>        | <pre> 1316       /// that may be melee-attacked (gunboats may not           -- §7.1), and no wall or 1317       /// thorn-hedge hexside blocks the attack (§7.2). 1318       pub fn can_melee(&amp;self, attacker: UnitId,           defender_hex: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 1319           let unit = self.unit_or_err(attacker)?; 1320   </pre> |

Proven by: [omdurman-types::src::lib::unit\\_kind\\_melee\\_capability\\_law\\_is\\_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::can\\_melee\\_gates\\_phase\\_adjacency\\_and\\_kind](#)

## §7.5 – Cavalry/camel retreat before melee

**implemented**

manual page unknown

Cavalry and camel units may retreat two hexes from an infantry melee attack. Note, however, that only one retreat per unit per turn is permitted. Thus, if their retreat places them adjacent to enemy units whose melee attacks have not yet been resolved, those enemy units may elect to attack the retreating unit(s).

| File                                                                                                      | Symbol                            | Code Snippet                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/effect.rs:148</a><br>GH:omdurman-rules/src/effects/effect.rs:148   | <b>RetreatBeforeMelee</b>         | <pre> 146       /// 147       /// **Postconditions:** Unit position moved to           `to`. 148       RetreatBeforeMelee { unit_id: UnitId, to:           HexCoord }, 149   150       /// An attacking unit advances into a hex vacated           by combat (rulebook §6.82 </pre>                                                                               |
| <a href="#">omdurman-rules/src/effects/dispatch.rs:33</a><br>GH:omdurman-rules/src/effects/dispatch.rs:33 | <b>apply_retreat_before_melee</b> | <pre> 31           GameEffect::ResolveMelee =&gt;               apply_resolve_melee(state), 32           GameEffect::RetreatBeforeMelee { unit_id, to }           =&gt; { 33               apply_retreat_before_melee(state, *unit_id,                   *to) 34           } 35           GameEffect::AdvanceAfterCombat { unit_id, to }           =&gt; { </pre> |
| <a href="#">omdurman-types/src/lib.rs:952</a>                                                             | <b>may_retreat_before_melee</b>   | <pre> 950       /// Cavalry and camel units may retreat two hexes           from an infantry melee 951       /// attack (§7.5). 952       pub fn may_retreat_before_melee(self) -&gt; bool { </pre>                                                                                                                                                               |

|                                                                                         |                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GH:omdurman-types/src/lib.rs:952                                                        |                                 | <pre> 953           matches!(self, UnitKind::Cavalry { .. }   UnitKind::Camel { .. }) 954       }</pre>                                                                                                                                                                                                                                                                                                                             |
| omdurman-rules/src/lib.rs:187<br>GH:omdurman-rules/src/lib.rs:187                       | <b>HexDistance</b>              | <pre> 185       /// (rulebook §6.22, §7.5). 186       #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Hash, Debug)] 187       pub struct HexDistance(u16); 188   189       impl HexDistance {</pre>                                                                                                                                                                                                   |
| omdurman-rules/src/effects/state.rs:1863<br>GH:omdurman-rules/src/effects/state.rs:1863 | <b>can_retreat_before_melee</b> | <pre> 1861           /// two hexes away and empty. (Does not verify the attacker is infantry -- 1862           /// the caller offers the retreat only in response to one.) 1863           pub fn can_retreat_before_melee(&amp;self, unit_id: UnitId, to: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 1864               let unit = self.unit_or_err(unit_id)?; 1865               if !matches!(self.phase, Phase::Melee) {</pre> |

Proven by: [omdurman-rules::src::effects::resolve\\_melee\\_is\\_atomic](#) [omdurman-rules::src::effects::advance\\_phase\\_is\\_atomic](#)  
[omdurman-types::src::lib::unit\\_kind\\_melee\\_capability\\_law\\_is\\_exact](#)

Covered by tests: [omdurman-rules::src::effects::tests::retreat\\_before\\_melee\\_only\\_cavalry\\_two\\_hexes](#)  
[omdurman-rules::src::effects::tests::retreat\\_opens\\_window\\_only\\_when\\_hex\\_empties](#)  
[omdurman-rules::src::effects::tests::rejected\\_resolve\\_melee\\_keeps\\_the\\_declaration](#)

## §7.6 – Advance after melee

**implemented**

manual page unknown

If a melee attack eliminates all of the defenders in an adjacent hex, the Dervish player **MUST** advance into the vacated hex. To be eligible to advance, the Dervish units must have been adjacent to the vacated hex and participated in the melee attack that eliminated the defenders. All surviving eligible Dervish units **MUST** advance, up to the stacking limit. The Anglo-Egyptian player may advance if desired. Note that only attacking units may advance.

| File                                                                                      | Symbol                            | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects/effect.rs:162<br>GH:omdurman-rules/src/effects/effect.rs:162   | <b>AdvanceAfterCombat</b>         | <pre> 160           /// **Postconditions:** Unit position moved to `to`, `vacated_by_combat` 161           /// entry consumed. 162           AdvanceAfterCombat { unit_id: UnitId, to: HexCoord }, 163   164           // -- Unit state changes -----</pre>                                                                                                                                                                                                                     |
| omdurman-rules/src/effects/dispatch.rs:36<br>GH:omdurman-rules/src/effects/dispatch.rs:36 | <b>apply_advance_after_combat</b> | <pre> 34           } 35           GameEffect::AdvanceAfterCombat { unit_id, to } =&gt; { 36               apply_advance_after_combat(state, *unit_id, *to) 37           } 38           GameEffect::RecoverUnit { unit_id } =&gt; apply_recover_unit(state, *unit_id),</pre>                                                                                                                                                                                                     |
| omdurman-rules/src/effects/state.rs:1932<br>GH:omdurman-rules/src/effects/state.rs:1932   | <b>can_advance_after_combat</b>   | <pre> 1930           /// player's unit, not artillery, adjacent to `to`, and `to` now empty. 1931           /// Wall/khor hexside restrictions are not enforced (no hexside map data). 1932           pub fn can_advance_after_combat(&amp;self, unit_id: UnitId, to: HexCoord) -&gt; Result&lt;(), RuleError&gt; { 1933               let unit = self.unit_or_err(unit_id)?; 1934               // §6.7: there is no advance after combat as a result of defensive fire.</pre> |

Proven by: `omdurman-rules::src::effects::advance_phase_is_atomic`

Covered by tests: `omdurman-rules::src::effects::tests::dervish_advance_after_melee_is_mandatory`  
`omdurman-rules::src::effects::tests::rejected_advance_phase_keeps_vacated_windows`  
`omdurman-rules::src::effects::tests::melee_mandatory_advance_leaders_are_free_stacking`

## §7.7 – Melee modifiers

**implemented**

manual page unknown

To resolve melee, both the attacker and the defender roll on the Combat Results Table and apply the applicable melee modifier to their die roll. The Dervish player receives a +2 melee modifier, the Anglo-Egyptian player receives a +1 melee modifier. No terrain modifiers are applied to melee combat (Exception: Zariba hexsides in the historical scenario and the campaign game, if constructed — see 9.23). Melee losses must be taken from meleeing units first!

See also: §9.23

| File                                                                                | Symbol                                      | Code Snippet                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:1019</a><br>GH:omdurman-rules/src/lib.rs:1019 | <b>MeleeModifier</b>                        | <pre>1017   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1018   1019   pub enum MeleeModifier { 1020       /// +2 to all Dervish melee rolls (§7.7). 1021       DervishStandard,</pre>                                                                                     |
| <a href="#">omdurman-rules/src/lib.rs:1042</a><br>GH:omdurman-rules/src/lib.rs:1042 | <b>MeleeAttack</b>                          | <pre>1040   /// A melee attack: simultaneous, both sides roll on the Combat Results Table (§7.3, §7.7). 1041   #[derive(Serialize, Deserialize, Clone, Debug, PartialEq, Eq)] 1042   pub struct MeleeAttack { 1043       pub attacker_player: Player, 1044       pub attacker_hex: HexCoord,</pre> |
| <a href="#">omdurman-rules/src/lib.rs:1023</a><br>GH:omdurman-rules/src/lib.rs:1023 | <b>MeleeModifier::AngloEgyptianStandard</b> | <pre>1021       DervishStandard, 1022       /// +1 to all Anglo-Egyptian melee rolls (§7.7). 1023       AngloEgyptianStandard, 1024       /// Inverted to -2 when Dervish units melee-attack across a trench into 1025       /// an entrenched defender (§9.232).</pre>                            |
| <a href="#">omdurman-rules/src/lib.rs:1026</a><br>GH:omdurman-rules/src/lib.rs:1026 | <b>MeleeModifier::DervishVsDefender</b>     | <pre>1024       /// Inverted to -2 when Dervish units melee-attack across a trench into 1025       /// an entrenched defender (§9.232). 1026       DervishVsTrenchedDefender, 1027   } 1028   }</pre>                                                                                              |
| <a href="#">omdurman-rules/src/lib.rs:1021</a><br>GH:omdurman-rules/src/lib.rs:1021 | <b>MeleeModifier::DervishStandard</b>       | <pre>1019   pub enum MeleeModifier { 1020       /// +2 to all Dervish melee rolls (§7.7). 1021       DervishStandard, 1022       /// +1 to all Anglo-Egyptian melee rolls (§7.7). 1023       AngloEgyptianStandard,</pre>                                                                          |

Proven by: `omdurman-rules::src::lib::melee_modifier_keeps_roll_legal`

Covered by tests: `omdurman-rules::src::effects::tests::melee_resolves_simultaneously`  
`omdurman-rules::src::effects::tests::melee_modifiers_are_engine_derived_and_mismatches_rejected`

2/3 implemented

## §8 – Night Game Turns

### §8 – Night Game Turns

descriptive

manual page unknown

#### Night Game Turns

### §8.1 – Night effects

implemented

manual page unknown

The effects of night game turns are: a) all Anglo-Egyptian movement allowances are halved (round down), b) there is no Anglo-Egyptian howitzer fire, and c) all fire ranges are halved for both sides (round down, but range 1 stays range 1). Range effects on fire combat are the same as during day game turns. For example, an Anglo-Egyptian infantry unit firing at night will be doubled at range 1, normal at range 2, and may not fire at range 3 or greater.

**\*\*8.2) Dervish Desertion Roll:\*\*** Once each campaign game, during the first night turn of the game, the Dervish player rolls one die to see how many of his units desert. The roll is made during the movement phase and the number of deserting Dervish units is equal to 1½ times the roll of one die. The Dervish player may choose which units desert by merely removing them from the mapsheet. The KHALIFA unit, gunboats, artillery units, and forts are the only Dervish units that may not be chosen. No victory points are awarded to the Anglo-Egyptian player for deserting Dervishes.

(see manual for full text)

See also: §8.2

| File                                                                                                | Symbol                             | Code Snippet                                                                                                                                                                                                                                                                                                          |
|-----------------------------------------------------------------------------------------------------|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:156</a><br>GH:omdurman-rules/src/lib.rs:156                   | <b>MovementAllowance::halve</b>    | <pre>154   impl MovementAllowance { 155       /// Night movement allowance = halved (round down)           (rulebook §8.1, §5.11). 156       pub fn halve(self) -&gt; Self { 157           let v = self.value() / 2; 158           MovementAllowance::try_from(v).expect("halved value always a named variant")</pre> |
| <a href="#">omdurman-types/src/lib.rs:779</a><br>GH:omdurman-types/src/lib.rs:779                   | <b>DayNight</b>                    | <pre>777   /// (rulebook §8.1). 778   #[derive(Serialize, Deserialize, Clone, Copy,       PartialEq, Eq, Debug)] 779   pub enum DayNight { 780       Day, 781       Night,</pre>                                                                                                                                      |
| <a href="#">omdurman-rules/src/range_effects.rs:71</a><br>GH:omdurman-rules/src/range_effects.rs:71 | <b>night_max_range</b>             | <pre>69   70   /// The halved maximum range at night (§8.1): round       down, minimum 1. 71   pub fn night_max_range(weapon: WeaponClass, ae: bool) -       &gt; u8 { 72       let day = max_day_range(weapon, ae); 73       if day &lt;= 1 { 1 } else { day / 2 }</pre>                                             |
| <a href="#">omdurman-rules/src/lib.rs:1514</a><br>GH:omdurman-rules/src/lib.rs:1514                 | <b>effective_movement_at_night</b> | <pre>1512   /// Apply night-turn movement halving for Anglo-       Egyptian units (§8.1): all 1513   /// Anglo-Egyptian movement allowances are halved       (round down). 1514   pub fn effective_movement_at_night( 1515       allowance: MovementAllowance, 1516       player: Player,</pre>                       |

Proven by: `omdurman-rules::src::lib::movement_allowance_half_never_panics`  
`omdurman-rules::src::lib::night_halving_is_ae_only_and_day_neutral`  
`omdurman-rules::src::range_effects::night_cap_is_halved_day_max_floored_at_one`  
`omdurman-rules::src::range_effects::night_range_effects_gates_the_day_band_by_the_cap`  
`omdurman-rules::src::effects::fire::night_capped_distance_is_some_exactly_within_the_night_max`

Covered by tests: `omdurman-rules::src::range_effects::night_max_ranges` `omdurman-rules::src::range_effects::night_max_ranges_remaining`  
`omdurman-rules::src::range_effects::ae_rifle_at_night_matches_rulebook_example`  
`omdurman-rules::src::range_effects::max_day_range_all_combos`  
`omdurman-rules::src::effects::tests::night_movement_overlay_allowance_halved`

## §8.2 – Dervish Desertion Roll

implemented

manual page unknown

| File                                                                                                 | Symbol                  | Code Snippet                                                                                                                                                                                                                                                                                               |
|------------------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>omdurman-rules/src/effects/effect.rs:190</code><br>GH:omdurman-rules/src/effects/effect.rs:190 | <b>DervishDesertion</b> | <pre>188   // the effect. The Khalifa, gunboats, artillery, 189   // chosen. 190   DervishDesertion { 191       roll: DieRoll, 192       deserters: Vec&lt;UnitId&gt;,</pre>                                                                                                                               |
| <code>omdurman-rules/src/turn_track.rs:63</code><br>GH:omdurman-rules/src/turn_track.rs:63           | <b>DervishDesertion</b> | <pre>61   None, 62   // Dervish desertion roll (§8.2) -- occurs on the      first night turn. 63   DervishDesertion, 64   // Dervish reinforcements are available. 65   DervishReinforcements,</pre>                                                                                                       |
| <code>omdurman-rules/src/turn_track.rs:60</code><br>GH:omdurman-rules/src/turn_track.rs:60           | <b>TurnEvent</b>        | <pre>58   // Special events that occur on specific turns      (rulebook §8.2, §9.112, §9.113). 59   #[derive(Serialize, Deserialize, Clone, Copy,      PartialEq, Eq, Debug)] 60   pub enum TurnEvent { 61       None, 62   // Dervish desertion roll (§8.2) -- occurs on the      first night turn.</pre> |

Proven by: `omdurman-rules::src::effects::desertion_count_is_one_and_a_half_times_the_roll`  
`omdurman-rules::src::turn_track::desertion_event_is_unique_to_the_first_night_turn`

Covered by tests: `omdurman-rules::src::effects::tests::desertion_count_is_floor_one_and_a_half`  
`omdurman-rules::src::turn_track::desertion_on_first_night`  
`omdurman-rules::src::effects::tests::desertion_roll_required_before_first_night_movement_ends`

23/33 implemented

## §9 – The Scenarios

### §9 – The Scenarios

descriptive

manual page unknown

The Scenarios

### §9.1 – The Campaign Game

descriptive

manual page unknown

The Campaign Game

### §9.2 – The Historical Scenario

descriptive

manual page unknown

The Historical Scenario

Players should note that the historical scenario is an exercise in futility for the Dervish player. It is, however, an interesting demonstration of the absolute imbecility of the Khalifa's generalship and vividly shows the superiority of entrenched firepower over traditional tribal arms in the colonial period.

### §9.3 – Bonus Game: Fall of Khartoum

descriptive

manual page unknown

Bonus Game: FALL OF KHARTOUM

### §9.11 – Set Up (Campaign)

descriptive

manual page unknown

Set Up

### §9.12 – Scenario Length (Campaign)

implemented

manual page unknown

Scenario Length

6:00 am, Sept. 1 through 8:00 am, Sept. 3, 22 Game Turns.

| File                                                | Symbol   | Code Snippet                                                                                                                                                               |
|-----------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/turn_track.rs:11<br>GH:omdurman- | GameTime | <pre>9   /// starts at one of these twelve times.<br/>10   #[derive(Serialize, Deserialize, Clone, Copy,<br/>    PartialEq, Eq, Debug)]<br/>11   pub enum GameTime {</pre> |

|                                                                                                 |                     |                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/<br>turn_track.rs:11                                                                  |                     | <div>12   SixAM,</div> <div>13   EightAM,</div>                                                                                                                                                                                                                                                                                                                                                                        |
| <a href="#">omdurman-rules/src/turn_track.rs:47</a><br>GH:omdurman-rules/src/turn_track.rs:47   | TurnEntry           | <div>45   /// A single entry on the Turn Record Track (rulebook §9.12, §9.22).</div> <div>46   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]</div> <div>47   pub struct TurnEntry {</div> <div>48       /// 1-based turn number.</div> <div>49       pub turn: u8,</div>                                                                                                                        |
| <a href="#">omdurman-rules/src/turn_track.rs:95</a><br>GH:omdurman-rules/src/turn_track.rs:95   | CAMPAIGN_TURN_TRACK | <div>93   /// is turn 9, which carries the once-per-game Dervish Desertion Roll (§8.2) --</div> <div>94   /// the printed track prints "Dervish Desertion Roll / NIGHT" on that cell.</div> <div>95   pub const CAMPAIGN_TURN_TRACK: [TurnEntry; 22] = [</div> <div>96       // Row 1, left-&gt;right: Sept 1, 6 am -&gt; 8 pm, then the first NIGHT.</div> <div>97       entry(1, SixAM, Day, TurnEvent::None),</div> |
| <a href="#">omdurman-rules/src/turn_track.rs:242</a><br>GH:omdurman-rules/src/turn_track.rs:242 | TurnLabel           | <div>240   /// printed cell; `Blank` is for unused positions in the 9×3 grid.</div> <div>241   #[derive(Serialize, Deserialize, Debug, Clone, Copy, PartialEq, Eq)]</div> <div>242   pub enum TurnLabel {</div> <div>243       Blank,</div> <div>244       /// "SEPT. 1\n6:00 am" -- day header plus time (turn 1).</div>                                                                                              |

Proven by: [omdurman-rules::src::turn\\_track::scenario\\_turn\\_indexing\\_is\\_exact\\_for\\_every\\_scenario](#)  
[omdurman-rules::src::turn\\_track::turn\\_labels\\_agree\\_with\\_the\\_rule\\_bearing\\_track](#)

Covered by tests: [omdurman-rules::src::turn\\_track::campaign\\_track\\_22\\_turns](#) [omdurman-rules::src::turn\\_track::desertion\\_on\\_first\\_night](#)  
[omdurman-rules::src::turn\\_track::campaign\\_track\\_label\\_and\\_day\\_night\\_agree](#)  
[omdurman-rules::src::turn\\_track::game\\_time\\_display\\_all\\_variants](#) [omdurman-rules::src::turn\\_track::turn\\_label\\_display](#)  
[omdurman-rules::src::turn\\_track::turn\\_label\\_out\\_of\\_range\\_is\\_none](#)  
[omdurman-rules::src::effects::tests::game\\_over\\_after\\_campaign\\_turns](#)

### §9.13 – Special Rules (Campaign)

[descriptive](#)  
manual page unknown

|               |
|---------------|
| Special Rules |
| None.         |

### §9.14 – Victory Conditions (Campaign)

[implemented](#)  
manual page unknown

Victory Conditions

The Mahdi’s Tomb in Omdurman was not only the tallest structure in the entire Sudan in 1898, it was also a Dervish holy shrine. Its loss or destruction would be a severe blow to the Mahdist cause. It is accordingly assigned 25 victory points which are awarded to the player who controls it at the conclusion of play. The Dervish player controls it at the start of play. As a tactical note, the Anglo-Egyptian player will find a decisive victory almost impossible unless he takes the Mahdi’s Tomb from the Dervish player. To take the Tomb hex, it must be occupied by one British leader plus any one non-“Friendlies” Anglo-Egyptian combat unit (both undisrupted) at the conclusion of play.

Additional victory points are awarded as follows:

**\*\*Dervish Player receives:\*\***

- 10 pts: each British leader eliminated

(see manual for full text)

See also: §5.21

| File                                                                                | Symbol                                        | Code Snippet                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:1168</a><br>GH:omdurman-rules/src/lib.rs:1168 | <b>VpSource</b>                               | <pre> 1166   /// the manual and the engine. 1167   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1168   pub enum VpSource { 1169       // ----- Anglo-Egyptian player receives: 1170       /// Mahdi's Tomb control at conclusion of play (\$9.14). </pre>                                                                |
| <a href="#">omdurman-rules/src/lib.rs:1194</a><br>GH:omdurman-rules/src/lib.rs:1194 | <b>VpSource::points</b>                       | <pre> 1192   impl VpSource { 1193       /// VP awarded to `who_scores()` (rulebook §9.14). 1194       pub fn points(self) -&gt; VictoryPoints { 1195           match self { 1196               VpSource::MahdisTomb =&gt; VictoryPoints::new(25), </pre>                                                                                         |
| <a href="#">omdurman-rules/src/lib.rs:1209</a><br>GH:omdurman-rules/src/lib.rs:1209 | <b>VpSource::who_scores</b>                   | <pre> 1207   1208       /// Which player receives these victory points (rulebook §9.14). 1209       pub fn who_scores(self) -&gt; Player { 1210           match self { 1211               VpSource::MahdisTomb </pre>                                                                                                                            |
| <a href="#">omdurman-rules/src/lib.rs:1244</a><br>GH:omdurman-rules/src/lib.rs:1244 | <b>VictoryLedger</b>                          | <pre> 1242   /// Cumulative victory ledger for one scenario (rulebook §9.14). 1243   #[derive(Serialize, Deserialize, Clone, Debug, Default)] 1244   pub struct VictoryLedger { 1245       pub events: Vec&lt;VpEvent&gt;, 1246   } </pre>                                                                                                       |
| <a href="#">omdurman-rules/src/lib.rs:1250</a><br>GH:omdurman-rules/src/lib.rs:1250 | <b>VpEvent</b>                                | <pre> 1248   /// A single victory-point scoring event (rulebook §9.14). 1249   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1250   pub struct VpEvent { 1251       pub turn: GameTurnIndex, 1252       pub source: VpSource, </pre>                                                                                      |
| <a href="#">omdurman-rules/src/lib.rs:1257</a><br>GH:omdurman-rules/src/lib.rs:1257 | <b>VictoryLedger::total_for</b>               | <pre> 1255   impl VictoryLedger { 1256       /// Total victory points earned by a given player (rulebook §9.14). 1257       pub fn total_for(&amp;self, player: Player) -&gt; VictoryPoints { 1258           VictoryPoints( 1259               self.events </pre>                                                                                |
| <a href="#">omdurman-rules/src/lib.rs:1269</a><br>GH:omdurman-rules/src/lib.rs:1269 | <b>VictoryLedger::superiority</b>             | <pre> 1267       /// Net superiority: positive = Anglo-Egyptian ahead, negative = Dervish ahead 1268       /// (rulebook §9.14). 1269       pub fn superiority(&amp;self) -&gt; VictoryPoints { 1270           VictoryPoints( 1271               self.total_for(Player::AngloEgyptian).value() - self.total_for(Player::Dervish).value(), </pre> |
| <a href="#">omdurman-rules/src/lib.rs:1289</a><br>GH:omdurman-rules/src/lib.rs:1289 | <b>CampaignVictoryLevel</b>                   | <pre> 1287   /// Campaign-game victory levels (§9.14). 1288   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 1289   pub enum CampaignVictoryLevel { 1290       Draw, 1291       Marginal(Player), </pre>                                                                                                                   |
| <a href="#">omdurman-rules/src/lib.rs:1298</a><br>GH:omdurman-rules/src/lib.rs:1298 | <b>CampaignVictoryLevel::from_superiority</b> | <pre> 1296   impl CampaignVictoryLevel { 1297       /// Assign a level from the net superiority (\$9.14). 1298       pub fn from_superiority(s: VictoryPoints) -&gt; Self { 1299           let net = s.0; </pre>                                                                                                                                 |

|                                                                                                         |                              |                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                         |                              | <div>1300   // Positive -&gt; Anglo-Egyptian thresholds:<br/>15/30/50</div>                                                                                                                                                                       |
| <div>omdurman-rules/src/effects/movement.rs:194<br/>GH:omdurman-rules/src/effects/movement.rs:194</div> | <div>score_elimination</div> | <div>192   .find_unit(leader)<br/>193   .is_some_and( u <br/>u.profile.identity.is_gordon());<br/>194   score_elimination(state, leader,<br/>ElimCause::Combat);<br/>195   state.units.retain( u  u.id != leader);<br/>196   if is_gordon {</div> |
| <div>omdurman-rules/src/lib.rs:239<br/>GH:omdurman-rules/src/lib.rs:239</div>                           | <div>VictoryPoints</div>     | <div>237   /// (rulebook §9.14).<br/>238   #[derive(Serialize, Deserialize, Clone, Copy,<br/>PartialEq, Eq, PartialOrd, Ord, Debug, Default)]<br/>239   pub struct VictoryPoints(i32);<br/>240  <br/>241   impl VictoryPoints {</div>             |

Proven by:

omdurman-rules::src::lib::campaign\_victory\_levels\_match\_manual\_superiority\_table  
omdurman-rules::src::lib::vp\_source\_points\_and\_scorer\_match\_the\_printed\_schedule  
omdurman-rules::src::effects::victory::vp\_source\_for\_routes\_every\_elimination\_to\_the\_printed\_source  
omdurman-rules::src::effects::victory::score\_elimination\_records\_exactly\_what\_it\_scores

Covered by tests:

omdurman-rules::src::effects::tests::friendlies\_bank\_scores\_by\_side  
omdurman-rules::src::effects::tests::mahdis\_tomb\_not\_scored\_without\_a\_leader  
omdurman-rules::src::effects::tests::mahdis\_tomb\_scores\_for\_anglo\_egyptian\_when\_held  
omdurman-rules::src::lib::vp\_source\_attributes

### §9.21 – Set Up (Historical)

descriptive

manual page unknown

Set Up

### §9.22 – Scenario Length (Historical)

implemented

manual page unknown

Scenario Length

6:00 am, September 2 through 12:00 noon, September 2. Four game turns.

| File                                                                                        | Symbol                     | Code Snippet                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div>omdurman-rules/src/turn_track.rs:132<br/>GH:omdurman-rules/src/turn_track.rs:132</div> | <div>HISTORICAL_TURN</div> | <div>130  <br/>131   /// Historical scenario track (§9.22 -- 4 turns, Sept 2<br/>6:00 am -&gt; 12:00 pm).<br/>132   pub const HISTORICAL_TURN_TRACK: [TurnEntry; 4] = [<br/>133       TurnEntry {<br/>134           turn: 1,</div> |

Proven by:

omdurman-rules::src::turn\_track::scenario\_turn\_indexing\_is\_exact\_for\_every\_scenario

Covered by tests:

omdurman-rules::src::turn\_track::historical\_turn\_all\_four\_turns

### §9.23 – Special Rule: The Zariba

descriptive

manual page unknown

Special Rule: “The Zariba”

**\*\*9.231)** Thorn hedge hexsides:**\*\*** -2 to die roll on all Dervish fire attacks; may not melee across in either direction; may not advance after combat across in either direction.

**\*\*9.232)** Trench hexsides:**\*\*** -4 to die roll on all Dervish fire attacks vs. entrenched units only; -2 (instead of +2) melee modifier to Dervish units melee attacking an entrenched unit; entrenched units may be fired “over” in both directions (i.e. they do not block line of sight); units are considered “entrenched” if they are directly adjacent to (and on the Nile River side of) a trench hexside.

(see manual for full text)

See also: §9.231, §9.232

§9.24 – Victory Conditions (Historical)

**implemented**  
manual page unknown

**Victory Conditions**

Victory Levels are based solely on eliminating enemy units while conserving your own force as much as possible.

| Victory Level | Anglo-Egyptian Player (Dervish units eliminated) | Dervish Player (Anglo-Egyptian units eliminated)  
| |—|—|—| | 5 — DECISIVE | 100+ | 30+ | | 4 — STRATEGIC | 60–99 | 15–29 | | 3 — TACTICAL | 45–59 | 10–14 | | 2 — MARGINAL | 30–44 | 5–9 | | 1 — DRAW | 0–29 | 0–4 |

The lower value victory level is then subtracted from the higher level to determine a player’s net victory. For example, if the Anglo-Egyptian player eliminates 104 Dervish units (decisive victory) but loses 18 units doing it (Dervish Strategic), the Anglo-Egyptian player only nets out with a draw (decisive worth 5 minus strategic worth 4 = 1, draw).

(see manual for full text)

| File                                                                                | Symbol                        | Code Snippet                                                                                                                                                                                                      |
|-------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:1328</a><br>GH:omdurman-rules/src/lib.rs:1328 | <b>HistoricalVictoryLevel</b> | <pre>1326   /// draw"). 1327   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, PartialOrd, Ord, Debug)] 1328   pub enum HistoricalVictoryLevel { 1329       Draw = 1, 1330       Marginal = 2,</pre> |

Proven by: [omdurman-rules::src::lib::historical\\_victory\\_ladders\\_match\\_manual\\_bands](#)

Covered [omdurman-rules::src::lib::historical\\_victory\\_level\\_for\\_dervish](#) by [omdurman-rules::src::lib::historical\\_victory\\_level\\_for\\_anglo\\_egyptian](#) tests:

§9.31 – Bonus game map

**implemented**  
manual page unknown

Only the small FALL OF KHARTOUM scenario map is used for this game.

| File                                                                                          | Symbol                      | Code Snippet                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/board_data.rs:37</a><br>GH:omdurman-rules/src/board_data.rs:37 | <b>fall_of_khartoum_map</b> | <pre>35   36   /// The Fall-of-Khartoum board (§9.3). 37   pub fn fall_of_khartoum_map_data() -&gt; MapData { 38       FOK.get_or_init(   parse(FOK_RON)).clone() 39   }</pre> |

Covered by tests: `omdurman-app::src::tests::scenario_maps_to_board` `omdurman-app::src::tests::start_game_scenario_selects_board`

§9.32 – Set Up (Bonus)

`descriptive`  
manual page unknown

Set Up

§9.33 – Scenario Length (Bonus)

`implemented`  
manual page unknown

Scenario Length  
Variable, see victory conditions (9.35). Rarely lasts five turns.

See also: §9.35

| File                                                                                                      | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                              |
|-----------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>omdurman-rules/src/turn_track.rs:172</code><br><code>GH:omdurman-rules/src/turn_track.rs:172</code> | <code>FALL_OF_KHARTOUM</code> | <pre>170   /// (the rulebook fixes none); only `day_night` is 171   rule-bearing (night halves 172   /// Anglo-Egyptian movement and ranges and bars 173   howitzer fire, §8.1). 174   pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] =       [           TurnEntry {               turn: 1,</pre> |

Proven by: `omdurman-rules::src::turn_track::scenario_turn_indexing_is_exact_for_every_scenario`  
`omdurman-rules::src::turn_track::fall_of_khartoum_night_window_is_turns_one_and_two`

Covered by tests: `omdurman-rules::src::turn_track::fall_of_khartoum_turn_one_is_night`  
`omdurman-rules::src::turn_track::fall_of_khartoum_turns_3_to_8_are_day`

§9.34 – Special Rules (Bonus)

`descriptive`  
manual page unknown

Special Rules

§9.35 – Victory Conditions (Bonus)

`implemented`  
manual page unknown

Victory Conditions  
Victory is determined by how many turns it takes the Dervish player to eliminate the GORDON leader unit and how many Dervish units are eliminated:

- Dervish decisive: eliminate GORDON turn four or sooner.
- Dervish tactical: eliminate GORDON turn five.
- Dervish marginal: eliminate GORDON turn six.
- British marginal: GORDON survives end of turn six.
- British tactical: GORDON survives end of turn seven.
- British decisive: GORDON survives end of turn eight.

(see manual for full text)

| File                                                                                | Symbol                          | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:1446</a><br>GH:omdurman-rules/src/lib.rs:1446 | <b>FoKVictoryLevel::resolve</b> | <pre> 1444       /// rulebook: GORDON dies turn 5 (tactical) with 1445       24 Dervish losses (-2 1446       /// levels) nets a British marginal. 1447       pub fn resolve(gordon_died_turn: Option&lt;u8&gt;, 1448       scenario_end_turn: u8, dervish_lost: i16) -&gt; Self { 1449           let base = Self::base(gordon_died_turn, 1450           scenario_end_turn); 1451           let base_idx = Self::LADDER </pre> |

Proven by: [omdurman-rules::src::lib::fok\\_victory\\_ladder\\_penalties\\_shift\\_monotonically](#)

Covered

by tests: [omdurman-rules::src::lib::fok\\_victory\\_level\\_worked\\_example](#) [omdurman-rules::src::lib::fok\\_victory\\_level\\_gordon\\_died\\_early](#) [omdurman-rules::src::lib::fok\\_victory\\_level\\_late\\_gordon\\_death](#) [omdurman-rules::src::lib::fok\\_victory\\_level\\_gordon\\_survived](#)

## §9.111 – Dervish set up (Campaign)

**implemented**

manual page unknown

Dervish player sets up first, moves second.

- Isa Zachneih infantry unit: anywhere on the east bank, in or south of El Debeba.
- KHALIFA ABDULLAH: in the walled city of Omdurman, in either palace hex.
- 3 artillery units, and all Taiasha units: anywhere in the walled city of Omdurman.
- 17 forts: anywhere on the mapsheet south of the Khor Shambat on the west bank, and/or south of all Halfaya hut hexes on the east bank and Nile River islands.
- 2 gunboats: any south edge Nile River hexes.

**\*\*9.112) Dervish reinforcements:\*\*** all reinforcements enter on the west edge of the mapsheet, south of the Khor Shambat. Each unit pays the terrain cost of the hex through which it enters, no matter how many units enter through

(see manual for full text)

See also: §9.112

| File                                                                                                  | Symbol                | Code Snippet                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state.rs:243</a><br>GH:omdurman-rules/src/effects/state.rs:243 | <b>setup_complete</b> | <pre> 241       /// currently shares the same "both sides deployed" 242       gate; when a scenario 243       /// needs a different minimum, branch on 244       `self.scenario` here. 245       pub fn setup_complete(&amp;self) -&gt; Result&lt;(), 246       RuleError&gt; { 247           let has =  player  { 248               self.units </pre> |

Covered

by

tests:

[omdurman-app::src::scenario\\_setup::campaign\\_has\\_no\\_fixed\\_placements](#) [omdurman-rules::src::unit\\_profiles::hadendowa\\_first\\_cell\\_is\\_isa\\_zachneih](#) [omdurman-rules::src::effects::tests::campaign\\_deployment\\_is\\_boat\\_land\\_exclusive](#) [omdurman-rules::src::effects::tests::campaign\\_setup\\_rejects\\_non\\_initial\\_force](#)

## §9.112 – Dervish reinforcements (Campaign)

**implemented**

manual page unknown

| File                                                                     | Symbol                     | Code Snippet                                                                                                                     |
|--------------------------------------------------------------------------|----------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/effect.rs:182</a><br>GH:omdurman- | <b>PlaceReinforcements</b> | <pre> 180       // -- Reinforcement / placement 181       ----- 182       /// Place reinforcements onto the map (rulebook </pre> |

|                                                                                                           |                                |                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rules/src/effects/<br>effect.rs:182                                                                       |                                | <pre> \$9.112, \$9.113). 182       PlaceReinforcements(Vec&lt;UnitPlacement&gt;), 183   184       // -- Scenario-specific ----- </pre>                                                                                                                                                                                                      |
| omdurman-<br>rules/src/effects/<br>dispatch.rs:44<br>GH:omdurman-<br>rules/src/effects/<br>dispatch.rs:44 | <b>apply_place_reinforceme</b> | <pre> 42           GameEffect::Demolition { unit_id, target } =&gt; apply_demolition(state, *unit_id, *target), 43           GameEffect::PlaceReinforcements(placements) =&gt; { 44               apply_place_reinforcements(state, placements) 45           } 46           GameEffect::DervishDesertion { roll, deserters } =&gt; { </pre> |
| omdurman-<br>rules/src/reinforcements.rs:69<br>GH:omdurman-<br>rules/src/reinforcements.rs:69             | <b>dervish_campaign_sched</b>  | <pre> 67   /// All reinforcements enter on the west edge, south of the Khor Shambat. 68   /// Each unit pays terrain cost of the hex it enters through. 69   pub fn dervish_campaign_schedule() -&gt; ReinforcementSchedule { 70       ReinforcementSchedule { 71           player: Player::Dervish, </pre>                                 |
| omdurman-<br>types/src/<br>lib.rs:524<br>GH:omdurman-<br>types/src/lib.rs:524                             | <b>Location</b>                | <pre> 522   /// Named map landmarks (rulebook mapsheet, \$9.111, \$9.113, \$9.212 scenarios). 523   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 524   pub enum Location { 525       FortMakran, 526       NorthFort, </pre>                                                                        |
| omdurman-<br>types/src/<br>lib.rs:604<br>GH:omdurman-<br>types/src/lib.rs:604                             | <b>SetupLetter</b>             | <pre> 602   /// Each letter marks a specific hex where a Dervish leader is placed. 603   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug, strum::Display)] 604   pub enum SetupLetter { 605       Y, 606       K, </pre>                                                                                                 |
| omdurman-<br>types/src/<br>lib.rs:737<br>GH:omdurman-<br>types/src/lib.rs:737                             | <b>Faction</b>                 | <pre> 735   /// `Some(BrigadeId::friendlies())`. 736   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)] 737   pub enum Faction { 738       Dervish { 739           tribe: DervishTribe, </pre>                                                                                                                          |

Proven by: [omdurman-rules::src::reinforcements::wave\\_for\\_turn\\_answers\\_exactly\\_on\\_the\\_printed\\_schedule](#)

Covered by tests: [omdurman-rules::src::reinforcements::dervish\\_schedule\\_has\\_three\\_waves](#)  
[omdurman-rules::src::reinforcements::dervish\\_wave\\_one\\_has\\_baggaara\\_and\\_three\\_leaders](#)  
[omdurman-rules::src::reinforcements::dervish\\_wave\\_two\\_has\\_hadendowa](#)  
[omdurman-rules::src::reinforcements::dervish\\_wave\\_three\\_is\\_all\\_remaining](#)  
[omdurman-rules::src::reinforcements::wave\\_for\\_turn\\_returns\\_correct\\_wave](#)  
[omdurman-rules::src::effects::tests::campaign\\_reinforcements\\_gate\\_by\\_wave](#)  
[omdurman-rules::src::effects::tests::campaign\\_setup\\_rejects\\_non\\_initial\\_force](#)

## \$9.113 – Anglo-Egyptian reinforcements (Campaign)

**implemented**

manual page unknown

The Anglo-Egyptian player moves first. There are no Anglo-Egyptian units on the mapsheet at start. The GORDON unit is not used in this scenario.

- The leader units KITCHENER, GATACRE, and HUNTER may enter anytime during the first four game turns and do not count against the 12 units per turn limit. All three leaders must be in play by the end of turn four!

- All gunboats enter through any north edge Nile River hex, paying one movement point for the first hex entered. The “Friendlies” brigade enters through the Abu Alim hut hex on the east bank, paying eight movement points per unit. All other Anglo-Egyptian units enter through the west bank “ANGLO-EGYPTIAN ENTRANCE AREA”, each unit paying one movement point to enter the mapsheet.
- Turn 1) Any three gunboats; “Friendlies” brigade; Egyptian Cavalry; Horse Artillery; and two infantry brigades

(see manual for full text)

| File                                                                                                    | Symbol                         | Code Snippet                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/reinforcements.rs:126</a><br>GH:omdurman-rules/src/reinforcements.rs:126 | <b>anglo_egyptian_campaign</b> | <pre> 124   /// - "Friendlies" enter via Abu Alim hut on the east       bank (8 MP per unit). 125   /// - All other AE units enter via the Anglo-Egyptian       Entrance Area (1 MP). 126   pub fn anglo_egyptian_campaign_schedule() -&gt;       ReinforcementSchedule { 127       let free_leaders = vec![ 128           CampaignLeader::British(BritishLeader::Kitchener), </pre> |

Proven by: [omdurman-rules::src::reinforcements::wave\\_for\\_turn\\_answers\\_exactly\\_on\\_the\\_printed\\_schedule](#)

Covered by tests:

- [omdurman-rules::src::reinforcements::anglo\\_egyptian\\_schedule\\_has\\_four\\_waves](#)
- [omdurman-rules::src::reinforcements::anglo\\_egyptian\\_leaders\\_available\\_each\\_wave](#)
- [omdurman-rules::src::reinforcements::anglo\\_egyptian\\_turn\\_four\\_is\\_all\\_remaining](#)
- [omdurman-rules::src::effects::tests::campaign\\_reinforcement\\_cap\\_and\\_double\\_entry](#)
- [omdurman-rules::src::effects::tests::campaign\\_gunboats\\_quota\\_three\\_per\\_turn](#)
- [omdurman-rules::src::effects::tests::reinforcement\\_rejected\\_onto\\_enemy\\_occupied\\_hex](#)
- [omdurman-rules::src::effects::tests::campaign\\_setup\\_rejects\\_non\\_initial\\_force](#)

## §9.211 – Anglo-Egyptian set up first, moves second (Historical)

**implemented**

manual page unknown

The Anglo-Egyptian player sets up first, and moves second:

- Not in play: GORDON leader unit, “Friendlies” brigade.
- Gunboats start in any Nile River hexes adjacent to the Zariba.
- Camel Corps, Egyptian Cavalry, and Horse Artillery start in the village of Kerreri hut hexes.
- All remaining Anglo-Egyptian units set up in the 13 hexes of the Zariba.

| File                                                                                                        | Symbol              | Code Snippet                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/dispatch.rs:346</a><br>GH:omdurman-rules/src/effects/dispatch.rs:346 | <b>first_player</b> | <pre> 344   345   /// The player who moves first in a scenario (§4,       §9.113, §9.212, §9.322). 346   pub fn first_player(scenario: Scenario) -&gt; Player { 347       match scenario { 348           Scenario::Campaign =&gt; Player::AngloEgyptian, </pre> |

Covered by tests: [omdurman-rules::src::effects::tests::historical\\_setup\\_rejects\\_not\\_in\\_play\\_units](#)

## §9.212 – Dervish set up (Historical) – deployment zones and leader-to-hex mapping

**implemented**

manual page unknown

The Dervish player sets up second, and moves first.

- Not in play: Isa Zachneih, gunboats, and forts.

- All Dervish units must be set up out of the line of sight of all Anglo-Egyptian units.
- Dervish leaders start on the lettered hexes:
  - A: Ali Wad Helu
  - D: Sheik El Din
  - Y: Yakub
  - K: Khalifa Abdullah
  - S: Sherif

(see manual for full text)

| File                                                                                                  | Symbol                             | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state.rs:353</a><br>GH:omdurman-rules/src/effects/state.rs:353 | <b>in_deployment_zone</b>          | <pre> 351       /// plan / UI rather than this hex predicate.           Documented, not silently 352       /// dropped. 353       pub fn in_deployment_zone(&amp;self, player: Player,           hex: HexCoord, is_boat: bool) -&gt; bool { 354           // No board attached -&gt; permissive (unit tests,               unbound session). 355           if self.board.terrain.is_empty() { </pre> |
| <a href="#">omdurman-rules/src/lib.rs:387</a><br>GH:omdurman-rules/src/lib.rs:387                     | <b>DervishLeader::setup_letter</b> | <pre> 385       /// (§9.212): A→Ali Wad Helu, D→Sheik El Din,           Y→Yakub, K→Khalifa Abdullah, 386       /// S→Sherif, O→Osman Digna. Inverse of           [`dervish_leader_for_setup_letter`]. 387       pub fn setup_letter(self) -&gt; SetupLetter { 388           match self { 389               DervishLeader::AliWadHelu =&gt;                   SetupLetter::A, </pre>                  |
| <a href="#">omdurman-rules/src/lib.rs:403</a><br>GH:omdurman-rules/src/lib.rs:403                     | <b>dervish_leader_for_setup</b>    | <pre> 401       /// inherent impl here, so the mapping is a free           function -- the bijective 402       /// inverse of [`DervishLeader::setup_letter`]. 403       pub fn dervish_leader_for_setup_letter(letter:           SetupLetter) -&gt; DervishLeader { 404           match letter { 405               SetupLetter::A =&gt; DervishLeader::AliWadHelu, </pre>                           |

Covered by tests:

- [omdurman-rules::src::effects::tests::deploy\\_rejected\\_outside\\_zone](#)
- [omdurman-rules::src::unit\\_profiles::embedded\\_leaders\\_resolve\\_from\\_their\\_host\\_section](#)
- [omdurman-app::src::scenario\\_setup::historical\\_places\\_all\\_six\\_leaders\\_when\\_anchors\\_present](#)
- [omdurman-app::src::scenario\\_setup::missing\\_anchor\\_is\\_reported\\_not\\_dropped\\_silently](#)
- [omdurman-rules::src::effects::tests::setup\\_letter\\_dervish\\_leader\\_roundtrip](#)
- [omdurman-rules::src::effects::tests::setup\\_letter\\_to\\_dervish\\_leader\\_known\\_values](#)
- [omdurman-rules::src::effects::tests::historical\\_setup\\_rejects\\_not\\_in\\_play\\_units](#)

## §9.231 – Thorn hedge hexsides

**implemented**

manual page unknown

| File                                                                              | Symbol                  | Code Snippet                                                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/lib.rs:939</a><br>GH:omdurman-rules/src/lib.rs:939 | <b>ZaribaThornHedge</b> | <pre> 937       Terrain(i16), 938       /// -2 thorn-hedge defensive modifier (§9.231). 939       ZaribaThornHedge, 940       /// -4 trench defensive modifier (§9.232). Only           applies vs. "entrenched" 941       /// units (those Nile-side of the trench hexside). </pre> |
| <a href="#">omdurman-types/src/lib.rs:206</a><br>GH:omdurman-types/src/lib.rs:206 | <b>ZaribaThornHedge</b> | <pre> 204       Crest, 205       /// Historical-scenario thorn-hedge segment of the           Zariba (§9.231). 206       ZaribaThornHedge, 207       /// Historical-scenario trench segment of the           Zariba (§9.232). 208       ZaribaTrench, </pre>                         |

|                                                                               |                               |                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/<br>board.rs:284<br>GH:omdurman-rules/src/<br>board.rs:284 | <b>has_zariba_thorn_hedge</b> | <pre> 282       /// hexside on its perimeter – i.e. whether the           ZaribaThornHedge modifier 283       /// applies (§9.231). 284       pub fn has_zariba_thorn_hedge(&amp;self, hex: HexCoord)           -&gt; bool { 285           for n in hex.neighbors() { 286               if let Some(kind) = self.hexside_between(hex, n) </pre> |
|-------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Covered by tests: [omdurman-rules::src::effects::tests::zariba\\_fire\\_penalties\\_apply\\_to\\_dervish\\_fire\\_only](#)

## §9.232 – Trench hexsides

**implemented**

manual page unknown

| File                                                                  | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------------------------------------------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/<br>lib.rs:942<br>GH:omdurman-rules/src/lib.rs:942 | <b>ZaribaTrenchEntrenched</b> | <pre> 940       /// -4 trench defensive modifier (§9.232). Only           applies vs. "entrenched" 941       /// units (those Nile-side of the trench hexside). 942       ZaribaTrenchEntrenched, 943   } 944   </pre>                                                                                                                         |
| omdurman-types/src/<br>lib.rs:208<br>GH:omdurman-types/src/lib.rs:208 | <b>ZaribaTrench</b>           | <pre> 206       ZaribaThornHedge, 207       /// Historical-scenario trench segment of the           Zariba (§9.232). 208       ZaribaTrench, 209       /// One of the two end hexsides of a Zariba trench           segment that connect to 210       /// the Nile River (§9.233). Units may only enter/           leave the Zariba via </pre> |

Proven by: [omdurman-rules::src::lib::melee\\_modifier\\_keeps\\_roll\\_legal](#)

Covered by tests: [omdurman-rules::src::effects::tests::trench\\_entrenched\\_units\\_take\\_trench\\_modifiers](#)

## §9.233 – Zariba entry/exit costs

**implemented**

manual page unknown

Units may only enter and/or leave the Zariba via the two end hexsides that connect to the Nile River, paying +2 movement points to cross (Exception: advance after combat across an entrenched hexside).

| File                                                                          | Symbol                        | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-types/src/<br>lib.rs:258<br>GH:omdurman-types/src/lib.rs:258         | <b>blocks_movement</b>        | <pre> 256       /// `omdurman-rules`). The trench *end* variants           are therefore intentionally 257       /// not blocking. 258       pub fn blocks_movement(self) -&gt; bool { 259           matches!( 260               self, </pre>                                                                                                                                                    |
| omdurman-rules/src/<br>board.rs:299<br>GH:omdurman-rules/src/<br>board.rs:299 | <b>zariba_entry_surcharge</b> | <pre> 297       /// movement points to cross"). Returns 2 when the           edge between `from` and 298       /// `to` is one of the two trench ends, else 0. 299       pub fn zariba_entry_surcharge(&amp;self, from:           HexCoord, to: HexCoord) -&gt; i16 { 300           match self.hexside_between(from, to) { 301               Some(k) if k.is_zariba_trench_end() =&gt; 2, </pre> |
| omdurman-rules/src/effects/<br>state.rs:856                                   | <b>movement_cost_for</b>      | <pre> 854       /// 855       /// §5.42: entering or leaving an enemy ZOC adds no           MP cost. </pre>                                                                                                                                                                                                                                                                                      |

|                                                                                           |                        |                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GH:omdurman-rules/src/effects/state.rs:856                                                |                        | <pre> 856           pub fn movement_cost_for( 857               &amp;self, 858               unit: &amp;UnitPlacement, </pre>                                                                                                                                                           |
| omdurman-rules/src/effects/movement.rs:31<br>GH:omdurman-rules/src/effects/movement.rs:31 | <b>apply_move_unit</b> | <pre> 29   /// the true terrain cost (\$5.11) and enforces gunboat      upstream/downstream 30   /// allowances (\$5.24); otherwise it falls back to the      caller-supplied `cost`. 31   pub fn apply_move_unit( 32       state: &amp;mut GameState, 33       unit_id: UnitId, </pre> |

Proven by: [omdurman-types::src::lib::zariba\\_trench\\_ends\\_differ\\_only\\_in\\_the\\_entry\\_rule](#)

Covered by tests: [omdurman-rules::src::effects::tests::zariba\\_end\\_hexside\\_costs\\_extra\\_mp](#)  
[omdurman-rules::src::effects::tests::zariba\\_thorn\\_hedge\\_blocks\\_movement](#)

## §9.321 – British set up (Bonus)

**implemented**

manual page unknown

The British player sets up first, moves second:

- General GORDON leader unit in the palace.
- Two old style (unnamed) gunboats in any Nile River hexes.
- Set up in any building or hut hexes of Khartoum, Forts Makran and/or Buri, and/or adjacent to any wall hex:
  - one Egyptian Battalion artillery unit
  - two British infantry units (represents Caucasian troops)
  - three Egyptian infantry units (represents Cairo “Bazouks”)
  - four Sudan infantry units (represents Sudanese blacks)
  - four “Friendlies” units (represents the Shaggyeh)

(see manual for full text)

| File                                                                                  | Symbol                  | Code Snippet                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/scenario_setup.rs:42<br>GH:omdurman-rules/src/scenario_setup.rs:42 | <b>FALL_OF_KHARTOUM</b> | <pre> 40   /// North Fort uses a campaign HadendowaForts counter      (one of the spare fort 41   /// sprites). 42   pub const FALL_OF_KHARTOUM_SETUP: &amp;[FixedPlacement] =      &amp;[ 43       FixedPlacement { 44           section: SectionName::BritishBoats, </pre> |

Covered by tests: [omdurman-rules::src::effects::tests::confirm\\_ready\\_rejected\\_below\\_scenario\\_target](#)  
[omdurman-rules::src::effects::tests::remove\\_deployed\\_unit\\_happy\\_path](#)  
[omdurman-app::src::scenario\\_setup::fall\\_of\\_khartoum\\_places\\_gordon\\_in\\_the\\_palace](#)  
[omdurman-app::src::scenario\\_setup::fall\\_of\\_khartoum\\_reports\\_missing\\_palace](#)  
[omdurman-app::src::scenario\\_setup::fall\\_of\\_khartoum\\_fort\\_landmarks\\_sit\\_at\\_the\\_correct\\_hexes](#)  
[omdurman-rules::src::effects::tests::fok\\_ae\\_gunboat\\_deploys\\_only\\_on\\_nile](#)  
[omdurman-rules::src::effects::tests::fok\\_ae\\_land\\_unit\\_rejected\\_on\\_nile](#)  
[omdurman-rules::src::effects::tests::fok\\_setup\\_complete\\_requires\\_full\\_oob](#)  
[omdurman-rules::src::effects::tests::deploy\\_via\\_real\\_sprite\\_resolution\\_matches\\_engine](#)  
[omdurman-rules::src::effects::tests::british\\_boats\\_named\\_vs\\_old\\_gunboat\\_detection](#)  
[omdurman-rules::src::effects::tests::fok\\_order\\_of\\_battle\\_british](#) [omdurman-rules::src::effects::tests::fok\\_order\\_of\\_battle\\_dervish](#)  
[omdurman-rules::src::effects::tests::fok\\_caps\\_bind\\_across\\_counter\\_variants](#)

## §9.322 – Dervish enters turn one (Bonus)

**implemented**

manual page unknown

Dervish player moves first: enters turn one through any hexes on the south or east edge of the map.

- 32 Mulazmin units (represents combined forces of Wad El Nejumi, Abu Girgeh, and Sheik El Obeid)
- 2 Hadendowa; 6 Kehena; 5 Degheim (represents Mahdi's combined west bank forces)
- 3 Dervish artillery units.

| File                                                                                                  | Symbol                     | Code Snippet                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/unit_profiles.rs:389</a><br>GH:omdurman-rules/src/unit_profiles.rs:389 | <b>ali_wad_helu</b>        | <pre>387   /// §9.2/§9.3) unplaceable -- the FoK order-of-battle 388   table 389   pub fn ali_wad_helu(col: u32, row: u32) -&gt;     Option&lt;Classification&gt; { 390       match (col, row) { 391           (0, 0) =&gt;     dervish_leader(DervishLeader::AliWadHelu),</pre>                                                                                 |
| <a href="#">omdurman-types/src/lib.rs:836</a><br>GH:omdurman-types/src/lib.rs:836                     | <b>sections_for_picker</b> | <pre>834   /// counter of those two forces and leave the 835   §9.322 entry force -- and 836   /// with it the setup Ready gate (§9.2/§9.3) --     unplaceable. 837   pub fn sections_for_picker(self) -&gt; Option&lt;&amp;'static     [SectionName]&gt; { 838       match self { 839           Scenario::Campaign   Scenario::Historical     =&gt; None,</pre> |

Covered by tests: [omdurman-rules::src::unit\\_profiles::ali\\_wad\\_helu\\_block\\_resolves\\_leader\\_and\\_degelim\\_tribes](#)  
[omdurman-rules::tests::fok\\_setup\\_flow::degheim\\_counters\\_resolve\\_to\\_the\\_degheim\\_tribe](#)  
[omdurman-rules::src::effects::tests::fok\\_setup\\_complete\\_requires\\_full\\_oob](#)  
[omdurman-rules::src::effects::tests::fok\\_dervish\\_land\\_unit\\_rejected\\_on\\_nile](#)  
[omdurman-types::src::lib::fok\\_picker\\_allowlist\\_has\\_dervish\\_entry\\_force\\_blocks](#)  
[omdurman-rules::src::effects::tests::fok\\_order\\_of\\_battle\\_dervish](#)  
[omdurman-rules::src::effects::tests::fok\\_dervish\\_east\\_edge\\_on\\_diamond\\_board](#)

## §9.341 – Turn 1 is always a night turn (Bonus)

**implemented**

manual page unknown

Turn 1 is always a night turn (see 8.1).

See also: §8.1

| File                                                                                            | Symbol                  | Code Snippet                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/turn_track.rs:172</a><br>GH:omdurman-rules/src/turn_track.rs:172 | <b>FALL_OF_KHARTOUM</b> | <pre>170   /// (the rulebook fixes none); only `day_night` is 171   rule-bearing (night halves 172   /// Anglo-Egyptian movement and ranges and bars     howitzer fire, §8.1). 173   pub const FALL_OF_KHARTOUM_TURN_TRACK: [TurnEntry; 8] =     [ 174       TurnEntry {         turn: 1,</pre> |

Proven by: [omdurman-rules::src::turn\\_track::scenario\\_turn\\_indexing\\_is\\_exact\\_for\\_every\\_scenario](#)  
[omdurman-rules::src::turn\\_track::fall\\_of\\_khartoum\\_night\\_window\\_is\\_turns\\_one\\_and\\_two](#)

Covered by tests: [omdurman-rules::src::turn\\_track::fall\\_of\\_khartoum\\_turn\\_one\\_is\\_night](#)

## §9.342 – All hexes are playable, including half hexes (Bonus)

**implemented**

manual page unknown

All hexes are playable, including hexes showing up half or less.

| File                                                                                          | Symbol                      | Code Snippet                                                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/board_data.rs:37</a><br>GH:omdurman-rules/src/board_data.rs:37 | <b>fall_of_khartoum_map</b> | <pre>35  <br/>36   /// The Fall-of-Khartoum board (§9.3).<br/>37   pub fn fall_of_khartoum_map_data() -&gt; MapData {<br/>38       FOK.get_or_init(   parse(FOK_RON)).clone()<br/>39   }</pre> |

Covered by tests: [omdurman-rules::src::effects::tests::fall\\_of\\_khartoum\\_board\\_excludes\\_no\\_hexes](#)

## §9.343 – Both players use the Dervish Range Effects Table (Bonus)

implemented

manual page unknown

Both players must use the Dervish Range Effects Table.

| File                                                                                              | Symbol                | Code Snippet                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/fire.rs:53</a><br>GH:omdurman-rules/src/effects/fire.rs:53 | <b>range_band_for</b> | <pre>51   /// in FALL OF KHARTOUM *both* players use the Dervish<br/>     Range Effects Table<br/>52   /// (§9.343).<br/>53   pub fn range_band_for(<br/>54       scenario: Scenario,<br/>55       player: Player,</pre> |

Proven by: [omdurman-rules::src::effects::fire::range\\_band\\_for\\_routes\\_to\\_the\\_right\\_faction\\_table](#)  
[omdurman-rules::src::effects::fire::range\\_table\\_player\\_for\\_routes\\_friendlyes\\_and\\_fok\\_to\\_the\\_dervish\\_table](#)

Covered by tests: [omdurman-rules::src::effects::tests::fok\\_both\\_players\\_use\\_dervish\\_range\\_table](#)

## §9.344 – Dervish controls the North Fort (Bonus)

implemented

manual page unknown

The Dervish player controls the “North Fort” and may fire its guns.

| File                                                                                                    | Symbol                    | Code Snippet                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/scenario_setup.rs:42</a><br>GH:omdurman-rules/src/scenario_setup.rs:42   | <b>FALL_OF_KHARTOUM</b>   | <pre>40   /// North Fort uses a campaign HadendowaForts counter<br/>     (one of the spare fort<br/>41   /// sprites).<br/>42   pub const FALL_OF_KHARTOUM_SETUP: &amp;[FixedPlacement] =<br/>     &amp;[<br/>43       FixedPlacement {<br/>44           section: SectionName::BritishBoats,</pre>                                                                            |
| <a href="#">omdurman-rules/src/effects/state.rs:1400</a><br>GH:omdurman-rules/src/effects/state.rs:1400 | <b>hex_has_enemy_fort</b> | <pre>1398   /// may neither occupy an enemy fort nor advance<br/>     after combat into one<br/>1399   /// (forts are never captured -- only destroyed,<br/>     §6.62/§6.53/§7.6).<br/>1400   pub fn hex_has_enemy_fort(&amp;self, hex: HexCoord,<br/>     mover: Player) -&gt; bool {<br/>1401       self.units.iter().any( u  {<br/>1402           u.position == hex</pre> |

Covered by tests: [omdurman-app::src::scenario\\_setup::fall\\_of\\_khartoum\\_places\\_gordon\\_in\\_the\\_palace](#)  
[omdurman-app::src::scenario\\_setup::fall\\_of\\_khartoum\\_fort\\_landmarks\\_sit\\_at\\_the\\_correct\\_hexes](#)  
[omdurman-app::src::scenario\\_setup::placement\\_done\\_gate\\_matches\\_by\\_identity\\_not\\_allocated\\_id](#)

omdurman-rules::src::effects::tests::fok\_order\_of\_battle\_dervish  
omdurman-types::src::lib::fok\_picker\_allowlist\_has\_dervish\_entry\_force\_blocks

## §9.345 – Gunboat White Nile <-> Blue Nile crossing (Bonus)

implemented

manual page unknown

The British gunboats may move from the White Nile to the Blue Nile and vice-versa at an off-board movement cost of six “upstream” movement points.

| File                                                                                                    | Symbol                          | Code Snippet                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state.rs:1384</a><br>GH:omdurman-rules/src/effects/state.rs:1384 | <b>is_nile_mouth_crossing</b>   | <pre>1382       /// must be named on the board, else this is 1383       `false` and the move falls 1384       /// through to the ordinary contiguous-Nile rules. 1385       pub fn is_nile_mouth_crossing(&amp;self, from: 1386       HexCoord, to: HexCoord) -&gt; bool { 1387           let white = self 1388           .board</pre>                       |
| <a href="#">omdurman-types/src/lib.rs:540</a><br>GH:omdurman-types/src/lib.rs:540                       | <b>Location::WhiteNileMouth</b> | <pre>538       /// The off-board mouth of the White Nile branch 539       (FALL OF KHARTOUM §9.345) -- 540       /// a British gunboat may cross to the Blue Nile 541       mouth for 6 upstream MP. 542       WhiteNileMouth, 543       /// The off-board mouth of the Blue Nile branch 544       (FALL OF KHARTOUM §9.345). 545       BlueNileMouth,</pre> |

Covered by tests: omdurman-rules::src::effects::tests::fok\_gunboat\_crosses\_between\_nile\_mouths

## §9.346 – GORDON immobile, eliminated only at the Palace (Bonus)

implemented

manual page unknown

The GORDON leader unit starts in the palace and may not move during the scenario. He may only be eliminated by a Dervish unit passing through or occupying the palace hex (as normal movement or as advance after combat).

| File                                                                                                        | Symbol                         | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/dispatch.rs:482</a><br>GH:omdurman-rules/src/effects/dispatch.rs:482 | <b>check_gordon_palace</b>     | <pre>480       /// after combat). Records the turn (which fixes the 481       §9.35 victory level) and 482       /// ends the game. A no-op outside FoK, or once GORDON 483       is already gone. 484       pub fn check_gordon_palace(state: &amp;mut GameState) { 485           if state.scenario != Scenario::FallOfKhartoum    486           state.gordon_eliminated_turn.is_some() { 487               return; 488           }</pre> |
| <a href="#">omdurman-rules/src/lib.rs:644</a><br>GH:omdurman-rules/src/lib.rs:644                           | <b>UnitIdentity::is_gordon</b> | <pre>642       /// Whether this is the GORDON leader unit (§9.32, 643       §9.346) -- the immobile 644       /// palace defender whose elimination ends FALL OF 645       KHARTOUM (§9.35). 646       pub fn is_gordon(&amp;self) -&gt; bool { 647           matches!( 648               self,</pre>                                                                                                                                      |

Covered by tests: omdurman-rules::src::unit\_profiles::gordon\_is\_an\_immobile\_british\_leader

omdurman-rules::src::effects::tests::gordon\_survives\_means\_no\_elimination

omdurman-app::src::scenario\_setup::fall\_of\_khartoum\_places\_gordon\_in\_the\_palace

7/10 implemented

## §10 – Optional Rules

### §10 – Optional Rules

descriptive

manual page unknown

Optional Rules (Campaign game only)

It is suggested that the most intriguing employment of the following two options is to permit the Dervish player to have either one or the other, but the Anglo-Egyptian player doesn't know which one until he stumbles onto it. Players are advised that the employment of both optionals in the same game is not recommended.

### §10.1 – River Mines

descriptive

manual page unknown

River Mines

The Khalifa twice tried (unsuccessfully) to submerge a powerful mine in the Nile to sink or damage British gunboats. This option assumes that both attempts were successful.

### §10.2 – River Chain

descriptive

manual page unknown

River Chain

The Khalifa also tried (also unsuccessfully) to string a heavy chain across the Nile to stop or slow down the British gunboats. This option assumes the chain was emplaced.

### §10.11 – Secretly record mine hexes

implemented

manual page unknown

Prior to the commencement of play the Dervish player secretly records two Nile River hexes to be mined (the mines may not both be placed in the same hex). These hexes must be south of the E-W hexrow in which the Khor Shambat empties into the Nile.

| File                                                                                                  | Symbol                  | Code Snippet                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/setup.rs:330</a><br>GH:omdurman-rules/src/effects/setup.rs:330 | <b>apply_place_mine</b> | <pre>328   /// Lay a river mine during setup (§10.11). Validated       by 329   /// [<code>GameState::can_place_mine`</code>]. 330   pub fn apply_place_mine(state: &amp;mut GameState, hex:       HexCoord) -&gt; Result&lt;(), RuleError&gt; { 331       state.can_place_mine(hex)?; 332       state.mines.push(MinePlacement {</pre> |
| <a href="#">omdurman-rules/src/effects/state.rs:174</a><br>GH:omdurman-rules/src/effects/state.rs:174 | <b>GameState::mines</b> | <pre>172           friendlies_transport: None, 173           optional_rules: Vec::new(), 174           mines: Vec::new(), 175           chain: None, 176           board: BoardInfo::default(),</pre>                                                                                                                                   |
| <a href="#">omdurman-rules/src/</a>                                                                   | <b>OptionalRule</b>     | <pre>314   /// two should be in play (rulebook §10). 315   #[derive(Serialize, Deserialize, Clone, Copy,</pre>                                                                                                                                                                                                                          |

|                                                |  |                                                                                                           |
|------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------|
| lib.rs:316<br>GH:omdurman-rules/src/lib.rs:316 |  | PartialEq, Eq, Debug))<br>316   pub enum OptionalRule {<br>317       RiverMines,<br>318       RiverChain, |
|------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------|

Covered by tests: `omdurman-rules::src::effects::tests::mine_and_chain_limits_enforced_in_setup`  
`omdurman-rules::src::effects::tests::mines_and_chain_require_their_optional_rule`

## §10.12 – Mine resolution

implemented

manual page unknown

When a British gunboat enters a mined hex, the Dervish player must order it to stop as it has struck a mine. The Dervish player then resolves the effect of the mine's blast by rolling the ten-sided die:

- 1–4: No effect
- 5–7: Gunboat damaged, lost use of its engines and must drift two hexes per turn (with the current) for the rest of the game. No effect on guns or Maxims unless they drift out of range.
- 8–10: Gunboat sunk!

| File                                                                                    | Symbol                  | Code Snippet                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects/effect.rs:200<br>GH:omdurman-rules/src/effects/effect.rs:200 | <b>RiverMine</b>        | 198   // -- Optional rules<br>-----<br>199   /// River mine resolution (rulebook §10.12).<br>200   RiverMine {<br>201       gunboat_id: UnitId,<br>202       hex: HexCoord,                                                  |
| omdurman-rules/src/effects/river.rs:146<br>GH:omdurman-rules/src/effects/river.rs:146   | <b>apply_river_mine</b> | 144  <br>145   /// Apply a river-mine resolution (rulebook §10.12).<br>146   pub fn apply_river_mine(<br>147       state: &mut GameState,<br>148       gunboat_id: UnitId,                                                   |
| omdurman-rules/src/lib.rs:1120<br>GH:omdurman-rules/src/lib.rs:1120                     | <b>MineResult</b>       | 1118   /// British gunboat enters a mined hex.<br>1119   #[derive(Serialize, Deserialize, Clone, Copy, PartialEq, Eq, Debug)]<br>1120   pub enum MineResult {<br>1121       /// Roll 1-4: no effect.<br>1122       NoEffect, |

Proven by: `omdurman-rules::src::lib::mine_result_bands_match_the_printed_rule`  
`omdurman-rules::src::effects::river::river_mine_resolution_matches_the_printed_bands_and_fires_once`

Covered by tests: `omdurman-rules::src::effects::tests::mine_fires_once_and_spare_dervish`

## §10.13 – Mines consumed after both rolled for

implemented

manual page unknown

After both mines have been rolled for, no more are available.

| File                                                    | Symbol                  | Code Snippet                                                                                              |
|---------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------|
| omdurman-rules/src/effects/river.rs:146<br>GH:omdurman- | <b>apply_river_mine</b> | 144  <br>145   /// Apply a river-mine resolution (rulebook §10.12).<br>146   pub fn apply_river_mine(<br> |

|                                    |  |                                                           |
|------------------------------------|--|-----------------------------------------------------------|
| rules/src/effects/<br>river.rs:146 |  | 147   state: &mut GameState,<br>148   gunboat_id: UnitId, |
|------------------------------------|--|-----------------------------------------------------------|

Proven by: `omdurman-rules::src::effects::river::river_mine_resolution_matches_the_printed_bands_and_fires_once`  
`omdurman-rules::src::effects::river::a_triggered_mine_never_fires_again`

Covered by tests: `omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish`

§10.14 – Dervish gunboats pass safely

implemented

manual page unknown

The Dervish player’s gunboats may pass through the mined hexes with no ill effect (he knows where the mines are).

| File                                                                                               | Symbol                        | Code Snippet                                                                                                                                                                                  |
|----------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>omdurman-rules/src/effects/river.rs:146</code><br>GH:omdurman-rules/src/effects/river.rs:146 | <code>apply_river_mine</code> | <pre>144  <br/>145   /// Apply a river-mine resolution (rulebook §10.12).<br/>146   pub fn apply_river_mine(<br/>147       state: &amp;mut GameState,<br/>148       gunboat_id: UnitId,</pre> |

Proven by: `omdurman-rules::src::effects::river::dervish_gunboats_pass_mined_hexes_unharmd`

Covered by tests: `omdurman-rules::src::effects::tests::mine_fires_once_and_spares_dervish`

§10.21 – Secretly record chain hexes

implemented

manual page unknown

Prior to the commencement of play the Dervish player secretly records a line of river hexes (not exceeding four hexes long) across which the chain is strung. The hexes must be south of the E–W hexrow in which the Khor Shambat empties into the Nile.

| File                                                                                                 | Symbol                         | Code Snippet                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>omdurman-rules/src/effects/state.rs:618</code><br>GH:omdurman-rules/src/effects/state.rs:618   | <code>can_place_chain</code>   | <pre>616   /// Read-only check of a river-chain placement in<br/>617   setup (§10.21): Setup phase<br/>618   /// and at most [<code>MAX_CHAIN_HEXES</code>] hexes.<br/>619   pub fn can_place_chain(&amp;self, hexes: &amp;[HexCoord]) -<br/>620   &gt; Result&lt;(), RuleError&gt; {<br/>621       self.require_setup_phase()?;<br/>622       // Optional-rule gate: the chain exists only<br/>623       when the River Chain option</pre> |
| <code>omdurman-rules/src/effects/setup.rs:341</code><br>GH:omdurman-rules/src/effects/setup.rs:341   | <code>apply_place_chain</code> | <pre>339   /// Lay (or replace) the river chain during setup<br/>340   (§10.21). Validated by<br/>341   /// [<code>GameState::can_place_chain`</code>].<br/>342   pub fn apply_place_chain(state: &amp;mut GameState, hexes:<br/>343   &amp;[HexCoord]) -&gt; Result&lt;(), RuleError&gt; {<br/>344       state.can_place_chain(hexes)?;<br/>345       state.chain = Some(ChainPlacement {</pre>                                            |
| <code>omdurman-rules/src/effects/state.rs:1851</code><br>GH:omdurman-rules/src/effects/state.rs:1851 | <code>MAX_CHAIN_HEXES</code>   | <pre>1849  <br/>1850   /// Maximum contiguous Nile hexes the river chain may<br/>1851   span (§10.21).<br/>1852   pub const MAX_CHAIN_HEXES: usize = 4;<br/>1853   //</pre>                                                                                                                                                                                                                                                                 |

Covered by tests: `omdurman-rules::src::effects::tests::mine_and_chain_limits_enforced_in_setup`  
`omdurman-rules::src::effects::tests::mines_and_chain_require_their_optional_rule`

## §10.22 – Gunboat stops on chained hex

implemented

manual page unknown

When a British gunboat enters a “chained” river hex it must stop and may move no further that turn.

| File                                                                                                  | Symbol                  | Code Snippet                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/state.rs:895</a><br>GH:omdurman-rules/src/effects/state.rs:895 | <b>can_move_gunboat</b> | <pre>893       /// upstream movement allowance is their maximum           for that turn." Chained Nile 894       /// hexes stop the gunboat (§10.22). 895       pub fn can_move_gunboat(               &amp;self, 896           unit_id: UnitId,</pre> |

Covered by tests: `omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk`

## §10.23 – Sinking the chain

implemented

manual page unknown

No gunboats (British or Dervish) may cross the chain until it has been sunk by the British player. He may sink the chain by a) having an infantry or cavalry unit spend one complete turn on either riverbank adjacent to a “chained” river hex, or b) firing at the chain with artillery and achieving a 3 or more on the Combat Results Table.

| File                                                                                                  | Symbol                  | Code Snippet                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/effects/river.rs:190</a><br>GH:omdurman-rules/src/effects/river.rs:190 | <b>apply_sink_chain</b> | <pre>188       /// Sink the river chain (rulebook §10.23). Marks the           placed chain cleared so it 189       /// no longer stops gunboats (§10.22). 190       pub fn apply_sink_chain(state: &amp;mut GameState) -&gt;               Result&lt;(), RuleError&gt; { 191           match state.chain.as_mut() { 192               Some(chain) if !chain.sunk =&gt; {</pre> |

Covered by tests: `omdurman-rules::src::effects::tests::chain_stops_gunboat_until_sunk`

0/1 implemented

## §11 – Historical Notes

### §11 – Historical Notes

descriptive

manual page unknown

#### Historical Notes

In 1881 Mohammed Ahmed Ibn Al-Sayid Abdullah, the son of an obscure carpenter in the hinterlands of the Sudan, proclaimed himself the “Mahdi” — the Messiah of the Islamic faith. His timing was propitious indeed. Since the early 1820’s a corrupt Egypt, with the Sultan of Turkey’s blessing, had incessantly raped the Sudan, taking ivory and flooding the slave markets with some half million captured Sudanese blacks. By 1880, nearly 40,000 Egyptian troops occupied outposts scattered throughout the Sudan, enforcing Egypt’s hold on this lucrative ivory and slave trade and squeezing the native population dry through vicious and corrupt tax officials. All was controlled from Khartoum via the office of Governor General of the Sudan. The title had been held by a succession of individuals, including General Charles Gordon, whose appointment was an attempt to reinstate some rudimentary justice in the Sudan after France and Britain assumed joint political control of a bankrupt Egypt.

By 1881, however, Gordon’s term had expired and a new Governor General, again corrupt and incompetent,  
(see manual for full text)

See also: §9.1, §9.2, §9.3

0/1 implemented

# Credits

## §Credits – Credits

**descriptive**

manual page unknown

1/1 implemented

## Combat Results Table (shared reference)

### §CRT – Combat Results Table (shared by §6.22 fire and §7.7 melee)

implemented

manual page unknown

| File                                                                                                              | Symbol                      | Code Snippet                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">omdurman-rules/src/combat_results_table.rs:10</a><br>GH:omdurman-rules/src/combat_results_table.rs:10 | <b>FireFactorRow</b>        | <pre>8       serde::Serialize, serde::Deserialize, Clone, Copy, 9       PartialEq, Eq, PartialOrd, Ord, Hash, Debug, 10   } 11   pub enum FireFactorRow { 12       /// 1-5 factors 13       Row01to05,</pre>                                                                                                                                 |
| <a href="#">omdurman-rules/src/combat_results_table.rs:46</a><br>GH:omdurman-rules/src/combat_results_table.rs:46 | <b>from_total</b>           | <pre>44       /// Determine which row a given total fire factor 45       falls into (rulebook §6.22). 46       pub fn from_total(total: u16) -&gt; Self { 47           match total { 48               0..=5 =&gt; FireFactorRow::Row01to05,</pre>                                                                                            |
| <a href="#">omdurman-rules/src/combat_results_table.rs:88</a><br>GH:omdurman-rules/src/combat_results_table.rs:88 | <b>combat_results_table</b> | <pre>86       /// D = `Disrupt` (1/2 of target units, round up) 87       /// 1...5 = `Eliminate(n)` (that many units removed) 88       pub fn combat_results_table(row: FireFactorRow, roll: 89           DieRoll) -&gt; CombatResult { 90           crate::tables_data::CRT[row.index()][roll.value() 91               - 1] as usize]</pre> |
| <a href="#">omdurman-rules/src/lib.rs:1008</a><br>GH:omdurman-rules/src/lib.rs:1008                               | <b>CombatResult</b>         | <pre>1006       /// * `--` -- no effect 1007       #[derive(Serialize, Deserialize, Clone, Copy, 1008           PartialEq, Eq, Debug)] 1009       pub enum CombatResult { 1010           NoEffect, 1011           Disrupt,</pre>                                                                                                             |

Proven by: [omdurman-rules::src::lib::disrupt\\_half\\_is\\_rounded\\_up](#)  
[omdurman-rules::src::combat\\_results\\_table::crt\\_eliminate\\_count\\_stays\\_within\\_printed\\_bounds](#)  
[omdurman-rules::src::combat\\_results\\_table::crt\\_is\\_monotone\\_in\\_the\\_die\\_roll\\_for\\_every\\_row](#)

Covered by tests: [omdurman-rules::src::combat\\_results\\_table::ae\\_combat\\_results\\_table\\_lowest\\_is\\_no\\_effect](#)  
[omdurman-rules::src::combat\\_results\\_table::ae\\_combat\\_results\\_table\\_highest\\_is\\_eliminate\\_5](#)  
[omdurman-rules::src::combat\\_results\\_table::ae\\_combat\\_results\\_table\\_progresses\\_with\\_roll](#)  
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## **| Reference – Charts and Tables**

### **\$Reference – Charts and Tables**

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